METRIC

MIL-STD-2204

31 October 1992

MILITARY STANDARD

SURVIVABLE ADAPTABLE FIBER OPTIC EMBEDDED NETWORK (SAFENET)

AMSC N/A AREA MCCR

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FOREWORD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, Code 231-2B2, Washington, D.C., 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This document provides requirements for the implementation of fiber optic local area networks which are intended for use in support of mission critical computer resources.

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MIL-S-24623 - Splice, Fiber Optic, Shipboard, General Specification for (Metric)

MIL-S-24725 - Switches, Fiber Optic, Shipboard, General Specification for (Metric)

FEDERAL INFORMATION PROCESSING STANDARDS

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

GOSIP 2.0 - U. S. Government Open Systems Interconnection Profile (GOSIP), Federal Information Processing Standards Publication (FIPS PUB) 146-1, 3 Apr 1991

(Applications for copies should be addressed to the U. S. Department of Commerce, National Technical Information Service (NTIS), 5285 Port Royal Road, Springfield, VA 22161)

STANDARDS

MILITARY

MS3475 - Connectors, Plug, Electric, Series 2, Crimp Type, Bayonet Coupling, Classes A, L, S and W

HANDBOOKS

MILITARY

MIL-HDBK-818-1 - SAFENET Network Development Guidance

MIL-HDBK-818-2 - SAFENET Conformance Test Guidance (DRAFT)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVSEA 6710781 - Cable, Fiber Optic, Shipboard General Specification for

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(Applications for copies should be addressed to PMS-400B3B, Naval Sea Systems Command, National Center 2, Room 10E24, Washington DC 20362-5101)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST Special Pub. 500-202 - Stable Implementation Agreements for Open Systems Interconnection Protocols, Version 5, Edition 1, December 1991

(Applications for copies should be addressed to the U. S. Department of Commerce, National Technical Information Service (NTIS), 5285 Port Royal Road, Springfield, VA 22161)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

FDDI SMT - Fiber Distributed Data Interface - Station Management - ANSI X3T9/92-067. Proposed American National Standard.

(Applications for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

IEEE 802 - IEEE Standards for Local and Metropolitan Area Networks - Overview and Architecture. IEEE Std. 802-1990

(Applications for copies should be addressed to the Institute of Electrical and Electronic Engineers, 445 Hoes Lane, Piscataway, NJ 08854-4150)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 6523 - Data Exchange - Structure for the Identification of Organizations. February 1984

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ISO 7498 - Information Processing Systems - Open Systems Interconnection - Basic Reference Model. October 1984

ISO 7498-4 - Information Processing Systems - Open Systems Interconnection - Basic Reference Model Part 4 - OSI Management Framework. November 1984

ISO 8073 - Information Processing Systems - Open Systems Interconnection - Connection Oriented Transport Protocol Specification. December 1988

ISO 8073/ADD2 Information Processing Systems - Open Systems Interconnection - Connection Oriented Transport Protocol Specification - Addendum 2: Class 4 Operation over Connectionless-mode Network Service. September 1989

ISO 8326 - Information Processing Systems - Open Systems Interconnection - Basic Connection Oriented Session Service Definition. August 1987

ISO 8326/ADD2 Information Processing Systems - Open Systems Interconnection - Basic Connection Oriented Session Service Definition - ADD 2 to ISO 8326 to Incorporate Unlimited User Data

ISO 8327 - Information Processing Systems - Open Systems Interconnection - Basic Connection Oriented Session Protocol Specification. August 1987

ISO 8327/ADD2 Information Processing Systems - Open Systems Interconnection - Basic Connection Oriented Session Protocol Specification - ADD 2 to ISO 8327 to Incorporate Unlimited User Data

ISO 8348/ADD2 Information Processing Systems - Data Communications - Network Service Definition - Addendum 2: Network Layer Addressing. March 1988

ISO 8473 - Information Processing Systems - Data Communications - Protocol for providing the Connectionless-mode Network Service. December 1988

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ISO/IEC TR 8509 - Information Processing Systems - Open Systems Interconnection - Service Conventions. September 1987

ISO 8571-1 - Information Processing Systems - Open Systems Interconnection - File Transfer, Access and Management Part 1: General Introduction. October 1988

ISO 8571-2 - Information processing systems - Open Systems Interconnection - File Transfer, Access and Management Part 2: Virtual File Store Definition. 1988 (e)

ISO 8571-3 - Information Processing Systems - Open Systems Interconnection - File Transfer, Access and Management Part 3: The File Service Definition. 1988 (e)

ISO 8571-4 - Information Processing Systems - Open Systems Interconnection - File Transfer, Access and Management Part 4: The File Protocol Specification. 1988 (e)

ISO 8602 - Information Processing Systems - Open Systems Interconnection - Protocol for providing the Connectionless-mode Transport Service. December 1987

ISO 8649 - Information Processing Systems - Open Systems Interconnection - Service Definition for the Association Control Service Element. December 1988

ISO 8650 - Information Processing Systems - Open Systems Interconnection - Protocol Specification for the Association Control Service Element. December 1988

ISO 8802-2 - Information Processing Systems - Open Systems Interconnection - Local Area Networks - Part 2: Logical Link Control (LLC).

ISO 8822 - Information Processing Systems - Open Systems Interconnection - Connection Oriented Presentation Service Definition. August 1988

ISO 8823 - Information Processing Systems - Open Systems Interconnection - Connection Oriented

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Presentation Protocol Specification. August 1988

ISO 8824 - Information Processing Systems - Open Systems Interconnection, Specification of Abstract Syntax Notation One (ASN.1). 1990 (e)

ISO 8825 - Information Processing Systems - Open Systems Interconnection, Specification of Basic Encoding Rules for Abstract Syntax Notation One (ASN.1). 1990 (e)

ISO/IEC 9072-1 Information Processing Systems - Text Communication - Remote Operations - Part 1: Model, Notation and Service Definition. November 1989

ISO/IEC 9072-2 Information Processing Systems - Text Communication - Remote Operations - Part 2: Protocol Specification. November 1989

ISO 9314-1 - Information Processing Systems - Fibre Distributed Data Interface (FDDI) - Part 1: Token Ring Physical Layer Protocol (PHY). April 1989

ISO 9314-2 - Information Processing Systems - Fibre Distributed Data Interface (FDDI) - Part 2: Token Ring Media Access Control (MAC). May 1989

ISO 9314-3 - Information Processing Systems - Fibre Distributed Data Interface (FDDI) - Part 3: Token Ring Physical Layer, Medium Dependent (PMD).

ISO 9542 - Information Processing Systems - Telecommunications and Information Exchange Between Systems - End System to Intermediate System Routing Exchange Protocol for use in Conjunction with the Protocol for Providing the Connectionless-mode Network Service. August 1988

ISO/IEC TR 9577 - Information Technology - Telecommunications and Information Exchange between Systems - Protocol Identification in the Network Layer. October 1990

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ISO/IEC 9594-1 Information Technology - Open Systems Interconnection - The Directory - Part 1: Overview of Concepts, Models, and Services. 1990 (e)

ISO/IEC 9594-2 Information Technology - Open Systems Interconnection - The Directory - Part 2: Models. 1990 (e)

ISO/IEC 9594-3 Information Technology - Open Systems Interconnection - The Directory - Part 3: Abstract Service Definition. 1990 (e)

ISO/IEC 9594-4 Information Technology - Open Systems Interconnection - The Directory - Part 4: Procedures For Distributed Operation. 1990 (e)

ISO/IEC 9594-5 Information Technology - Open Systems Interconnection - The Directory - Part 5: Protocol Specification. 1990 (e)

ISO/IEC 9594-6 Information Technology - Open Systems Interconnection - The Directory - Part 6: Selected Attribute Types. 1990 (e)

ISO/IEC 9594-7 Information Technology - Open Systems Interconnection - The Directory - Part 7: Authentication Framework. 1990 (e)

ISO/IEC 9594-8 Information Technology - Open Systems Interconnection - The Directory - Part 8: Authentication Framework. 1990 (e)

ISO/IEC 9595 - Information Processing Systems - Open Systems Interconnection - Common Management Information Service Definition. May 1990

ISO/IEC 9596-1 Information Processing Systems - Open Systems Interconnection - Common Management Information Protocol Specification, Part 1. November 1990

ISO/IEC TR 10000-1 Information Technology - Framework and Taxonomy of International Standardized Profiles - Part 1: Framework. May 1990

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Check the source to verify that this is the current version before use.

ISO/IEC DIS 10040 Information Processing Systems - Open Systems Interconnection - Systems Management Overview.

ISO/IEC 10164-1 - Information Processing Systems - Open Systems Interconnection - Systems Management - Part 1: Object Management Function. September 1991

ISO/IEC 10164-2 - Information Processing Systems - Open Systems Interconnection - Systems Management - Part 2: State Management Function. September 1991

ISO/IEC 10164-3 - Information Processing Systems - Open Systems Interconnection - Systems Management - Part 3: Attributes for Representing Relationships. September 1991

ISO/IEC 10164-4 - Information Processing Systems - Open Systems Interconnection - Systems Management - Part 4: Alarm Reporting Function. August 1991

ISO/IEC 10164-5 - Information Processing Systems - Open Systems Interconnection - Systems Management - Part 5: Event Reporting Function. August 1991

ISO/IEC 10164-6 - Information Processing Systems - Open Systems Interconnection - Systems Management - Part 6: Log Control Function. June 1991

ISO/IEC 10165-1 - Information Processing Systems - Open Systems Interconnection - Structure of Management Information - Part 1: Management Information Model. June 1991

ISO/IEC 10165-2 - Information Processing Systems - Open Systems Interconnection - Structure of Management Information - Part 2: Definition of Management Information. August 1991

ISO/IEC 10165-4 - Information Processing Systems - Open Systems Interconnection - Structure of Management Information - Part 4:

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Guidelines for the Definitions of Managed Objects. July 1991

ISO/IEC TR 10178 - Information technology - Telecommunications and Information Exchange between Systems - The Structure and Coding of Logical Link Control Addresses in Local Area Networks.

ISO 10589 - Information Technology - Telecommunications and Information Exchange Between Systems - Intermediate System to Intermediate System Routing Exchange Protocol for use in Conjunction with the Protocol for Providing the Connectionless-mode Network Service.

(Applications for copies of ISO documents should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)

NATO INDUSTRIAL ADVISORY GROUP (NIAG)

NATO Naval Intra-Ship Network Independent Interface Specification (Proposed Draft Standard) - ANEP 30, 2nd edition - NIAG Sub-Group 6: Compatibility of Naval Data Handling Equipment. Dec 1988

(Copies of NIAG documents are expected to be available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111. Until that time applications for copies should be addressed to Space and Naval Warfare Systems Command, SPAWAR 231-2C, Washington DC 20363-5100).

NORTH AMERICAN MAP/TOP USERS GROUP

MAP 3.0 - Manufacturing Automation Protocol Specification - Ver. 3.0, Aug 1988

(Applications for copies of MAP documents should be addressed to the Corporation for Open Systems, 1750 Old Meadow Road, Suite 400, McLean, VA 22102)

XTP FORUM

XTP 3.6 - Xpress Transfer Protocol Definition - Revision 3.6 - Protocol Engines, Inc. 11 Jan 1992

(Applications for copies should be addressed to the XTP Forum, 1900 State Street, Santa Barbara, CA 93101)

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INTERNET SPECIFICATIONS

RFC 1305 - Network Time Protocol (Version 3) Specification, Implementation, and Analysis. March 1992

(Applications for copies should be addressed to Network Information System Center, SRI International, 333 Ravenswood Ave. Room EJ291, Menlo Park, CA 94025. Online copies can be obtained by sending a request to nisc@nisc.sri.com or via anonymous FTP from the "rfc" directory on ftp.nisc.sri.com.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

ACSE - Association Control Service Element
ANSI - American National Standards Institute
ASN.1 - Abstract Syntax Notation One
CL - Connectionless
CLNP - Connectionless Network Protocol
CMISE - Common Management Information Service Element
CO - Connection-oriented
DAS - Dual Attachment Station
DSAP - Destination Service Access Point
ES/IS - End System-to-Intermediate System
FDDI - Fiber Distributed Data Interface
FOIC - Fiber Optic Interface Connector
FTAM - File Transfer, Access, and Management
IEEE - Institute of Electrical and Electronics Engineers
IS/IS - Intermediate System-to-Intermediate System
ISO - International Organization for Standardization
LAN - Local Area Network
LLC - Logical Link Control
MAC - Media Access Control
MAP - Manufacturing Automation Protocol
MIB - Management Information Base
MIL - Management Information Library
NE - N-layer Entity
NIAG - NATO Industrial Advisory Group
NIIF - Network Independent Interface
NIST - National Institute of Standards and Technology
NSAP - Network Service Access Point
NTP - Network Time Protocol
OIW - OSI Implementor's Workshop
OSI - Open Systems Interconnection
PDU - Protocol Data Unit
PHY - Physical Layer Protocol
PMD - Physical layer Medium Dependent
PSAP - Presentation Service Access Point
QOS - Quality Of Service
ROSE - Remote Operation Service Element
SAFENET - Survivable Adaptable Fiber Optic Embedded Network
SAP - Service Access Point
SAS - Single Attachment Station
SLA - SAFENET Lightweight Application
SMAE - System Management Application Entity
SMASE - System Management Application Service Element
SMT - Station Management

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SSAP - Source Service Access Point
STS - SAFENET Time Service
TCU - Trunk Coupling Unit
TPDU - Transport Protocol Data Unit
TP4 - Transport Protocol, Class 4
TSAP - Transport Service Access Point
UTC - Coordinated Universal Time
XTP - Xpress Transfer Protocol

3.2 Definitions used in this standard. The definitions used in this standard are shown below:

3.2.1 Application programming interface (API). A set of calls to request services with a complete definition of their syntax and semantics. A network API provides an interface between the communication resources of a network and application programs.

3.2.2 Application layer. The layer of the ISO basic reference model which provides the application processes access to and use of the network's communications resources. This layer is defined as layer 7 in SAFENET.

3.2.3. Binary millisecond. 1/1024 of a second. The number of binary milliseconds in a second can be expressed exactly by a binary number of ten bits.

3.2.4 Bypass. The physical isolation of a station from the ring resulting from the deactivation or non-activation of the insertion/bypass mechanism inside the trunk coupling unit.

3.2.5. Concentrator. An FDDI node that has additional ports beyond those required for its own attachment to the FDDI network. These additional ports are for the attachment of other FDDI nodes in a tree topology.

3.2.6 Connectionless (CL). A mode of information transfer between peer entities in which each data transfer is independent of and not coordinated with previous or subsequent transfers and in which no state information is maintained.

3.2.7 Connection-oriented (CO). A mode of information transfer between peer entities in which a logical association is established prior to the exchange of data and which is maintained for the lifetime of the exchange process.

3.2.8 Datagram. A unit of data that is transferred as a single, non-sequenced, unacknowledged unit.

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3.2.9 Data link layer. The layer of the ISO basic reference model which provides the functional and procedural means to transfer data between stations, and to detect and correct errors that may occur in the physical layer. This layer is defined as layer 2 in SAFENET.

3.2.10 Dual ring. A basic ring topology consisting of a pair of counter-rotating logical rings arranged such that two signal paths, one in each direction, exists in a ring topology.

3.2.11 Electrical interface. The electrical connection between a trunk coupling unit and an attached station. It is used by the station to control the state of the optical bypass switch within the TCU.

3.2.12 Interoperability. The capability, promoted but not guaranteed by joint conformance with a given set of standards, that enables heterogeneous equipment, generally built by various vendors, to work together in a network environment.

3.2.13 LAN node. A generic term applying to any FDDI ring attachment (station or concentrator).

3.2.14 LAN segment. That subset of a LAN within which all communications can occur among its stations without requiring the use of bridges, routers, or gateways.

3.2.15 LAN services. In the SAFENET architecture, the set of services provided by the local area network. These services reside at layers 1-2 of the ISO basic reference model.

3.2.16 Layer. A well-defined, logical and hierarchical subdivision of a network architecture. Each layer can use the services of the next lower layer, plus its own functions, to create services which are made available to the next higher layer.

3.2.17 Local area network (LAN). A communications network designed for a moderate size geographic area and characterized by moderate to high data transmission rates, low delay, and low bit error rates. As used herein, the LAN protocols cover ISO layer 1 and the MAC sublayer of ISO layer 2.

3.2.18 Logical LAN segment. One or more LAN segments which share a common network address. Nodes on a single logical LAN segment can communicate directly without using protocols above the data link layer.

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3.2.19 Logical link control (LLC). A sublayer of the data link layer that supports medium independent data link functions and uses the services of the media access control sublayer to provide services to the network layer.

3.2.20 Logical ring. The set of MACs serially connected to form a single ring.

3.2.21 Media access control (MAC). A sublayer of the data link layer that controls and mediates access to the underlying physical medium and supplies services to the LLC sublayer.

3.2.22 Multicast. An addressing mode in which a single application process sends the same message to one or more destinations in the network.

3.2.23 Network clock. Elements (software, hardware, or both) present in each system node which implement a time source that is synchronized to a higher strata clock via the Network Time Protocol.

3.2.24 Network layer. The layer of the ISO basic reference model which performs those routing and relaying services necessary to support data transmission over interconnected LAN segments. This layer is defined as layer 3 in SAFENET.

3.2.25 Network management. The policies, procedures, resources, and mechanisms which collectively provide support for specific management applications, and which allow such applications to manipulate network entities and attributes.

3.2.26 Open system. A non-proprietary design approach which allows components and systems manufactured by multiple vendors to interoperate.

3.2.27 Optical interface. The optical connection between the physical medium and an attached station.

3.2.28 Physical layer. The layer of the ISO basic reference model which provides the mechanical, electrical, functional, and procedural characteristics access to the transmission medium. This layer is defined as layer 1 in SAFENET.

3.2.29 Physical medium. The components and materials used for the transmission of signals from one station to another.

3.2.30 Presentation layer. The layer of the ISO basic reference model which frees the application processes from

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concern with differences in data representation. This layer is defined as layer 6 in SAFENET.

3.2.31 Primitive. An abstract, implementation independent representation of an elementary interaction between two entities in adjacent layers at a service access point. Primitives are the means by which information is passed between a service user and a service provider. Each primitive can have a number of associated parameters that convey additional specific information for the respective service.

3.2.32 Protocol. The rules and conventions under which interactions between peer entities are carried out over the network.

3.2.33 Protocol profile. A defined set of protocols within the ISO basic reference model. A profile specifies protocol options sufficiently such that interoperability among implementations of its components can be expected.

3.2.34 Real-time. An event which must be accomplished within an allotted amount of time or the accomplishment of the action has no, diminishing or negative value.

3.2.35 Reconfiguration. The reestablishment of full or partial network connectivity following a network fault.

3.2.36 Repeater. A physical layer relay which connects segments of the physical medium. A repeater is used to extend the size of a ring or ring segment while preserving signal quality.

3.2.37 Ring. A set of stations wherein information is passed sequentially between stations, each station in turn examining or copying the information, finally returning it to the originating station. In FDDI usage, the term "ring" refers to the dual (counter-rotating) ring.

3.2.38 Router. A device which relays data between interconnected LAN segments through layer 3 protocols.

3.2.39 SAFENET. An acronym for Survivable Adaptable Fiber-optic Embedded Network. SAFENET is based on the Fiber Distributed Data Interface (FDDI) token-ring standard and incorporates profiles for both ISO compatibility and real-time performance.

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3.2.40 SAFENET node. The collection of SAFENET defined services and protocols as instantiated in a system node. A SAFENET node is a single logical entity.

3.2.41 Service. The capabilities and features provided by an (N)-layer to the (N)-users at the boundary between the (N)-layer and the (N+1)-layer.

3.2.42 Service access point (SAP). A logical interface between two adjacent layers, specifically, between an (N)-entity and an (N+1)-entity. The services of a layer are provided at these points.

3.2.43 Session layer. The layer of the ISO basic reference model which provides the mechanisms for organizing and structuring the interaction between two entities. This layer is defined as layer 5 in SAFENET.

3.2.44 Station. An addressable logical and physical device which is connected directly to the local area network and is capable of transmitting and receiving information.

3.2.45 Sublayer. A subdivision of a layer. For example, the LLC and MAC are sublayers of the data link layer.

3.2.46 System node. A physical entity which serves as the source and sink of communication in a system. It commonly comprises a single computer, associated application programs, peripheral equipment, and human operators.

3.2.47 Token. A control signal consisting of a unique signalling sequence that circulates on a token ring. It is the means by which the authority to transmit is passed from one station to another.

3.2.48 Token ring. A media access control technique for rings. A station may transmit by seizing the token, inserting a packet onto the ring, and then retransmitting the token.

3.2.49 Transfer services. In the SAFENET architecture, the set of services which provides the communication mechanisms necessary to interface the SAFENET user services and the SAFENET LAN services. These services reside at layers 2-4 of the ISO basic reference model.

3.2.50 Transport layer. The layer of the ISO basic reference model which accomplishes the transparent transfer of data over the established link, providing an end-to-end service

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with high data integrity. This layer is defined as layer 4 in SAFENET.

3.2.51 Trunk coupling unit (TCU). A physical device that enables a station to connect to the physical medium. The trunk coupling unit contains the mechanism for effecting insertion of a station into the ring or bypassing of a station.

3.2.52 User services. In the SAFENET architecture, the set of services which provide the processing necessary to allow SAFENET users to interface with, manage, and respond to the underlying transfer services. These services reside at layers 5-7 of the ISO basic reference model.

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4. GENERAL REQUIREMENTS

4.1 Communications architecture. The SAFENET communications architecture employs a layered model which is based on the seven layer ISO reference model for open systems interconnection (OSI) (see ISO 7498). SAFENET specifies protocols at each layer of this model, defining the complete SAFENET profile. In SAFENET, this protocol profile is organized two different ways. One organization consists of breaking the ISO reference model down into three service partitions, as described in 4.1.1. The other is to separate the protocols into defined profiles, as described in 4.1.2.

4.1.1 Service partitions. The SAFENET communications architecture divides the protocol profile into three service partitions, each of which includes a portion of the seven layer ISO reference model. These SAFENET partitions are called user services, transfer services, and local area network (LAN) services. These partitions are not intended to imply any specific implementation requirements. Figure 1 shows the seven layers of the ISO reference model on the left, the major elements of the SAFENET profile in the center, and the service partitions on the right.

4.1.1.1 SAFENET user services. The user services partition is that portion of the SAFENET profile through which a user interacts with the network. It corresponds to the session, presentation, and application layers of the ISO reference model (layers 5-7). These services provide a SAFENET user with the capability to interact with, manage, and respond to the underlying transfer services.

4.1.1.2 SAFENET transfer services. The transfer services partition is in the center of the SAFENET profile. It corresponds to the Logical Link Control (LLC) sublayer of the data link layer, as well as the network and transport layers of the ISO reference model (layers 2-4). These services provide reliable communication mechanisms to the SAFENET user.

At the boundary between the user services and the transfer services is the transfer services interface. This interface may be conceptual, or it may consist of a set of well-defined communication formats and primitives, depending on how the particular SAFENET system is implemented.

4.1.1.3 SAFENET LAN services. The LAN services partition is that part of the SAFENET profile which performs the actual data transfer. It corresponds to the physical layer of the ISO

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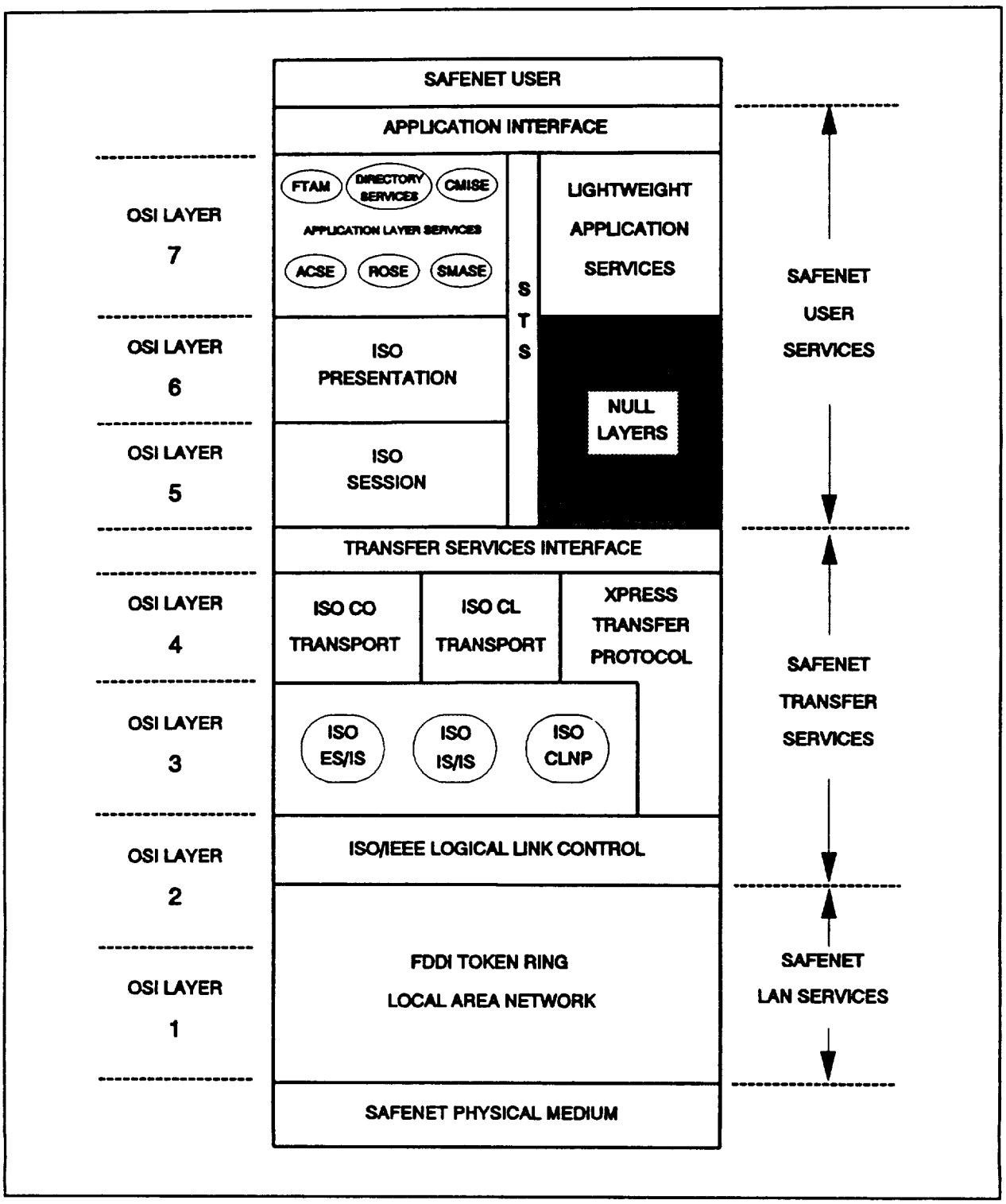


Figure 1. SAFENET Protocol Profile

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reference model, as well as the media access control sublayer of the data link layer (layers 1-2). The LAN services consist of those provided by the LAN, that is, the ability to get data on and off the physical medium in a controlled manner.

4.1.2 Profiles. The SAFENET communications architecture may also be viewed in terms of its different profiles. In SAFENET, these are the OSI profile, the lightweight profile, and the combined profile. These profiles define the three distinct implementation classes permitted in SAFENET. In these profiles some of the services and protocols are common to all implementation classes and others are used only in the OSI profile or the lightweight profile.

4.1.2.1 OSI profile. The OSI profile provides full OSI compliant networking to systems which require it. This profile's protocols and services are based upon ISO standards as further defined in this document or NIST Special Publication 500-202.

The OSI profile consists of MAP Private Communications (Private Comm), ISO File Transfer, Access, and Management (FTAM), Directory Services, Association Control Service Element (ACSE), Remote Operations Service Element (ROSE), System Management Application Service Element (SMASE), Common Management Information Service Element (CMISE), presentation and session layers, Connection-Oriented Transport Protocol (COTP), and those protocols and services listed in 4.1.2.4. Figure 2 shows an overview of the OSI profile.

4.1.2.2 Lightweight profile. The lightweight profile provides process to process message passing services applicable to homogeneous systems where control of timing is critical. These message passing services support point to point, multicast, and remote procedure call (transaction) styles of service. Multicast capability is limited to a single logical LAN segment. This profile comprises lightweight application services, the Xpress Transfer Protocol (XTP), and those protocols and services listed in 4.1.2.4. This profile provides a limited set of network management capabilities. This set includes managed object class definitions and layer management capabilities. Figure 3 shows an overview of the lightweight profile.

4.1.2.3 Combined profile. The combined profile is essentially the union of the OSI and lightweight profiles. This profile therefore includes all the protocols, services, and capabilities of those profiles. In addition, this profile provides network management protocols and services for those protocol entities contained within the lightweight profile. Figure 4 shows an overview of the combined profile.

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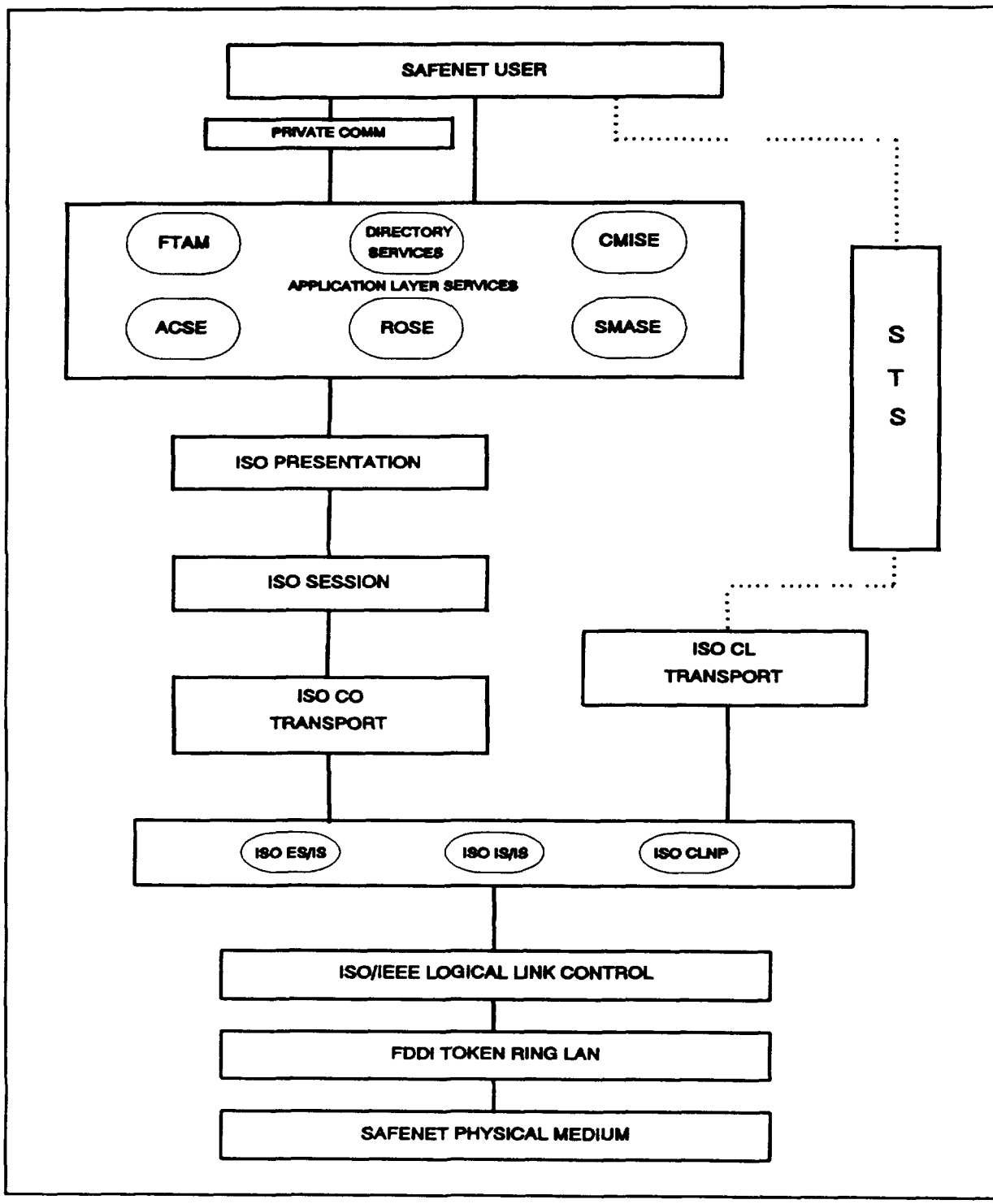


Figure 2. Overview of the OSI Profile

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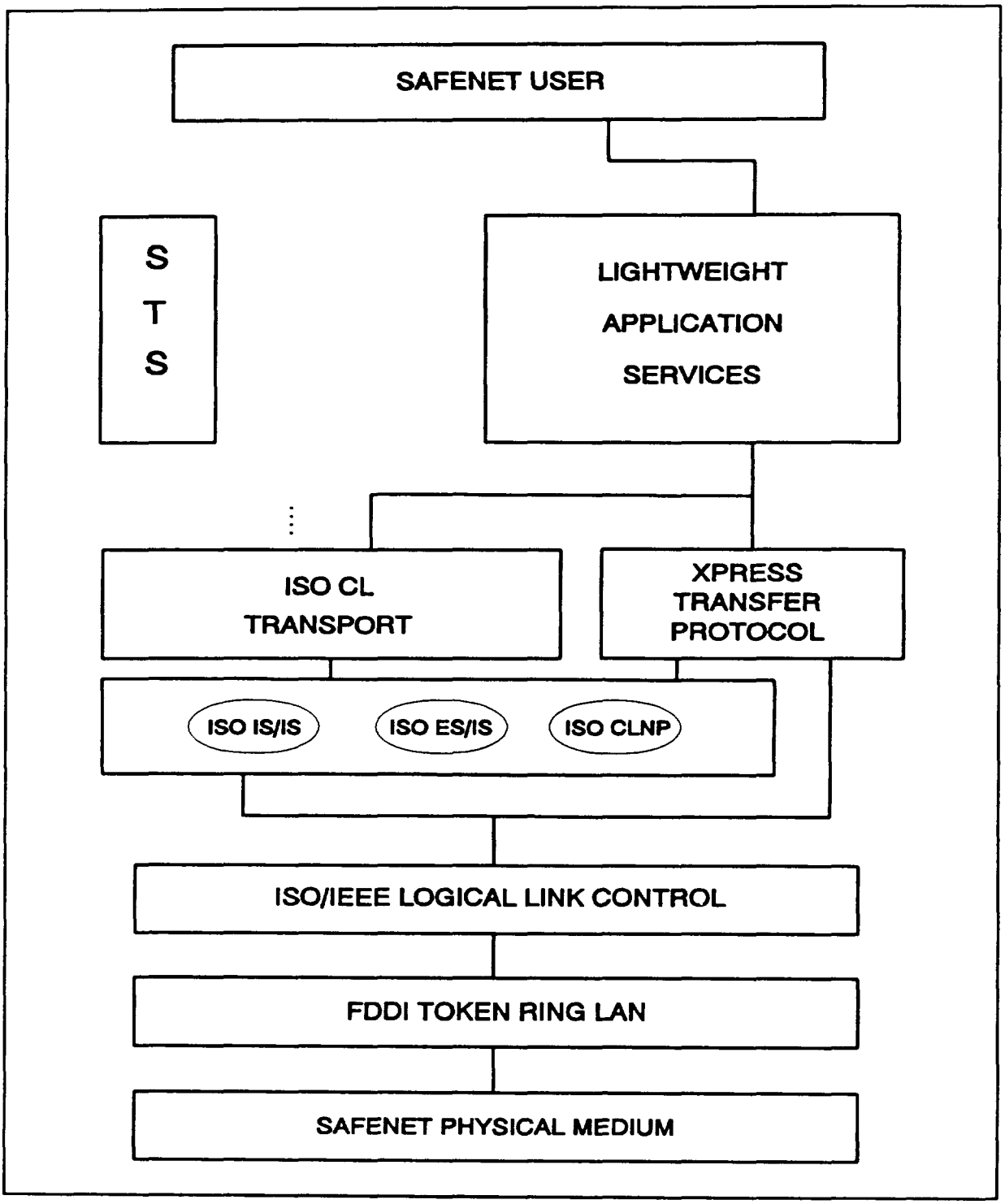


Figure 3. Overview of the Lightweight Profile

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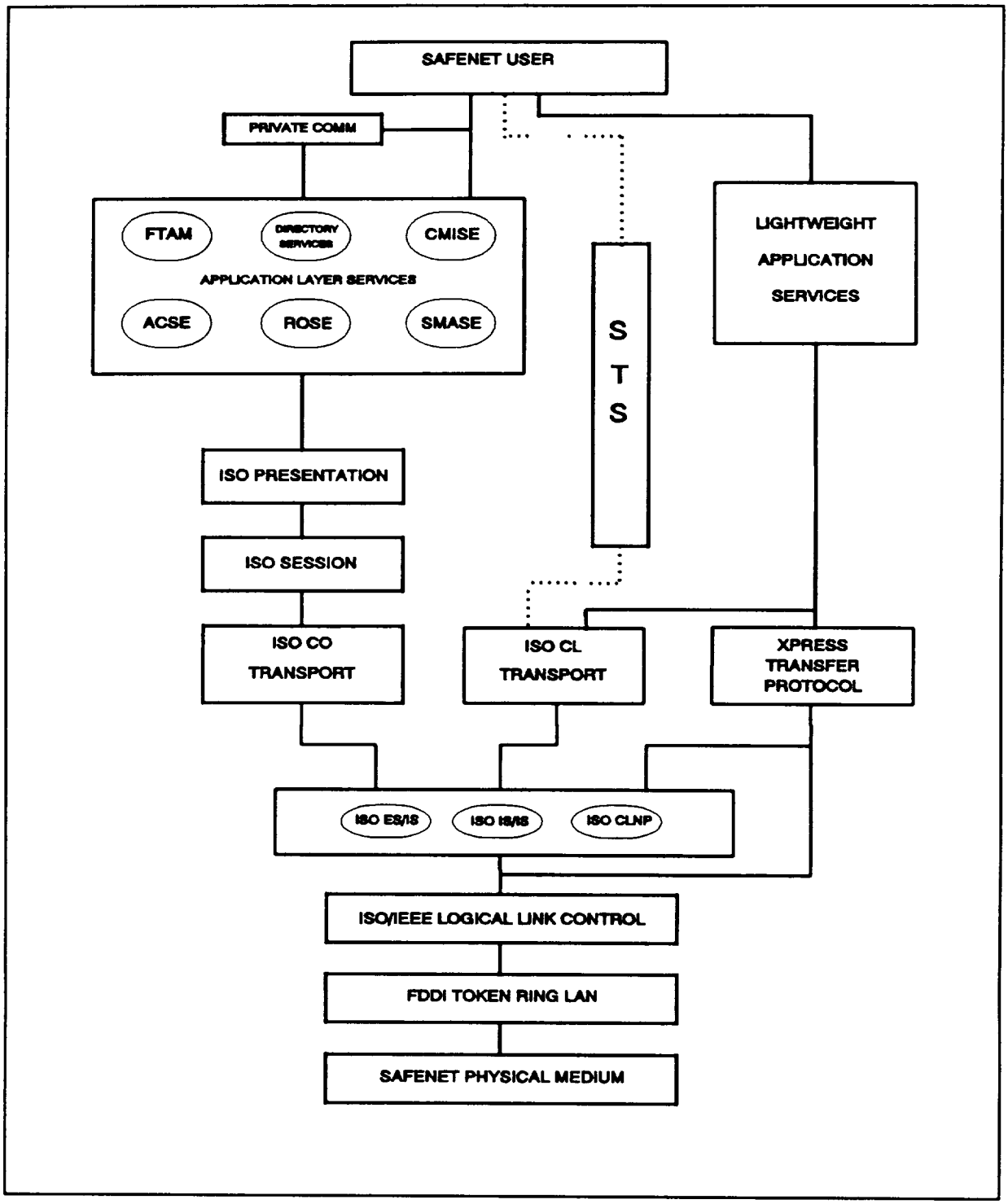


Figure 4. Overview of the Combined Profile

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Figures 2, 3, and 4 are included to illustrate how the protocols within SAFENET fit together and how they relate to the ISO reference model. They are not intended to provide a complete description of SAFENET. In each of these figures the SAFENET Time Service (STS) (see 4.8) is included because it is a service required in each profile, not because it is part of a data communication path.

4.1.2.4 Services common to all profiles. A portion of the SAFENET profile consists of protocols and services which are common to all profile implementations. They are the ISO Connectionless Transport and Network Protocols (CLTP and CLNP), ES/IS routing exchange protocol, IS/IS intra-domain routing protocol (optional in the lightweight profile), LLC protocol, and the Fiber Distributed Data Interface (FDDI) protocols. The STS is also required for all implementations. The STS is based on the Internet Network Time Protocol, Version 3 (NTP V3) as further specified in paragraphs 4.8 and 5.6. As shown in figures 2, 3, and 4, the STS makes use of ISO CLTP.

4.2 SAFENET profile. The following paragraphs briefly describe the communication protocols which comprise the SAFENET protocol profile.

4.2.1 User services. The SAFENET user services partition consists of two sets of protocols and services, one for the OSI profile and one for the lightweight profile. Each of these protocol sets shall be implemented to interface with the SAFENET user at the application layer, and with the transfer services partition through the transfer services interface.

4.2.1.1 OSI user services. The protocols and services of the OSI user services are based on the ISO OSI standards for network management and the application, presentation, and session layers, as further specified by NIST Special Publication 500-202. The OSI user services make use of the OSI connection oriented transport standards.

OSI user services shall be implemented whenever the OSI profile is specified for procurement, either alone or as part of the combined profile. The protocols which comprise OSI user services are described in 5.1.1.1.

4.2.1.2 Lightweight user services. The lightweight user services portion of the SAFENET profile permits an implementer to develop a set of communication services which support the performance and architectural requirements of a specific system. The lightweight user services shall be implemented using the services of both XTP and ISO CLTP.

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Lightweight user services shall be implemented whenever the lightweight profile is specified for procurement, either alone or as part of the combined profile. The services which comprise lightweight user services are described in MIL-HDBK-818-1.

4.2.1.2.1 Lightweight presentation and session layers. The presentation and session layers are considered null within the lightweight profile (see 5.1.1.2.2 and 5.1.1.2.3).

4.2.2 Transfer services. The SAFENET transfer services partition consists of the transfer services interface and three sets of protocols. One of these protocol sets is for the OSI profile, one is for the lightweight profile, and the last is the set of transfer service protocols which is common to all profiles. The SAFENET transfer services shall be implemented to interface with the user services partition through the transfer services interface, and with the LAN services partition through the LLC protocol.

4.2.2.1 Transfer services interface. The transfer services interface defines the boundary between the SAFENET user services and transfer services partitions. When user services and transfer services are implemented together this interface is conceptual only and need not be defined or made accessible to the user. However, when user services and transfer services are implemented independently, SAFENET requires that a specific set of communication formats and primitives be defined and implemented between the session and transport layers. These formats and primitives are collectively called the transfer services interface. The transfer services interface shall be defined fully so that access may be provided to independently developed software and hardware. This requirement is made in order to permit the SAFENET implementation to be more easily modified and maintained. SAFENET requirements for the transfer services interface are defined in 5.1.2.1.

4.2.2.2 OSI transfer services. The transport protocol which constitutes OSI transfer services is based on standards developed by ISO. This protocol provides ISO standard connection-oriented transport service. The OSI transfer services are capable of supporting the SAFENET OSI user services and other application layer services defined in ISO standards but outside the scope of SAFENET. They use the connectionless network service which is common to all profiles.

OSI transfer services shall be implemented whenever the OSI profile is specified for procurement, either alone or as part of the combined profile. The protocol which comprises OSI transfer services is described in 5.1.2.2.

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4.2.2.3 Lightweight transfer services. The transport protocols which constitute the lightweight transfer services are the Connectionless Transport Protocol and XTP. The lightweight transfer services shall be implemented to support the communication services defined both in lightweight user services, and in the connectionless network service.

Lightweight transfer services shall be implemented whenever the lightweight profile is specified for procurement, either alone or as part of the combined profile. The profile which comprises lightweight transfer services is described in MIL-HDBK-818-1.

4.2.2.4 Transfer services common to all profiles. Within the transfer services partition, there are several protocols which are common to all profile implementations. These are the ISO CLTP, CLNP, ES/IS protocol, IS/IS protocol, and LLC protocol. The connectionless transport protocol supports both the lightweight user services and the STS, and it uses the connectionless network service. The connectionless network protocol supports the OSI transfer services, the connectionless transport protocol, and the non-multicast transfer services of XTP. All data communication within SAFENET uses the LLC protocol in order to gain access to the FDDI local area network.

These common transfer services shall be implemented in any SAFENET procurement. The protocols which are common to all profile implementations are described in 5.1.2.4.

4.2.3 Local area network services. The LAN used in any implementation of SAFENET shall conform to the FDDI token-ring standard. This token-ring LAN provides services at both the physical and data link layers. The LAN shall be implemented to operate with the LLC protocol within the data link layer, and with the SAFENET specified physical medium at its interface to the physical layer.

The LAN services shall be implemented in any SAFENET procurement. The protocols which comprise SAFENET LAN services are described in 5.1.3.

4.3 Interconnection of LAN segments. SAFENET supports the possibility of multi-segment interconnection (i.e., multiple SAFENET networks) through the use of network layer routers. The ISO connectionless network protocol provides relaying services for the OSI profile, the non-multicast portions of the XTP protocol, and the ISO connectionless transport protocol (which may be supporting either the lightweight profile or the STS). All routing decisions shall be made using the route determination

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capabilities of the ISO network layer protocols (see 5.1.2.4.2). The protocol requirements of a SAFENET router are defined in 5.3.

4.4 SAFENET survivability approach. SAFENET uses capabilities in the FDDI standard and defines a set of topological components from which survivable systems may be designed. These components require the use of certain options defined by FDDI, in some cases require more stringent performance parameters than those in the base FDDI standard, and place additional restrictions on cable plant implementation. The system designer shall choose from the complete set of permitted components and implement the system in such a way so as to meet system performance, survivability, and cost goals. This flexibility allows system designers to use the components provided by SAFENET in the most intelligent manner, according to the requirements of the specific system. MIL-HDBK-818-1 provides guidance on and examples of implementations of survivable systems.

4.5 Topology components. There are three types of topological components defined by SAFENET: network stations, cable plant connections, and physical medium components. Alone or in combination with standard FDDI components, any SAFENET physical topology can be constructed from these topological components. SAFENET defines a number of components and techniques which support the development of survivable tactical systems. These components and techniques may be used in place of those in the ISO FDDI standards, and may be applied in any SAFENET implementation.

4.5.1 Network stations. SAFENET defines several station types that may be used by system designers. Their use depends on the individual system survivability requirements. These station types are defined below.

4.5.1.1 Trunk dual attachment station. SAFENET defines a trunk dual attachment station (DAS) as a network station that offers a dual attachment to the FDDI network and is capable of accommodating a dual (counter-rotating) ring. A trunk dual attachment station may be attached to the dual fiber trunk via a trunk connection (see 4.5.3.2).

4.5.1.2 Dual homed dual attachment station. SAFENET defines a dual homed dual attachment station as an FDDI defined dual attachment station, where the A Port (see FDDI SMT) is connected to the M Port of a concentrator (see 4.5.1.5) and the B Port is connected to the M Port of a different concentrator.

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4.5.1.3 Single attachment station. SAFENET defines a single attachment station (SAS) as a network station that offers a single attachment to the FDDI network. This station shall be attached to a concentrator via a tree connection (see 4.5.3.3). It is important to note that this type of station is a single point of failure, since a fault to it, the concentrator to which it is attached, or the connection between the two will take the network station out of communication with the rest of the network.

4.5.1.4 Multiple network attachment station. SAFENET defines a multiple network attachment station as a station containing at least two independent subnetwork connections. DAS and SAS connections may be used for attachment. Each multiple DAS can use two or more trunk connections, two or more pairs of tree connections, or a combination of trunk connections and pairs of tree connections. Each multiple SAS can use a pair of tree connections.

4.5.1.5 Concentrator. SAFENET defines a concentrator as an FDDI defined dual attachment concentrator station. It can attach to the dual fiber trunk, to another concentrator via two separate connections, or can dual home. A concentrator is considered a SAFENET network station.

4.5.2 Repeater. SAFENET permits the use of repeaters, which shall include a retiming function to preserve signal integrity. Repeaters shall perform all the functions of the FDDI Physical Layer (PHY) and Physical Layer Medium Dependent (PMD) protocols (modified, if necessary, in accordance with 5.1.3.4) with the deletion of the PHY repeat filter. A repeater requires a source of external power, and shall be attached to the FDDI dual-ring by a trunk connection (see 4.5.3.2).

Repeaters shall be attached to TCUs so they can be bypassed in the event of a power loss. TCUs and repeaters shall be allowed with any network station. The TCU used to connect a repeater to the network shall be controlled such that when power is applied to the repeater, the TCU engages and the repeater is placed in the optical path. When the repeater is powered down, the TCU shall move to the bypassed state.

4.5.3 Cable plant connections. Any SAFENET physical topology is based on the repeated use of specific types of cable plant connections. A cable plant connection is a complete description of the attachment between network stations, trunk coupling units, and concentrators. SAFENET defines three types of cable plant connections for use in a SAFENET physical

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topology: dual fiber trunks, trunk connections, and tree connections. These connections are described below.

4.5.3.1 Dual fiber trunk connection. The dual fiber trunk is formed by connecting fiber pairs between TCUs such that a closed dual-ring is established, as shown in Figure 5. Network station access to the dual fiber is provided by a TCU (see 5.4.1.3), which encloses two optical bypass switches, via a trunk connection (see 4.5.3.2).

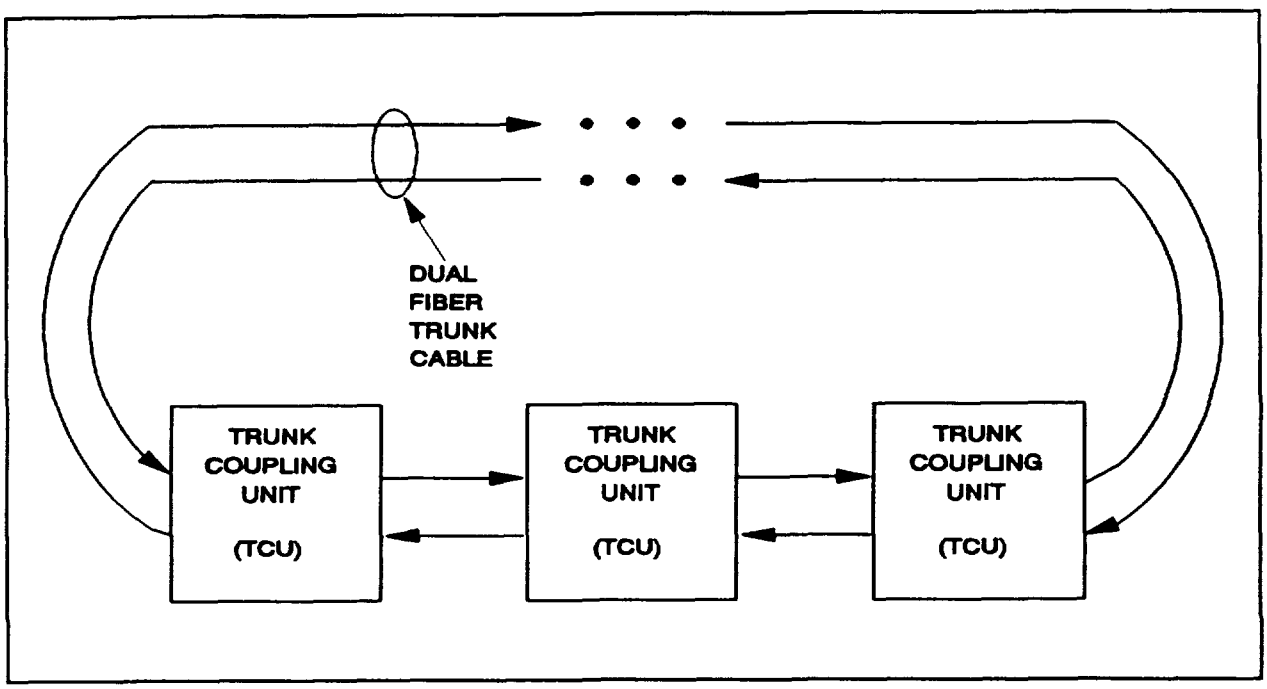


Figure 5. Dual Fiber Trunk

It is recommended that the dual fiber trunk cables be spliced to the TCUs rather than connectorized. If connectors are used the power budget will be adversely affected and the number of bypasses may be reduced. The maximum trunk cable run between adjacent TCUs shall be 200 meters.

4.5.3.2 Trunk connection. The trunk connection is used to attach a network station or repeater, via a TCU, to a dual fiber trunk. This connection consists of an optical interface and an electrical interface, as shown in Figure 6.

The optical interface of the trunk connection has two components, an optical interface cable (see 5.4.1.2.3) and a fiber optic interface connector (FOIC) (see 5.4.1.6). In this

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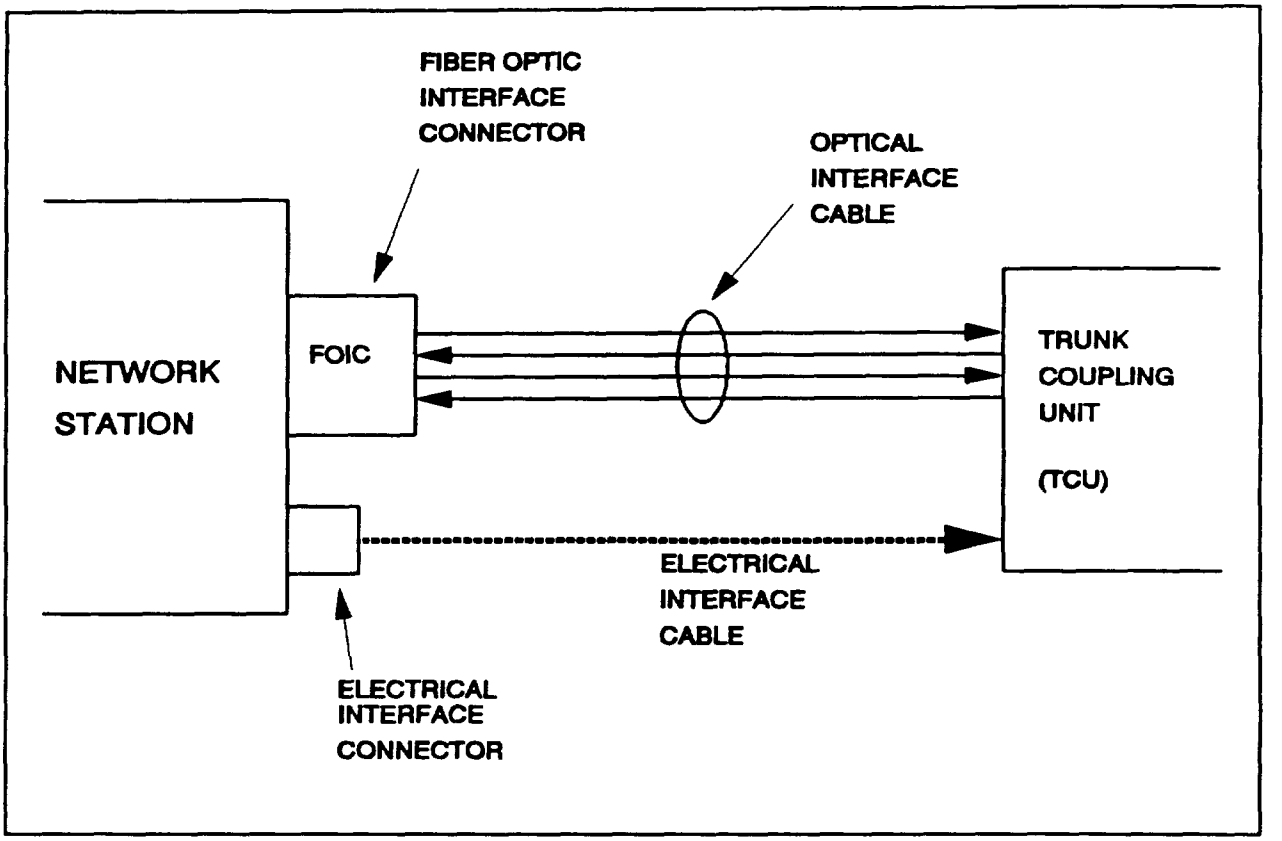


Figure 6. Trunk Connection

connection, the optical interface cable or cables must include at least four optical fibers. It is recommended that the optical interface cable be spliced to the TCU rather than connectorized. If connectors are used the power budget will be adversely affected and the number of bypasses may be reduced.

The electrical interface of the trunk connection also has two components, an electrical interface cable (see 5.4.2.1), and an electrical interface connector (see 5.4.2.2). This cable and connector are used to transmit the electrical control signal, which the station or repeater provides, to control the state of the TCU.

The maximum cable run for either an optical interface cable or an electrical interface cable shall not exceed 100 meters in the trunk connection. Any network station or repeater using a trunk connection shall be capable of supporting the SAFENET optical and electrical signal levels defined in 5.1.3.4.1.

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4.5.3.3 Tree connection. The tree connection is used to attach a network station to a concentrator. This connection consists solely of an optical interface, as shown in Figure 7.

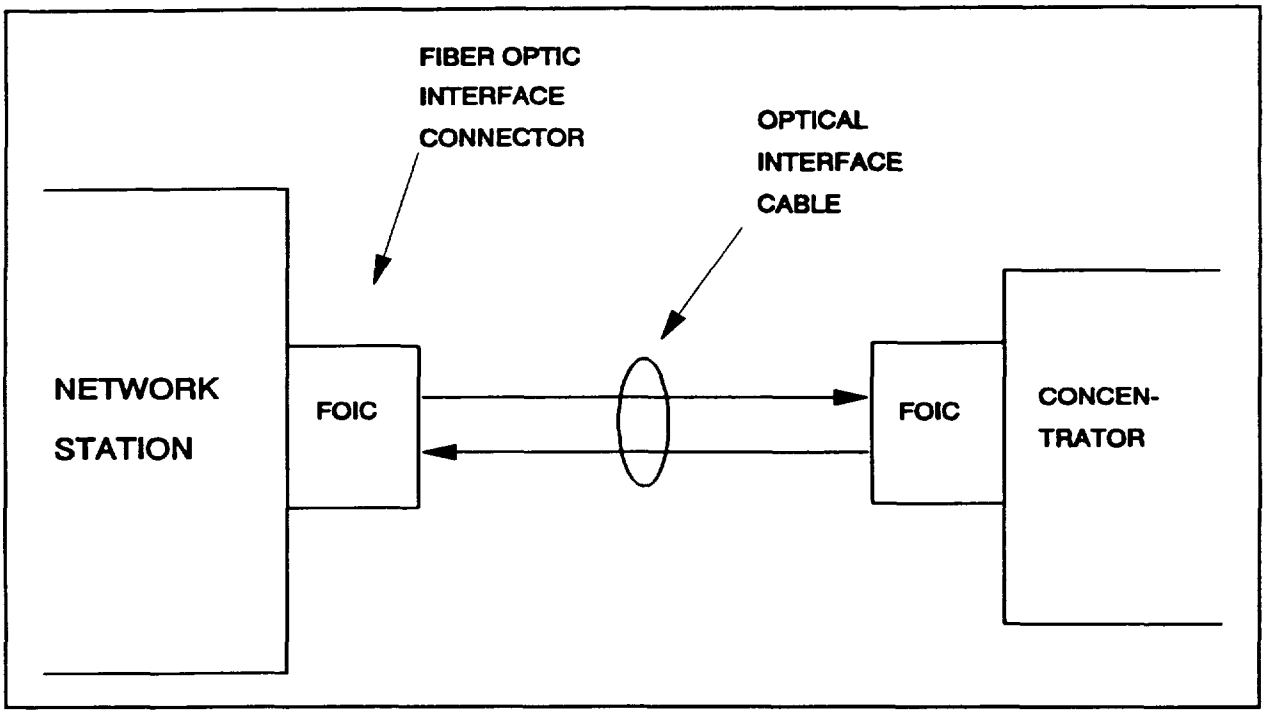


Figure 7. Tree Connection

The optical interface of the tree connection has three components, an optical interface cable (see 5.4.1.2.3) and two fiber optic interface connectors (FOICs) (see 5.4.1.6). In this connection, the optical interface cable shall include at least two optical fibers.

The maximum cable run for the optical interface cable shall not exceed 1 kilometer in the tree connection. Any network station using a tree connection shall be capable of supporting the optical signal levels defined in 5.1.3.4.2.

4.5.3.4 Direct trunk connection. This connection is similar to the dual fiber trunk connection, except that it is formed by connecting DAS's directly to the dual-fiber trunk ring without the use of TCUs. This type of connection will break the dual ring when power is not applied to these stations (forcing a wrap) and thus will segment the ring if multiple stations are not powered on. Thus this type of connection is only practical in special situations where very few stations are connected, when

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survivability is not needed, or when it is achieved in another way.

4.5.4 Physical medium components. The SAFENET physical medium consists of the following components: trunk coupling units, trunk cables, optical interface cables, fiber optic interface connectors, electrical interface cables, and electrical interface connectors. The requirements for these components are given in 5.4.

4.6 Network management. The SAFENET architecture includes a network management structure which supports the development of network performance measurement, fault tolerance, dynamic configuration control, and network security capabilities. This structure is defined in ISO/IEC 7498-4 Management Framework and ISO/IEC 10040 System Management Overview.

In accordance with this structure a complete set of management capabilities in a network system will require ISO system management functions, ISO layer management functions, and local or system node management functions. This last is outside the scope of the ISO standards and of this standard. Network system management in the ISO standards is based on the following concepts:

a. managed objects,
b. layer management,
c. managing applications,
d. management agent applications,
e. capability in each instantiation of a communication service (or equipment) to carry out management actions or provide management information,
f. protocols for communication between managing applications and agent applications,
g. interfaces between the agent application and layer management in the same system node.

The network management specific portions of this standard provides for the identification and definition of managed objects, the definition of required management agent application functions, and the protocols for communication between managing applications and management agent applications. Layer management capabilities are defined in the standards for the layer protocols and services called out by this standard. The definition of the functions of managing applications, the implementation of the capability in each instantiation of a communication service to carry out management actions or provide management information, and the details of the interface between the agent application

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and layer management functions in the same system node are local matters and beyond the scope of this standard.

4.6.1 OSI profile implementation requirements. In the OSI profile the following management capabilities shall be implemented:

a. CMISE,
b. ROSE,
c. SMASE,
d. ACSE,
e. Managed objects as defined in 5.5,
f. One or more applications in the managing role,
g. Managed objects as required by the layer standards.

The specific functions of the managing applications is a local matter.

4.6.2 Lightweight profile implementation requirements. In the lightweight profile the following management capabilities shall be implemented:

a. CLNP managed objects as defined in 5.5,
b. STS managed objects as defined in Appendix B,
c. Managed objects as required by the layer standards.

4.6.3 Combined profile implementation requirements. In the combined profile the following management capabilities shall be implemented:

a. CMISE,
b. ROSE,
c. SMASE,
d. ACSE,
e. Managed objects as defined in 5.5,
f. One or more applications in the managing role,
g. Managed objects as required by the layer standards.

The implementation of any other management capability is a local matter.

4.7 Network security. Security pertains to the protection of data that is transferred on the network. Although it may not be necessary for all implementations of SAFENET to provide all security services, the implementor shall conduct a risk analysis to determine which security services, if any, are required. Guidance as to how to conduct a risk analysis and how to satisfy security requirements is provided in MIL-HDBK-818-1.

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4.8 SAFENET time service. The synchronization of time among the components of a distributed real-time system is a critical requirement. To meet this requirement, SAFENET provides a service for the coordination of time information among all the participating nodes on a network. The SAFENET Time Service (STS) provides for the distribution of time information and the synchronization of distributed clocks within a system. As a result, each station in a SAFENET system has a stable, robust local clock with an identifiable accuracy bound.

The STS shall be implemented in all nodes in a SAFENET network. This service shall directly use the data transfer services of the OSI connectionless transport protocol for Protocol Data Unit (PDU) transfer. To support the STS each node is required to include a local time source, called a Network Clock.

4.8.1 Elements of the Time service. The Network Clock is defined by SAFENET to include both the software and hardware components necessary to implement a time source (see 5.6.1). The STS resident in each node uses the Network Clock to provide time to users in the local SAFENET node and to synchronize with other Network Clocks in the network. The value of the Network Clock in a particular node at any instant in time is referred to as Global Time.

The STS is partitioned into three functional areas: the clock synchronization service, the user time service, and the time management service. The clock synchronization service provides for the coordination of the distributed clocks in a SAFENET network (see 5.6.2). The user time service provides an attached SAFENET node with access to the current value of time for that node and a means to get basic information about the quality of the clock and the accuracy of the time value (see 5.6.3). Finally, the time management service includes a specification of the functions required and a definition of managed objects used for initialization, monitoring, and control of the STS (see 5.6.4).

4.9 Conformance requirements. Prior to claiming conformance to the SAFENET standard, an implementation shall be certified as conformant by a Navy designated Conformance Test Facility (CTF). The conformance test process implemented by the CTF is defined by all parts of ISO 9646, and is aligned with the certification process for NIST's U.S. GOSIP implementations.

The supplier of a SAFENET implementation which is claimed to conform to this standard shall complete a copy of the Protocol Implementation Conformance Statement (PICS) proforma provided in

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Appendix A. The completed PICS shall identify the capabilities of the implementation and shall provide the information necessary to identify both the supplier and the implementation. The PICS shall be submitted to the CTF prior to submission of the implementation for test.

Test related requirements will be identified by the CTF. These will include, as a minimum, a completed Protocol Implementation eXtra Information for Test (PIXIT) proforma and an embedded test responder for each submitted transport protocol. Detailed requirements for CTF submissions may be obtained from the CTF. Any information provided to the CTF solely for the support of testing will not be provided to any party outside the CTF without the implementors express approval.

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5. DETAILED REQUIREMENTS

5.1 Communication protocols. All communication protocol requirements for SAFENET are given below. Any optional features of the protocols referenced by this standard which are not specifically addressed are not required for an implementation to be SAFENET compliant.

5.1.1 User services. All requirements for SAFENET user services are given below. Additional guidance is found in MIL-HDBK-818-1.

5.1.1.1 OSI user services. All protocols and associated services used in the user services portion of the OSI profile shall be in accordance with NIST Special Pub 500-202. When the OSI profile is specified for a SAFENET procurement, either alone or as part of the combined profile, it shall be implemented to include all protocols and services defined in 5.1.1.1.1 through 5.1.1.1.9.

These requirements parallel the requirements of GOSIP Version 2.0 (FIPS PUB 146-1) for those protocols and services found in it. In some cases additional requirements or limitations are stated in this profile.

5.1.1.1.1 Private Communications interface specification. The Private Communications (Private Comm) application interface shall be implemented whenever the OSI profile is specified. This interface provides a means for gaining access to the Association Control Service Element (ACSE) and the presentation layer to enable the passing of messages. It shall be implemented in accordance with MAP 3.0, Appendix 7, Attachment 4: Private Communications Application Interface Specification.

5.1.1.1.2 File Transfer, Access, and Management. The File Transfer, Access, and Management (FTAM) protocol and services shall be implemented whenever the OSI profile is specified. FTAM provides services for transferring information in the form of files among application processes and filestores. It shall be implemented as specified in ISO 8571 parts 1 through 4, and as further specified in NIST Special Pub 500-202 Parts 9 and 10. Three levels of implementation are permissible. The user of this standard shall select the level appropriate for the system to which this standard is to be applied.

a. Limited: A limited purpose system provides the functions of simple file transfer and management. Limited purpose systems shall provide Implementation

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Profile T1 including the optional document type NBS-9 and Implementation profile M1 of Part 9 of NIST Special Pub 500-202.

b. Full: A full purpose FTAM system provides the functions of positional file transfer (including simple file transfer), simple file access, and management. Full purpose systems shall provide Implementation Profile T2 (Positional File Transfer) including the optional document types NBS-6 and NBS-9, Implementation Profile A1 (Simple File Access) and M1 (Management) of Part 9 of NIST Special Pub 500-202.

c. Extended: An extended purpose system shall provide all the capabilities of a full purpose system and add to them, as required, other document types or Implementation Profiles from either Part 9 or Part 10 of NIST Special Pub 500-202.

Limited, full, and extended purpose systems can interoperate at the intersection of their capabilities.

5.1.1.1.3 Directory services. Directory services shall be implemented whenever the OSI profile is specified. The primary function of directory services in the OSI profile is to provide a mapping from "user friendly" names (application entity titles) to the Presentation Service Access Point (PSAP) addresses required in instances of communications.

The implementation shall be in accordance with the structure, protocols, and services specified in ISO/IEC 9594 parts 1 through 8, and as further specified in the NIST Special Pub 500-202 Part 11. ISO/IEC 9594 and NIST Special Pub 500-202 provide for a broad range of capabilities. The implementation shall be tailored to meet the needs of the system in which it is to be used. At a minimum the implementation shall supply:

a. A Directory Service Agent (DSA) capable of registering application entity titles and PSAP addresses for use by FTAM and Private Comm. More detailed requirements are found in section 8.6 and the DSA column of Table 2 of NIST Special Pub 500-202. A real open system may contain one or more DSA's.

b. A Directory User Agent (DUA) instantiated in each SAFENET node capable of gaining access to the DSA's in support of local instantiations of FTAM and Private Comm. More detailed requirements are found in section 8.6 and the DUA column of table 2 of NIST Special Pub

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500-202. In addition to the required operations for a DUA of Table 2 at least the following operations from this table shall be implemented; Read, Compare, AddEntry, RemoveEntry, and ModifyEntry.

Implementation of authentication capabilities is not part of the minimum requirement.

5.1.1.1.4 Association Control Service Element. The ACSE and its protocol shall be implemented whenever the OSI profile is specified. ACSE provides services for the establishment and termination of application layer associations, and a standard service for application layer protocols to communicate common parameters. It shall be implemented in accordance with ISO 8649 and ISO 8650, and as further specified in NIST Special Pub 500-202, part 5, sections 5 and 13.

Connectionless ACSE protocol (NIST Special Pub 500-202 Part 5, section 5) is outside the scope of this standard. Peer entity authentication (NIST Special Pub 500-202 Part 5, section 5) is optional in this standard.

5.1.1.1.5 Remote Operations Service Element. The Remote Operations Service Element (ROSE) and its protocol shall be implemented whenever the OSI profile is specified. ROSE is used in support of the Common Management Information Service Element (CMISE) (see 5.1.1.1.7), and provides a simple, uniform service for remotely invoking operations and then receiving correlated replies to those operations. It shall be implemented in accordance with ISO/IEC 9072 parts 1 and 2, and as further specified in NIST Special Pub 500-202, part 5, sections 6 and 13.

5.1.1.1.6 System Management Application Service Element. The System Management Application Service Element (SMASE) and its protocols shall be implemented whenever the OSI profile is specified. SMASE specifies the management functions which are supported in a system node, and defines the semantics and abstract syntaxes of information transferred as relevant to OSI management. It uses CMISE for communication. It shall be implemented in accordance with ISO/IEC 10164 parts 1 through 6, and as further specified in NIST Special Pub 500-202, Part 18, section 5.

Additional parts of ISO/IEC 10164 may be implemented to provide expanded SMASE services, if they are included in follow-on editions to NIST Special Pub 500-202 at the time the implementation is made, and if the implementation is interoperable at the intersection of the capabilities provided.

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5.1.1.1.7 Common Management Information Service Element. The CMISE and its protocol shall be implemented whenever the OSI profile is specified. CMISE provides a common message framework for management procedures supplying both data-manipulation and notification/operation-oriented services. It shall be implemented in accordance with ISO/IEC 9595 and ISO/IEC 9596-1, and as further specified in NIST Special Pub 500-202, Part 18, section 6.

5.1.1.1.8 Presentation layer. The presentation layer services and protocol shall be implemented whenever the OSI profile is specified. They shall be implemented in accordance with ISO 8822 and ISO 8823, and as further specified in NIST Special Pub 500-202, part 5, sections 8 and 13. See 5.2.1 for Presentation addressing requirements.

5.1.1.1.9 Session layer. The session layer services and protocol shall be implemented whenever the OSI profile is specified. They shall be implemented in accordance with ISO 8326 and ISO 8327, and as further specified in NIST Special Pub 500-202, part 5, sections 9 and 13. See 5.2.2 for Session addressing requirements.

Connectionless session services are not required.

5.1.1.2 Lightweight user services. When the lightweight profile is specified for a SAFENET procurement, either alone or as part of the combined profile, it shall be implemented to include the services defined in 5.1.1.2.1 through 5.1.1.2.3.

5.1.1.2.1 Lightweight application services. A lightweight application service shall be implemented whenever the lightweight profile is specified. The details of this service are a local matter. The specification for a lightweight application service definition is provided in MIL-HDBK-818-1. This service definition consists of a set of communication service primitives which can be implemented to provide a user with direct, efficient data transfer capabilities. It permits implementation of a lightweight profile designed to meet the unique requirements of a specific system. These services shall not be implemented in a manner which requires the definition of communication protocols at the presentation or session layers. They may be implemented in such a manner that application layer protocols unique to the implementation are used.

5.1.1.2.2 Lightweight presentation layer. No presentation layer protocols or services are defined by SAFENET for the lightweight profile. Therefore, the lightweight presentation layer is considered null.

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5.1.1.2.3 Lightweight session layer. No session layer protocols or services are defined by SAFENET for the lightweight profile. Therefore, the lightweight session layer is considered null.

5.1.2 Transfer services. All requirements for SAFENET transfer services are given below. For further information refer to MIL-HDBK-818-1.

5.1.2.1 Transfer services interface. Whenever an implementor is required to provide a defined, accessible transfer services interface in the OSI profile, as per 4.2.2.1, the NATO Network Independent Interface (NIIF) shall be implemented. The NIIF shall be implemented in accordance with the interface specifications as defined in the Allied Naval Engineering Paper (ANEP) 30, second edition. An overview of the NIIF is given in MIL-HDBK-818-1.

5.1.2.2 OSI transfer services. When the OSI profile is specified for a SAFENET procurement, either alone or as part of the combined profile, it shall be implemented to include the protocol and associated services defined in 5.1.2.2.1 through 5.1.2.2.1.2.

5.1.2.2.1 Connection-oriented transport protocol. The ISO connection-oriented (CO) transport protocol shall be implemented whenever the OSI profile is specified. It provides the user with the ability to establish, use, and terminate connections with other CO transport service users. These connections provide for reliable sequential data transfer within the parameters established for the connection. The ISO CO transport protocol to be used in SAFENET shall be transport protocol class 4 (TP4). It shall be implemented in accordance with ISO 8073/ADD 2. See 5.2.3 for TP4 addressing requirements.

5.1.2.2.1.1 Additional TP4 requirements. The following requirements, from NIST Special Pub 500-202, shall be observed in any SAFENET implementation of TP4:

a. Any Connection Request (CR) or Connect Confirm (CC) Transport Protocol Data Unit (TPDU) which selects a protocol class other than 4 shall be rejected and a disconnect shall be initiated.
b. Both 7 bit and 31 bit sequence numbers shall be supported.
c. The protocol shall be capable of proposing and accepting the use or non-use of checksums.

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d. The protocol shall support the use of the acknowledgement time parameter, though its use on any particular connection is optional.
e. User data shall not be sent in the Connect Request or CC TPDUs. The disposition of any user data in a CR or CC TPDU is a local matter.
f. An unknown parameter in any received CR TPDU shall be ignored.
g. The protocol shall accept a Disconnect Request (DR) or Disconnect Confirm (DC) TPDU, with or without a checksum, in response to a CR or CC TPDU.
h. Transmitted DR TPDUs shall carry an appropriate disconnect reason code. A reason code of '0' (unspecified) shall be used if an appropriate reason code is not defined.
i. Unrecognized or not applicable bits within the additional options parameter shall be ignored.
j. A disconnect request shall be issued in response to a connect request when the maximum number of transport connections is reached or exceeded.
k. If present, the Transport Service Access Point (TSAP) ID field in the CR and CC TPDUs shall be encoded as a variable length field and interpreted as an octet string. The length of the string shall not exceed 32 octets.

5.1.2.2.1.2 Invalid parameter values. In any SAFENET implementation of TP4, known parameters with valid lengths but with invalid values in a CR TPDU shall be handled as follows:

a. TSAP ID: send DR TPDU
b. TPDU size: use default value
c. Version: use default value
d. Checksum: discard CR TPDU
e. Alternate protocol class: treat as protocol error

5.1.2.3 Lightweight transfer services. When the lightweight profile is specified for a SAFENET procurement, either alone or as part of the combined profile, it shall be implemented to include the protocol and associated services defined in 5.1.2.3.1 and 5.1.2.3.1.1.

5.1.2.3.1 Xpress Transfer Protocol. The Xpress Transfer Protocol (XTP) shall be implemented whenever the lightweight profile is specified. This protocol provides services which achieve increased efficiency and performance by combining the connection process with the data transfer process. XTP shall be implemented in accordance with the Xpress Transfer Protocol Version 3.6, produced by Protocol Engines, Inc., its associated

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addenda, and the XTP modifications defined in SAFENET. See 5.2.3 and 5.2.4 for XTP addressing requirements.

5.1.2.3.1.1 Additional Xpress Transfer Protocol requirements. The following requirements shall be observed in any SAFENET implementation of XTP:

a. Only the XTP SAFENET address format assigned by Protocol Engines, Inc., 0C (hex), shall be used. The allowable address field formats are defined in 5.2.3.2 and 5.2.3.3.
b. The sort service shall be implemented.
c. Only the FIFO sort policy shall be used.
d. A zero length data segment is denoted by filling the BTAG field with 1's.
e. The isochronous stream service is not supported.
f. ROUTE, MAINT, and MGMT packets shall be ignored and shall not be generated.
g. Context lookups shall be done on <SRC_NSAP,key> pairs instead of <srcid,key,route> triples as is done in XTP. SRC_NSAP is the source NSAP present in the connectionless network protocol (CLNP) packet which encapsulated the FIRST packet of a connection.
h. The address segment of an XTP packet shall be used only for Transport layer addresses. The formats of XTP SAFENET addresses are defined in 5.2.3.2. The network functions of XTP shall not be used.
i. Multicast addresses in XTP shall conform to the XTP SAFENET format defined in 5.2.3.1. Consequently, XTP multicast transmissions are restricted to a single logical LAN segment. XTP multicast packets are passed directly to LLC via SNAP encapsulation as defined in 5.2.5.
j. In multicast CNTL packets, the XKEY and XROUTE fields contain the MAC address of the originator of the FIRST packet (i.e., the group originator). The MAC address shall occupy the low-order bits of the field. The most significant unused bits shall be zero. (The MAC address is right justified within the ID field). The MAC address is used to identify unambiguously multicast control packets.
k. All non-multicast XTP packets shall be passed to CLNP for routing. Multicast XTP packets are passed to LLC as defined in 5.2.5.

5.1.2.4 Transfer services common to all profiles. All SAFENET networks shall be implemented to include the protocols and associated services defined in 5.1.2.4.1 through 5.1.2.4.4.

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5.1.2.4.1 Connectionless transport protocol. The ISO Connectionless (CL) transport protocol shall be implemented in all SAFENET networks. It provides the user with the ability to transmit a single unit of data, called a datagram, without the requirement of establishing a connection. It shall be implemented in accordance with ISO 8602. See 5.2.3 for CL transport protocol addressing requirements.

5.1.2.4.1.1 Multicast data transfer. SAFENET requires that the services to CL transport protocol be extended to provide a multicast data transfer capability. This capability allows one source transport service user to send a single unit of data to one or more destination transport service users. The delivery of these multicast data units to one or more transport service users at a single SAFENET station is a local matter. The destination TPDU selector field shall be used to indicate that the CL transport protocol or SAFENET XTP protocol is being used for a multicast transfer as opposed to a standard point-to-point transfer. A unique MAC group address from the SAFENET group address space shall be used to identify each multicast transport user group. The group MAC address shall be directly embedded in the TPDU selector field along with the multicast mode indicator described above. Use of the multicast data transfer capability of the extended CL transport protocol is constrained to the use of the inactive subset of the ISO connectionless network protocol, which limits the scope of transfers to users on a single LAN segment. Use of the multicast capability of the SAFENET XTP protocol is also limited to a single LAN segment since XTP packets are passed directly to LLC. See 5.2.3 for CL transport protocol multicast selection and addressing details.

5.1.2.4.2 Network protocol. The ISO CLNP shall be implemented in all SAFENET networks. It provides services for network routing and for the segmentation and reassembly of transport layer messages that are too large for the underlying communications service. It shall be implemented in accordance with ISO 8473. The inactive network layer protocol subset shall be supported in any SAFENET implementation of the CLNP. See 5.2.4 for CLNP addressing requirements.

5.1.2.4.2.1 Additional network protocol requirements. The following requirements, from NIST Special Pub 500-202, shall be observed in any SAFENET implementation of the CLNP:

a. The non-segmenting subset shall not be used.
b. The inactive subset shall not be used for PDU transfer by the TP4 protocol.
c. Any inactive subset PDU received shall be delivered to the CL transport protocol.

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d. Any PDU received from the CL transport protocol shall not be fragmented.
e. The protocol shall be capable of being configured to support the use or non-use of checksums. The default value shall be non-use.
f. The QOS maintenance, security, or priority parameters shall not be used for routing.
g. The life-time parameter shall be used as specified in ISO 8473. The parameter shall have an initial value of at least three times the network span or three times the maximum transit delay (in units of 500 milliseconds), whichever is greater.

5.1.2.4.2.2 Intra-domain routing protocol. In SAFENET networks where a protocol is needed for the dynamic determination of routes used to pass data between intermediate systems, the ISO Intermediate System-to-Intermediate System (IS/IS) intra-domain routing protocol shall be used. It shall be implemented in accordance with ISO 10589. This protocol is recommended if a SAFENET network includes three or more interconnected LAN segments.

5.1.2.4.2.3 Routing exchange protocol. The ISO End System-to-Intermediate System (ES/IS) routing exchange protocol shall be implemented in all SAFENET networks. It passes address information among all stations that are on the same LAN segment or through intermediate stations. The ES/IS protocol provides stations with the ability to associate a data link layer address with a given network layer address. It shall be implemented in accordance with ISO 9542. End system stations shall implement at least the ES portion of the protocol, and intermediate system stations shall implement at least the IS portion of the protocol. ES/IS is to be used in conjunction with CLNP.

5.1.2.4.2.4 Additional ES/IS requirements. The following requirements shall be observed in any SAFENET implementation of the ES/IS protocol:

a. Any valid Network Service Access Point (NSAP) format in conformance with ISO 8348/ADD 2 shall be supported.
b. Both the configuration information and route redirection information portions of the protocol shall be supported.
c. A mechanism to set the configuration and holding timer values shall be provided.
d. A mechanism to select the use or non-use of checksums shall be provided.
e. The QOS maintenance, security, or priority parameters shall not be used for routing.

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f. The configuration notification function for end systems and intermediate systems shall be supported. The protocol shall be capable of enabling or disabling this function.
g. The same Link Service Access Point (LSAP) as that used by the CLNP (FE hex) shall be employed.
h. The error report flag shall be set to zero (0) for NPDUs sent as a result of invoking the Query configuration function.

5.1.2.4.3 Logical Link Control. The Logical Link Control (LLC) protocol shall be implemented in all SAFENET networks. It provides services which include protocol stack selection and the ability to generate and respond to TEST and XID protocol data units. In SAFENET, LLC type 1 (connectionless) shall be used. It shall be implemented in accordance with ISO 8802-2. See 5.2.5 for LLC addressing requirements.

5.1.2.4.3.1 Sub-Network Access Protocol. The Sub-Network Access Protocol (SNAP) shall be supported in all SAFENET networks. It shall be implemented in accordance with IEEE 802. SAFENET uses SNAP for unambiguous identification of XTP. See 5.2.5 for SNAP addressing requirements.

5.1.3 Local area network services. All requirements for SAFENET LAN services are given below. Refer to MIL-HDBK-818-1 for a discussion of the SAFENET power budget.

5.1.3.1 Media access control protocol. The FDDI Media Access Control (MAC) protocol is required for all implementations of SAFENET. Its primary function is to manage each station's access to the physical medium in a deterministic manner using a token passing protocol. It shall be implemented in accordance with ISO 9314-2. All network stations connected to the FDDI network shall contain at least one MAC entity. In the normal configuration all data transfer shall be performed on the primary ring; the secondary ring shall not be used for data transfer except after reconfiguration caused by loss of integrity of the primary ring. See 5.2.6 for MAC addressing requirements.

5.1.3.1.1 Additional MAC protocol requirements. Any SAFENET implementation of the FDDI MAC protocol shall meet the following requirements:

a. SAFENET stations shall only transmit 48 bit addresses.
b. The protocol shall be capable of recognizing a minimum of 65,536 group addresses, as defined in 5.2.6.2.2.
c. Synchronous service, as specified in ISO 9314-2, is required.

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d. Asynchronous service, as specified in ISO 9314-2, is required.
e. The restricted token option shall not be used.
f. The immediate mode option shall not be used.

5.1.3.2 FDDI Station Management protocol. The FDDI Station Management (SMT) protocol is required for SAFENET stations. It shall be implemented in accordance with ANSI X3T9.5/84-49 Rev 7.2, and shall include the option for bypass switch control for those network stations capable of using trunk connections (see 4.5.3.2).

5.1.3.2.1 SMT bypass switch control option. The SMT bypass switch control option shall be implemented by any network station capable of using trunk connections. This option is needed to enable the station to control its TCUs through an electrical signal interface (see 5.1.3.4.4).

5.1.3.3 Physical layer protocol. The FDDI physical layer protocol (PHY) is required for all SAFENET stations. It performs clock recovery and symbol encoding and decoding. It shall be implemented in accordance with ISO 9314-1. The medium dependent portion of the physical layer is described separately in 5.1.3.4.

5.1.3.4 Physical Layer Medium Dependent. SAFENET specifies three optical interfaces from which to choose when designing a system. All are based on the FDDI Physical Layer Medium Dependent (PMD) standard, which provides the optical interface to the physical medium. The optical signals used by any station are for the transmission and reception of data, and are described as they shall appear at the fiber optic interface connector (FOIC) (see 5.4.1.6). Any of these three signalling levels may be used within the same network, though the implementor should carefully consider the survivability implications with regard to the available power budget before any design decisions are made.

5.1.3.4.1 Class one optical interface. This optical interface shall be in accordance with the requirements of ISO 9314-3.

5.1.3.4.2 Class two optical interface. This optical interface is a combination of the FDDI and the SAFENET-unique signalling levels.

5.1.3.4.2.1 Optical output signal. The station shall provide an optical output signal which conforms to ISO 9314-3, as described below:

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Average power:	minimum:	-20.0 dBm
	maximum:	-14.0 dBm

5.1.3.4.2.2 Optical input signal. The station's receiver shall operate according to the requirements of 5.1.3.4.3.2 below.

5.1.3.4.3 Class three optical interface. This optical interface is unique to SAFENET and provides additional available power budget for systems which require it.

5.1.3.4.3.1 Optical output signal. The station shall provide an optical output signal which conforms to ISO 9314-3 (measured in a 62.5/125 μm fiber), with the following exception:

Average power:	minimum:	-16.0 dBm
	maximum:	-9.0 dBm

5.1.3.4.3.2 Optical input signal. The station's receiver shall operate within the Bit Error Rate (BER) requirements of ISO 9314-3. It shall do so when presented with an input signal which conforms to ISO 9314-3, with the following exception:

Average power:	minimum:	-35.0 dBm
	maximum:	-10.0 dBm

5.1.3.4.3.3 Signal Detect function. The station shall provide a Signal_Detect parameter in accordance with ISO 9314-3, with the following exceptions:

a. Assert power (Pa):	maximum:	-35.0 dBm
	minimum:	per ISO 9314-3
b. Deassert power (Pd):	minimum:	-49.0 dBm, or Pb*

*as defined and applied by ISO 9314-3

5.1.3.4.3.4 Transmit Disable function. The optical transmitter shall respond to a Transmit_Disable parameter of SM_PM_CONTROL.request by producing an average output optical power of less than -49.0 dBm.

5.1.3.4.4 Electrical signal interface. Any station which is attached to TCUs shall observe the electrical signalling requirements shown in Table I. These electrical signals are used by the station to control the state of the optical bypass switches within the TCU. The TCU shall be capable of switching between two states, the bypassed state and the inserted state. The signal is described as it shall appear at the electrical interface connector (see 5.4.2.2). The electrical signal

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interface shall be short-circuit protected. Note that the minimum insertion voltage specified allows for a voltage drop in the electrical interface cable.

Table I. Electrical Signal Interface

PARAMETER	MIN	MAX	UNITS
Insertion Signal	4.7	5.3	Vdc
Bypass Signal	0.0	1.0	Vdc
Rise/Fall Time	---	0.2	ms
Current capability	340	---	ma

5.1.3.4.5 Tree connection PMD. Any network station which is implemented to use tree connections (see 4.5.3.3) shall observe the requirements of 5.1.3.4.1 in its implementation of PMD. Note that the optical signal requirements must be met at each end of the tree connection, and that the signals are described as they shall appear at the FOIC (see 5.4.1.6). Note also that this connection does not require an electrical signal interface.

5.1.3.4.5.1 Optical signalling. The optical signalling shall be in accordance with ISO 9314-3.

5.1.3.4.5.2 Signal Detect function. The Signal_Detect function shall be in accordance with ISO 9314-3.

5.2 Addressing requirements. All requirements for addressing in SAFENET are given below.

5.2.1 Presentation layer addresses. Presentation service users are addressed at PSAPs. A PSAP address shall consist of a Session Service Access Point (SSAP) address and a presentation-selector (P-selector). P-selectors used in SAFENET shall be a maximum of two octets in length, shall be encoded in binary in conformance with GOSIP 2.0, and shall use the values given below to indicate the applicable service/user binding.

PSAP Binding	P-Selector (hex)
a. Presentation : FTAM	01
b. Presentation : Private Comm	02
c. Presentation : Directory Services	03
d. Presentation : CMISE	04

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5.2.2 Session layer addresses. Session service users are addressed at SSAPs. A SSAP address shall consist of a transport service access point address and a session-selector (S-selector). One S-selector is specified for SAFENET implementations. The required S-selector shall be a maximum of two octets in length, shall be encoded in binary, and shall have a value of 01 (hex) as shown below to indicate the SSAP which binds the ISO Session protocol entity to the ISO Presentation protocol entity.

SSAP Binding	S-Selector (hex)
Session : Presentation	01

5.2.3 Transport layer addresses. Transport service users are addressed at TSAPs. A TSAP address consists of a NSAP address and a transport selector (T-selector), except that when the Inactive Subset of the CLNP is used or when XTP is bound directly to LLC for multicast transfers, the TSAP address consists solely of a T-selector.

The T-selector that binds the CO transport protocol entity with the session protocol entity shall be encoded as a variable-length octet string, shall have a maximum length of 32 octets, and shall have the value 01 (hex).

The T-selector that binds the CL transport protocol entity to the SAFENET Time Services (STS) protocol entity for both issuance and reception of non-multicast messages shall have a length of 4 octets, shall be encoded in binary, and shall have a value of 04 (hex).

An implementation of the lightweight suite shall contain at least one T-selector that binds the CL transport protocol entity to a lightweight service user for both the issuance and reception of non-multicast messages. This T-selector shall have a length of 4 octets, shall be encoded in binary, and shall have the value 08 (hex).

SAFENET allows, but does not specify the values of, other T-selectors for the binding of a lightweight service user to the CL transport protocol entity for non-multicast transfers, so long as the value of the selector is not one of those specified herein. If used, these additional T-selectors shall have a length of four octets and shall be encoded in binary.

An implementation of the lightweight suite shall contain at least one T-selector that binds the XTP entity to a lightweight service user for both the issuance and reception of non-multicast

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messages. This T-selector shall have a length of 4 octets, shall be encoded in binary, and shall have the value 12 (hex).

SAFENET allows, but does not specify the values of, other T-selectors for the binding of a lightweight service user to the XTP entity for non-multicast transfers, so long as the value of the selector is not one of those specified herein. If used, these additional T-selectors shall have a length of four octets and shall be encoded in binary.

5.2.3.1 Multicast addressing. The multicast services in SAFENET allow a lightweight service user to send a single unit of data to one or more destination transport service users. It also allows a STS protocol entity to send a multicast message to other STS protocol entities. The source T-selector field in a multicast transfer shall have the same structure as that in a unicast transfer. The destination T-selector field in a multicast packet shall consist of a special type field and an embedded 6-octet SAFENET MAC group address. This eight-octet T-selector identifies a TSAP associated with a specific multicast transport service user group.

1101 1100	1101 1100	01-40-4F-FF-FF-7F
1 octet	1 octet	6 octets

Figure 8. SAFENET Time Service Multicast T-Selector Format

The T-selector that binds the CL transport protocol entity to the STS protocol entity for reception of multicast messages shall have a length of 8 octets, shall be encoded in binary, and shall have the structure shown in Figure 8. The first, most significant two octets of the T-selector shall each have the value DC (hex). The remaining 6 octets shall contain an embedded SAFENET MAC group address with the value 01-40-4F-FF-FF-7F (IEEE 802 Standard illustrative representation in hexadecimal). The value DC shall not be used in either of the first two octets of the selector field of any other type of TPDU.

The T-selector that binds the CL transport protocol entity to the STS protocol entity for the issuance of multicast messages shall have a length of 4 octets, shall be encoded in binary, and shall have a value of 05 (hex).

T-selectors used to bind the CL transport protocol entity to a lightweight service user for the reception of multicast messages shall have a length of 8 octets, shall be encoded in

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binary, and shall have the structure shown in Figure 9. The first, most-significant two octets of the T-selector shall each have the value DA (hex). The remaining 6 octets shall contain an embedded SAFENET MAC group address. The value DA shall not be used in either of the first two octets of the selector field of any other type of TPDU.

1101 1010	1101 1010	SAFENET MAC Group Address
1 octet	1 octet	6 octets

Figure 9. Connectionless Transport Multicast T-Selector Format

The T-selector that binds the CL transport protocol entity to a lightweight service user for the issuance of multicast messages shall have a length of 4 octets, shall be encoded in binary, and shall have a value of 09 (hex).

SAFENET allows, but does not specify the values of, other T-selectors for the binding of a lightweight service user to the CL transport protocol entity for the issuance of multicast transfers, so long as the value of the selector is not one of those specified herein. If used, these additional source T-selectors shall have a length of four octets and shall be encoded in binary.

The T-selector that binds the XTP protocol entity to a lightweight service user for the reception of multicast messages shall have a length of 8 octets, shall be encoded in binary, and shall have the structure shown in Figure 10. The first, most-significant two octets of the T-selector shall each have the value DB (hex). The remaining 6 octets shall contain an embedded SAFENET MAC group address. The value DB shall not be used in either of the first two octets of the selector field of any other type of TPDU.

1101 1011	1101 1011	SAFENET MAC Group Address
1 octet	1 octet	6 octets

Figure 10. XTP Multicast T-Selector Format

The T-selector that binds the XTP protocol entity to a lightweight service user for the issuance of multicast messages

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shall have a length of 4 octets, shall be encoded in binary, and shall have a value of 13 (hex).

SAFENET allows, but does not specify the values of, other T-selectors for the binding of a lightweight service user to the XTP protocol entity for the issuance of multicast transfers so long as the value of the selector is not one of those specified herein. If used, these additional source T-selectors shall have a length of four octets and shall be encoded in binary.

5.2.3.2 Summary of T-Selector Values. Listed below are the specific T-selector values that are required to be recognized in SAFENET implementations.

TSAP Binding	T-Selector (Hex)
a. COTP : Session	01
b. CLTP : STS (non-multicast)	04
c. CLTP : STS (multicast issuance)	05
d. CLTP : STS (multicast reception)	DC-DC-01-40-4F-FF-FF-7F
e. CLTP : LW Application (non-multicast)	08
f. CLTP : LW Application (multicast issuance)	09
g. CLTP : LW Application (multicast reception)	DA-DA-Group Address
h. XTP : LW Application (non-multicast)	12
i. XTP : LW Application (multicast issuance)	13
j. XTP : LW Application (multicast reception)	DB-DB-Group Address

Note: In the list above the last six octets of the T-selector for the CLTP : STS multicast binding are specified using the IEEE 802 standard illustrative representation (hexadecimal notation), while the first two octets are specified in most-significant-bit form.

5.2.3.3 XTP address structure. By agreement with Protocol Engines, Inc., an XTP address syntax has been reserved exclusively for definition and allocation by SAFENET. The XTP aformat value for this syntax is 0C (hex). SAFENET introduces an additional format field to distinguish between different SAFENET network address formats used in XTP packets.

Figure 11 shows the complete XTP address field structure and associated aformat value for non-multicast XTP packets used in SAFENET. The two-octet SAFENET format field contains the value 0001 (hex) to indicate non-multicast addressing. The dtsel field contains the 4-octet destination transport selector. The stsel field contains the 4-octet source transport selector.

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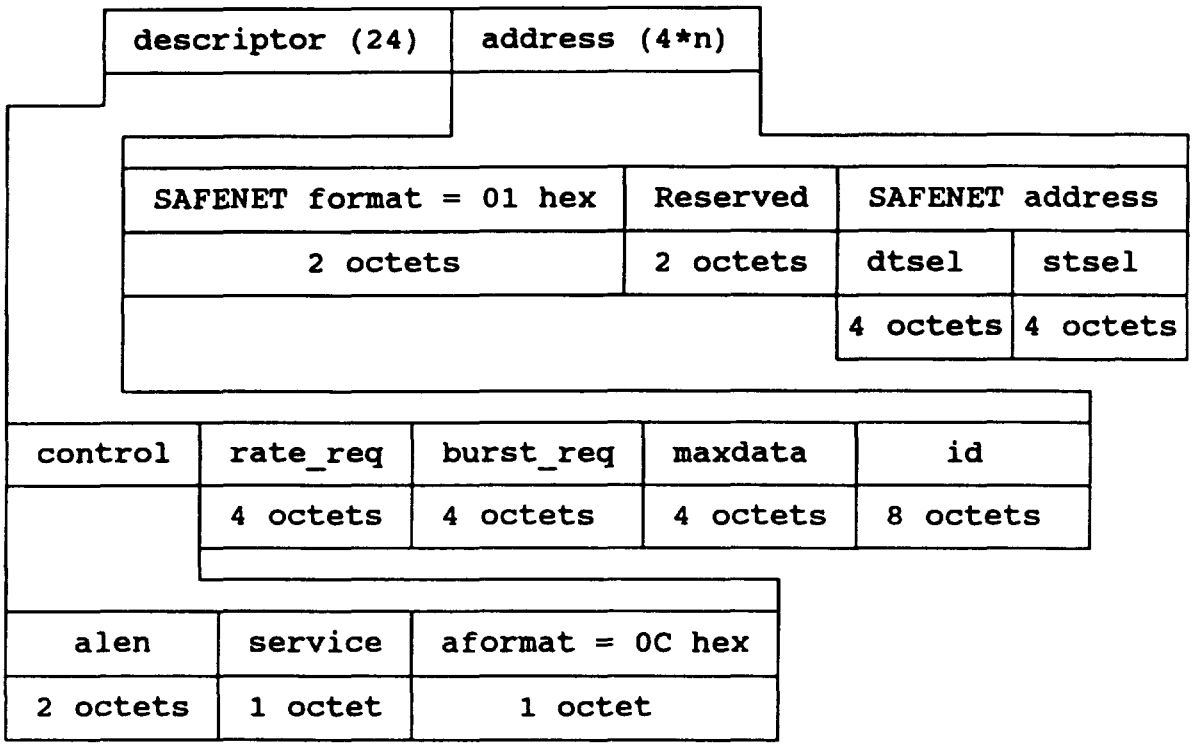


Figure 11. SAFENET Non-Multicast XTP Address Segment

Figure 12 shows the complete XTP address field structure and associated aformat value for multicast XTP packets used in SAFENET. The two-octet SAFENET format field contains the value 0002 (hex) to indicate multicast addressing. The dgtsel field contains the eight-octet destination multicast address. The most significant two octets of the dgtsel field each contain the value DB (hex). The remaining six octets shall contain a SAFENET MAC group address. This eight-octet T-selector identifies a TSAP associated with a specific multicast transport service user group. The stsel field contains the 4-octet source transport selector.

5.2.4 Network layer addresses. Network service users are addressed at NSAPs. SAFENET NSAP addresses shall be structured in accordance with ISO 8348/ADD 2 and with the additional specifications provided in this standard. The SAFENET NSAP address structure defined herein conforms with the Department of Defense OSI NSAP addressing structure and semantics that have been defined by the Defense Information Systems Agency. Actual assignment of SAFENET NSAP addresses and addressing subauthorities, including assignment of Routing Domain and Area

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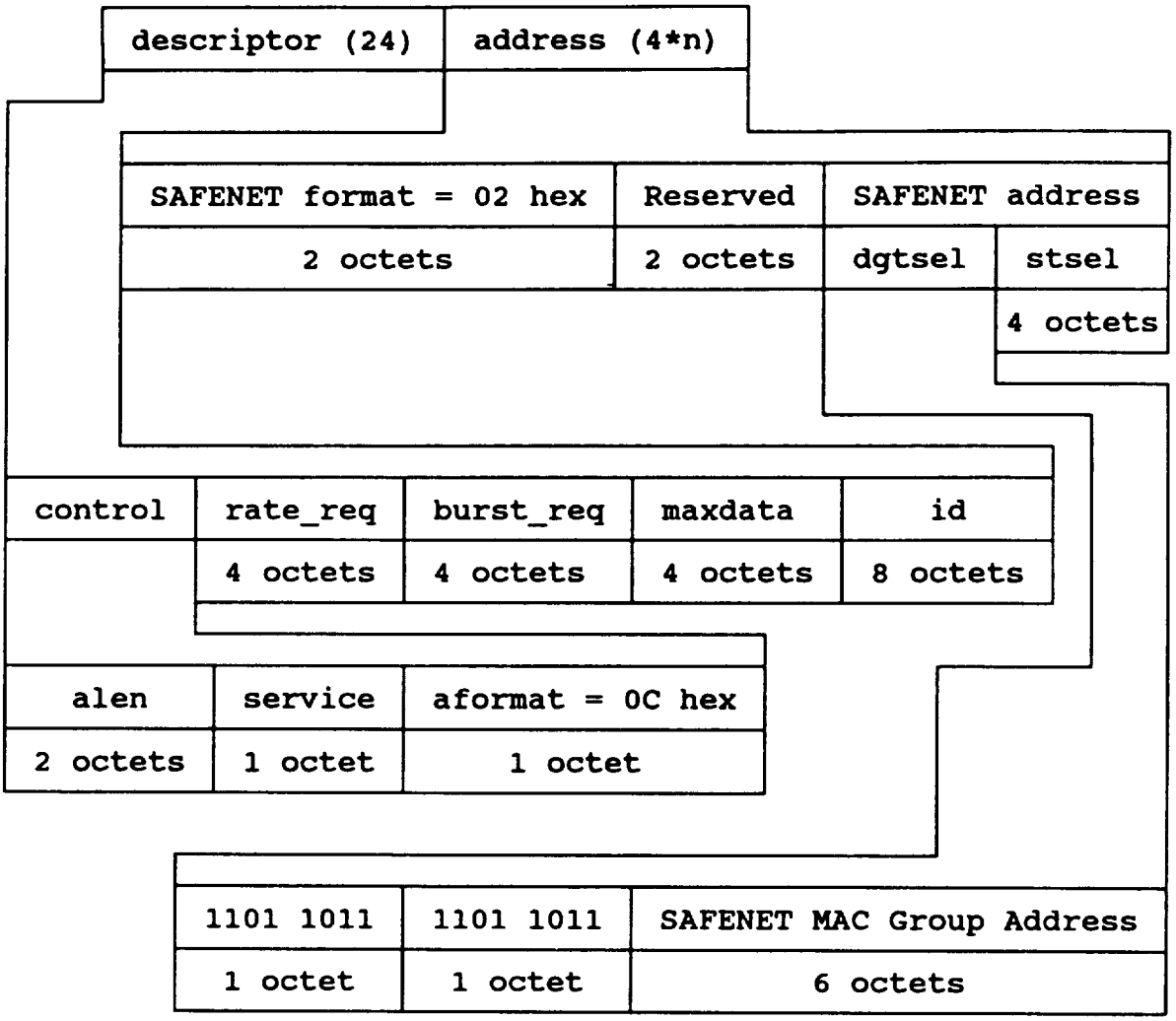


Figure 12. SAFENET Multicast XTP Address Segment

values or delegation of authority for the assignment thereof, shall be the responsibility of SPAWAR 231. SPAWAR 231 is the point of contact for those seeking actual NSAP addresses or NSAP address domains. Assignment of SAFENET NSAPs without the authorization of either SPAWAR 231 or an administrative authority designated by SPAWAR 231 for such purposes is prohibited.

SAFENET systems shall use the ISO 6523-ICD Initial Domain Identifier (IDI) format as defined in ISO 8348/ADD 2. The Domain Specific Part (DSP) shall be in the binary abstract syntax. The Authority and Format Identifier (AFI) shall have a value of

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decimal 47. The SAFENET NSAP address structure is shown in Figure 13.

IDP		DSP						
AFI	IDI	DFI	AA	RSVD	RD	Area	System	Sel
47	5	128	-	-	-	-	-	-
1	2	1	3	2	2	2	6	1

Note: The numbers in the bottom row of the figure are the octet lengths of the associated fields.

Abbreviations:
AA - Administrative Authority
AFI - Authority and Format Identifier
DFI - DSP Format Identifier
DSP - Domain Specific Part
IDI - Initial Domain Identifier
IDP - Initial Domain Part
RD - Routing Domain
Rsvd - Reserved
Sel - Selector

Figure 13. SAFENET NSAP Address Structure

SAFENET NSAP addresses shall have a fixed address length of 20 octets. Component field widths are as indicated in Figure 13.

The 1-octet AFI field shall contain the value decimal 47, encoded as two binary coded decimal digits (0100 0111), signifying the ISO 6523-ICD format as defined in ISO 8348 ADD/2. This means that the IDI part is interpreted as a four decimal digit International Code Designator (ICD) and that the DSP has a binary abstract syntax.

The 2-octet IDI field shall contain value decimal 5, encoded as four binary coded decimal digits (0000 0000 0000 0101), which is the International Code Designator associated with this NSAP structure.

The 1-octet DFI field is used to indicate the structure, semantics, and administrative requirements associated with the remainder of the Domain Specific Part of the NSAP address structure. The DFI shall contain the value decimal 128, encoded as a 1-octet binary integer (1000 0000), indicating that the overall DSP format is that defined in GOSIP 2.0.

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The 3-octet AA field is used to identify the entity responsible for the organization of IS's and ES's into Routing Domains and Areas, the allocation and assignment of the remaining portion of the DSP, and the policies that govern the dissemination of information within and external to the Administrative Domain. The top-level administrative authority for SAFENET NSAP addresses is the DoD. AA values have been assigned to the DoD for the administration of both classified and unclassified addressing domains. The AA value for unclassified domains shall be decimal 1792, encoded as a 3-octet binary integer (0000 0000 0000 0111 0000 0000). The AA value for classified domains shall be decimal 24064 encoded as a 3-octet binary integer (0000 0000 0101 1110 0000 0000).

The 2-octet RSVD field is positioned to be available for encoding higher level routing structures above those of the routing domain, or to be used to expand either the Administrative Authority or the Routing Domain fields in future DSP formats should the need arise. This 2-octet field will be set to zero in all SAFENET implementations.

The 2-octet RD field identifies a unique Routing Domain within an Administrative Domain. Values for these fields shall be assigned by SPAWAR 231. Assignment of an RD value by SPAWAR shall confer upon the assignee the authority for the designation and administration of areas and addresses within the associated routing domain. RD values will be encoded as 2-octet binary integers.

The 1-octet Area field identifies a unique subdomain of the Routing Domain. Area values shall be encoded as 2-octet binary integers.

The 6-octet System field identifies a unique system (End System or Intermediate System) within an Area. No two systems within the same area shall have the same system value. System values shall be encoded as 6-octet binary integers.

The 1-octet Selector field is used to identify a binding between a network layer protocol entity and a direct user of that protocol entity. The selector field value depends on the transport layer entity to which an NSAP is bound. See 5.2.4.1 for N-selector values to be used in SAFENET.

5.2.4.1 Network Service Access Point Selectors. The selector field of the NSAP address shall be used to identify the transport layer entity bound to the associated NSAP. The selector values shown below shall be used to indicate the applicable service/user binding. Note that there is no selector

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associated with the use of the Inactive Subset of the CLNP by the CL transport protocol. The CL transport protocol entity is the sole user of the Inactive Subset.

NSAP Binding	N-Selector (Hex)
a. Connectionless Network : Connection-oriented Transport	01
b. Connectionless Network : Connectionless Transport (non-multicast)	02
c. Connectionless Network : XTP (non-multicast)	08

5.2.4.2 Network entity titles. Each SAFENET router shall have only one Network Entity Title (NET) defined for it. The NET shall use the SAFENET NSAP address format defined in 5.2.4. All requirements governing SAFENET NSAP addresses apply to NETs, with the additional requirement that the NSAP selector field shall be set to zero. A zero NSAP-selector value is not allowed in any other case.

5.2.4.3 Inactive subset addressing. When the inactive subset of the ISO CLNP is used there shall be no NSAP address information required or conveyed in any PDU.

5.2.5 Logical Link Control addresses. LLC service users are addressed at LSAPs. SAFENET uses LSAP addresses for the purpose of selection between protocol entities which use the services of LLC. The two LSAP addressing fields within an LLC PDU are the destination service access point and the source service access point fields. Each of these fields shall be one octet in length. The values of these fields shall be the same within a given SAFENET LPDU. All SAFENET implementations shall implement and recognize the following standard LSAP addresses:

LSAP Address (hex)	Selected Protocol
a. (AA)	Sub-Network Access Protocol
b. (FE)	OSI CLNP

In accordance with ISO/IEC TR 10178 SAFENET specifies the LSAP value FE (hex) for protocols within the scope of ISO/IEC TR 9577. This includes the ISO CLNP, the Inactive Subset of the ISO CLNP, the ISO IS/IS Intradomain Routing Exchange Protocol, and the ISO ES/IS Routing Exchange Protocol.

In accordance with ISO/IEC TR 10178, SAFENET specifies the LSAP value AA (hex), the Sub Network Access Protocol Service Access Point (SNAP SAP), for the identification of protocols not identified by ISO/IEC TR 9577. In particular, SAFENET uses the SNAP SAP for the selection of XTP when used for the issuance and

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reception of multicast packets. (Note: Non-multicast XTP packets are encapsulated in ISO CLNP PDUs, which are conveyed through the LSAP associated with CLNP, namely LSAP FE (hex)).

Organizationally Unique Identifier	Subidentifier Field
01-40-4F	817D (hex)

Figure 14. SNAP SAP Protocol Identifier Fields

When an LLC PDU carries the SNAP SAP address in the destination service access point and source service access point fields, the LPDU information field is further defined such that it contains a protocol identification field and an protocol data field. The protocol identification field is a 40-bit field consisting of a 24-bit Organizationally Unique Identifier (OUI) and a 16-bit subidentifier field, as shown in Figure 14. The value of the OUI for SAFENET implementations is 01-40-4F (IEEE standard hexadecimal notation) as assigned to SAFENET by the IEEE standards office. The value of the subidentifier field, called the type field in the XTP definition, is 817D (hex) as specified in the XTP standard. This number is given as a big-endian, bit-order value. The protocol data field is used to contain the XTP packet being conveyed through the SNAP SAP.

5.2.6 Media Access Control addresses. Each MAC frame shall consist of two address fields: the destination station address field and source station address field. These fields contain the MAC addresses of the destination and source stations, respectively. All MAC addresses used in SAFENET shall be 48-bit Universal Addresses.

5.2.6.1 Individual addressing requirements. Each MAC in a SAFENET network shall have a universally-administered 48-bit address as defined in IEEE 802. Associated with every individual, universally-administered MAC address in a SAFENET implementation shall be a vendor/organization-specific IEEE standard OUI that has been obtained from the IEEE Standards Office. This OUI shall constitute the first 24 bits of the 48 bit address. Multiple OUIs may be found in a SAFENET system built with equipment from multiple vendors. All, however, are required to be obtained from the IEEE standards office. It is the responsibility of vendors/organizations to assure that MAC addresses they provide in a SAFENET system are globally unique.

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Note: The OUI assigned to SAFENET is reserved for special uses and is not available for providing universally-administered individual station addresses.

5.2.6.2 Group addressing requirements. Each SAFENET station shall be capable of recognizing a number of MAC group addresses. This means that a frame containing such a destination address shall be copied from the LAN and passed on to the appropriate protocol entity within the station. The MAC source address field in a frame addressed to a group of destination stations shall identify the single station originating the frame, and the Individual/Group bit in the source address field shall be 0. The group address recognition requirements in SAFENET are divided between those associated with universally-administered group addresses for standard protocols and those associated with the SAFENET multicast address space.

5.2.6.2.1 Standard group addresses. The following standard group addresses shall be recognized in all SAFENET implementations, with the exception of the first two addresses, which shall be required only in Intermediate systems which implement the ISO 10589 IS/IS Routing Exchange Protocol. All addresses are given using the IEEE 802 standard illustrative representation (in hexadecimal) for LAN MAC addresses.

a.	01-80-C2-00-00-14	Intermediate Systems (ISO 10589)
b.	01-80-C2-00-00-15	Intermediate Systems (ISO 10589)
c.	01-80-C2-00-01-00	FDDI SMT Directed Beacon
d.	01-80-C2-00-01-10	FDDI SMT Status Report Frame
e.	09-00-2B-00-00-04	All End System Network Entities (ISO 9542)
f.	09-00-2B-00-00-05	All Intermediate System Network Entities (ISO 9542)

Note: When a group address is used in conjunction with the FDDI Synchronous Bandwidth Allocation option, the following group address needs to be recognized:

01-80-C2-00-01-30	FDDI SMT Synch Bandwidth Allocation

5.2.6.2.2 SAFENET group address space. In addition to the requirement to recognize the standard, universally-administered group addresses specified above, a SAFENET implementation shall also be capable of registering to receive from any address within the SAFENET group address space. This address space consists of 65,536 contiguous universally-administered group addresses and has been established in support of the multicast services that are available when SAFENET lightweight services are implemented. This space is divided into two subspaces, one for user

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assignment, the other reserved for administration and assignment by SPAWAR 231. Implementations are required to be able to register for and recognize any group addresses defined within the entire 65,536 SAFENET group address range. SAFENET does not specify how or where this registration and recognition capability is performed within a SAFENET implementation. The means by which this capability is provided is left to the implementor.

The first subspace, the User-controlled Subspace, is a set of 8,191 group addresses that are available for a SAFENET user/implementer to use. SAFENET places no restrictions on the use of these addresses. It should be noted that while these addresses should not occur in non-SAFENET systems that correctly implement IEEE standard addresses, there may be duplication within and between SAFENET implementations. It is the responsibility of implementors and system administrators to develop policies for the use and control of group addresses from this general user pool.

The second subspace, the SAFENET controlled subspace, is a contiguous set of 57,343 group addresses. These addresses are not available for user allocation and assignment. Allocation and assignment of these addresses is reserved solely for SPAWAR 231. An address from this subspace has been defined to be used in conjunction with the STS.

Figure 15 depicts the SAFENET group address space. The figure shows the relative addresses within the 65,536 element address block bounded by the following absolute addresses (shown in left-to-right media transmission order):
1000 0000 0000 0010 1111 0010 1111 1111 0000 0000 0000 0000 and
1000 0000 0000 0010 1111 0010 1111 1111 1111 1111 1111 1111
inclusive.

Global-uniqueness of SAFENET group addresses is assured by conformance of SAFENET to the address administration standards of IEEE 802. SAFENET has been assigned an OUI by the IEEE. The OUI is 24 bits in length and has the following bit pattern:

x000 0000 0000 0010 1111 0010.

The first bit of the OUI, the Individual/Group (I/G) bit, indicates whether an associated address is an individual address (I/G = 0) or a group address (I/G = 1). The second bit of the OUI is always set to 0 indicating universal address administration. In the bit representation shown, the bits are displayed in the order transmitted on the SAFENET LAN medium, from left to right. Thus the left-most bit displayed is the I/G bit.

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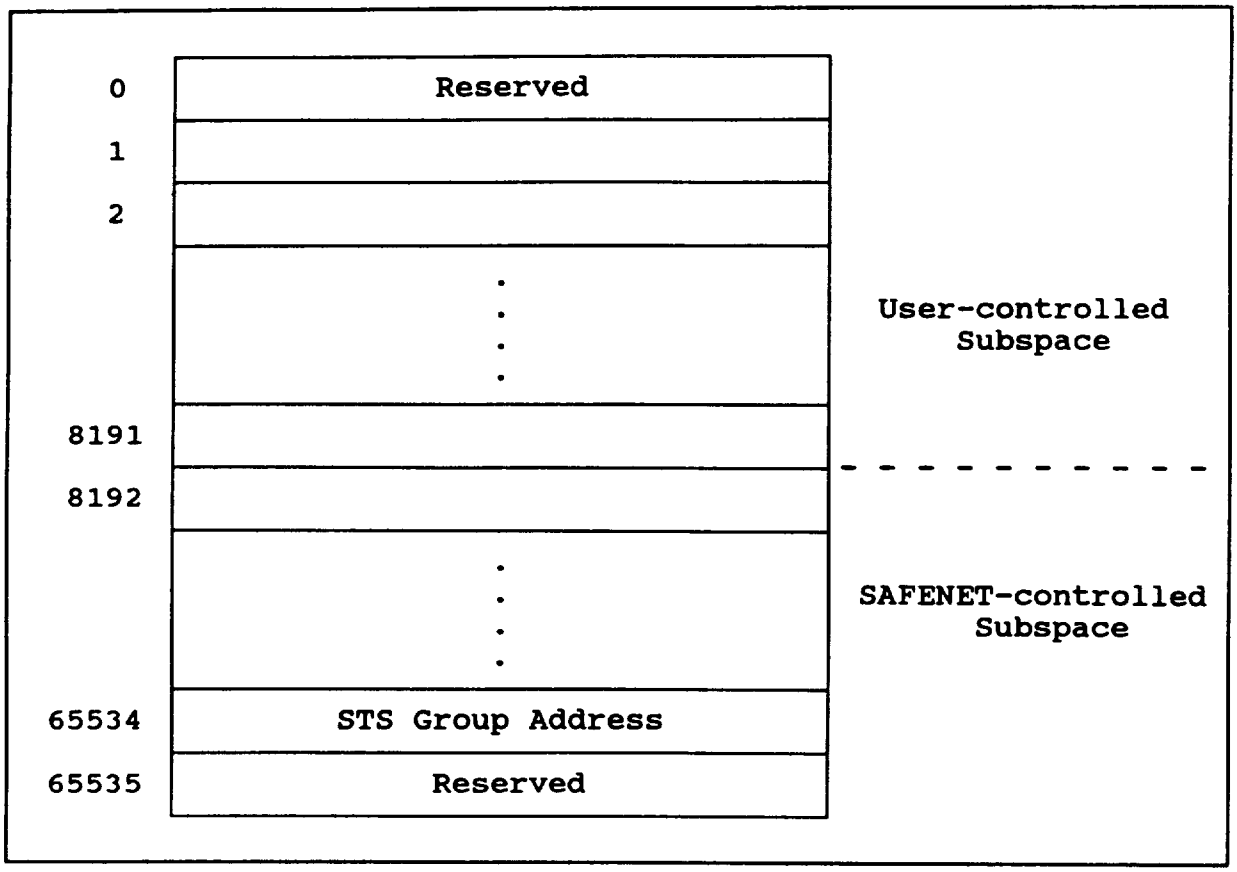


Figure 15. SAFENET Group Address Space

When used in conjunction with group addressing, the I/G bit = 1 and the OUI can be represented using the IEEE 802 standard illustrative representation (in hexadecimal) as 01-40-4F. This corresponds to the left-to-right transmission-order bit pattern:

1000 0000 0000 0010 1111 0010.

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The SAFENET OUI shall constitute the first three octets of any 48-bit SAFENET group address. This is followed by a fixed 8-bit field consisting of all 1's. This is followed by a 16-bit field containing the non-OUI portion of a SAFENET group address from either the User-controlled Subspace or the SAFENET-controlled Subspace. Figure 16 shows the complete SAFENET group address format.

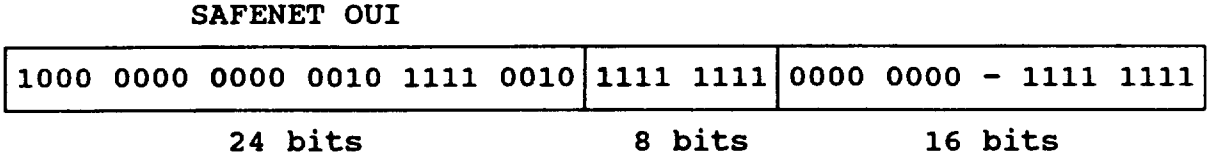


Figure 16. SAFENET Group Address Format

5.3 Router protocol requirements. A SAFENET network layer router shall be capable of relaying packets between interconnected LAN segments for any conforming SAFENET profile. A SAFENET router shall meet the ISO definition of an intermediate system (IS). It may be self contained (i.e., serve solely as an IS) or it may be a subset of a complete SAFENET end system. The minimum set of protocols required to implement a self contained router in SAFENET shall be as follows:

a. ISO CLNP (see 5.1.2.4.2)
b. ISO ES/IS routing exchange protocol (see 5.1.2.4.2.3)
c. IEEE logical link control protocol (see 5.1.2.4.3)
d. ANSI FDDI protocols (see 5.1.3)

In addition, if dynamic route determination between intermediate systems is required, the ISO IS/IS intra-domain routing protocol shall be implemented (see 5.1.2.4.2.2).

5.4 Physical medium. Implementation of the SAFENET physical medium shall be in accordance with ISO 9314-3 or with the requirements given in this section. The components and techniques defined below provide the implementor with additional options for a more survivable tactical system. The requirements in this section are only for those systems where additional survivability is required, and are not requirements for every SAFENET physical medium implementation.

5.4.1 Optical components. All specifications and requirements for the optical components used in the SAFENET physical medium are given below. A discussion of the SAFENET power budget is provided in MIL-HDBK-818-1.

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5.4.1.1 Optical fiber. The optical fiber used in SAFENET shall meet the requirements of MIL-F-49291. It shall be multi-mode, graded index, radiation hardened fiber. The required characteristics of this fiber shall be as follows:

a.	Dimensions:	core dia:	62.5 μm ± 3.0 μm
		cladding dia:	125 μm ± 2.0 μm
		coating dia:	250 μm ± 15 μm
b.	Numerical aperture:	0.275 ± 0.015	

5.4.1.2 Fiber optic cable. The fiber optic cables used in SAFENET shall meet the requirements of MIL-C-85045 or NAVSEA 6710781. All multi-mode optical fibers used in these cables shall meet the requirements of 5.4.1.1. The required performance characteristics of the cabled multi-mode fiber shall be as follows:

a.	Attenuation:	1.0 dB/km @ 1.30μm (MAX)
b.	Bandwidth:	350 MHz-km @ 1.30μm (MIN)

5.4.1.2.1 Recommended cable type. For shipboard use, it is recommended that the cable be of the optical fiber cable component (OFCC) type.

5.4.1.2.2 Trunk cables. The trunk cables are those fiber optic cables which are used to create the rings of an FDDI dual fiber trunk. Dual fiber trunk cables are used to construct the combined dual-rings of any SAFENET physical topology. Trunk cables shall be of the type specified in 5.4.1.2, and they should be of the type recommended in 5.4.1.2.1.

5.4.1.2.3 Optical interface cables. Optical interface cables are those fiber optic cables which connect network stations, TCUs, and repeaters. Optical interface cables used in the tree connection (see 4.5.3.3) require two optical fibers, while those used in the trunk connection (see 4.5.3.2) require four optical fibers. Optical interface cables shall be of the type specified in 5.4.1.2, and they should be of the type recommended in 5.4.1.2.1.

5.4.1.3 Trunk coupling unit. The trunk coupling unit (TCU) is a device which enables a network station or repeater to insert or de-insert itself onto the dual fiber trunk via a trunk connection. It shall be controlled by an electrical signal from the attached station, and shall not have any other source of electrical power. It consists of a pair of 2x2 optically independent bypass switches, packaged together, which are electrically connected in parallel to a single electrical interface cable. This pair of bypass switches shall be

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configured such that one switch provides access to the primary ring and the other provides access to the secondary ring, as shown in Figure 17. Any optical switch used in SAFENET shall meet the requirements of MIL-S-24725. The required performance characteristics of this switch shall be as shown in Table II (losses measured in a 62.5/125 μm fiber @ 1.30μm).

5.4.1.4 Fiber optic splices. All fiber optic splices used in SAFENET, both for installation and repair, shall meet the requirements of MIL-S-24623. Optical power losses cited for splices are measured in a 62.5/125 μm fiber @ 1.30μm.

5.4.1.4.1 Installation splice. An installation splice is a splice which is used to connect a TCU pigtail to either a trunk cable or an optical interface cable. This splice shall be a part of the initial SAFENET installation. Installation splices shall be protected by an interconnection box as defined in 5.4.1.5. The required performance characteristic of this splice shall be as follows:

Optical power loss: 0.6 dB untuned (MAX)
0.2 dB tuned (MAX)

Table II. Optical Bypass Switch Performance Characteristics

CHARACTERISTIC	REQUIRED PERFORMANCE
Optical power loss	Bypass path: 0.8 dB (MAX) Transmit path: 0.8 dB (MAX) Receive path: 0.8 dB (MAX) Loop-back path: 1 - 6 dB
Channel isolation	40 dB (MIN)
Switching time	Insertion: 25 ms (MAX) Bypass: 25 ms (MAX)
Control voltage	Insertion: 4.0 - 5.3 Vdc Bypass: 0.0 - 1.0 Vdc
Operating current	170 mA (MAX)

5.4.1.4.2 Installation repair splice. An installation repair splice is a splice which is used to replace an installation splice. Installation repair splices shall be

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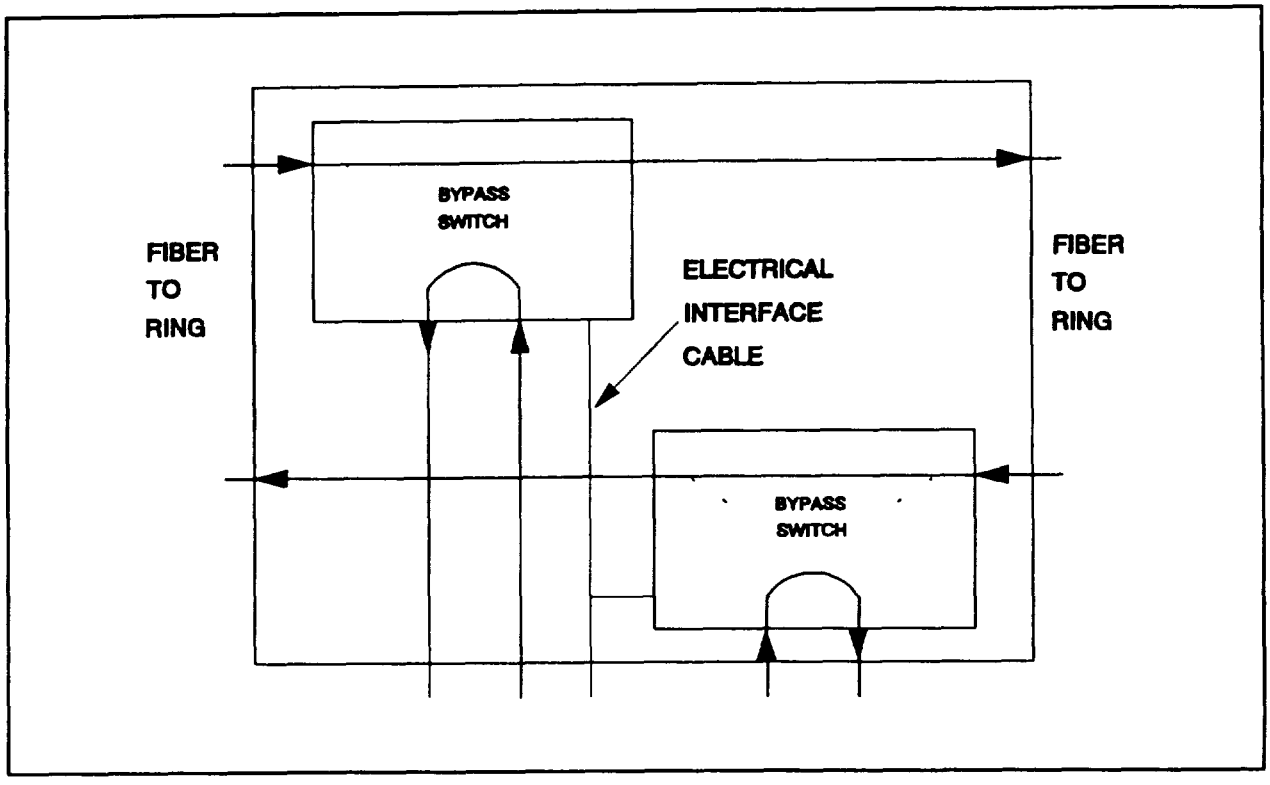


Figure 17. Trunk Coupling Unit

protected by an interconnection box as defined in 5.4.1.5. The required performance characteristic of this splice shall be as follows:

Optical power loss: 0.6 dB untuned (MAX)
0.2 dB tuned (MAX)

5.4.1.4.3 Cable repair splice. A cable repair splice is a splice which is used to repair a cable break which is outside an interconnection box. This splice shall include its own protective enclosure. The required performance characteristic of this splice shall be as follows:

Optical power loss: 0.6 dB untuned (MAX)
0.2 dB tuned (MAX)

5.4.1.5 Interconnection boxes. The splices described in 5.4.1.4.1 and 5.4.1.4.2 shall be protected by an interconnection box. This interconnection box shall meet the requirements of MIL-I-24728.

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5.4.1.6 Fiber optic interface connector. The FOIC used in SAFENET shall meet the optical requirements of MIL-C-28876, specification sheets #1 or #11. This is a circular, heavy duty, multi-fiber connector receptacle. The receptacle part number shall be B1S1 (shell size 13, insert arrangement 1, socket termini, key position 1). The mating plug connector shall be intermatable with the above receptacles. The required performance characteristic of this connector shall be as follows (loss measured in a 62.5/125 μm fiber @ 1.30μm):

Mated optical power loss: 1.0 dB (MAX)

The pin definition for the FOIC shall be as follows:

Pin 1: Ring A In
Pin 2: Ring A Out
Pin 3: Ring B In
Pin 4: Ring B Out

5.4.1.7 Optional fiber optic connector. If single fiber optical connectors are used instead of optical splices they shall meet the requirements of MIL-C-83522/16. Single fiber optical connectors shall be protected by an interconnection box as defined in 5.4.1.5. The required performance characteristic of this connector shall be as follows (loss measured in a 62.5/125 μm fiber @ 1.30μm):

Optical power loss: 1.0 Db (MAX)

5.4.2 Electrical components. All specifications and requirements for the electrical components used in the SAFENET physical medium are given below.

5.4.2.1 Electrical interface cable. The electrical interface cable used in this network shall meet the requirements of MIL-C-24640 or MIL-C-24643. The cable shall have at least two conductors, and it shall be shielded. The required performance characteristic of this cable shall be as follows:

Maximum resistance: 8.2 ohms/Km (2.5 ohms/Kft)

Further information on the layout and installation of electrical interface cables is given in 4.5.3.

5.4.2.2 Electrical interface connector. The electrical interface connector shall meet the requirements of MIL-C-26482, military standard MS3476. This is a three pin, bayonet coupling connector. The backshell used shall meet the requirements of MIL-C-85049.

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Intermatability with the following part number is be required for SAFENET conformance: MS3475 W 14 - 4 P.

5.5 Management Information Base (MIB). Object class definitions for the MIB in a SAFENET implementation shall be taken, in order of precedence:

a) from those defined in the base standard,
b) from the NIST Special Pub 500-202 Management Information Library (MIL),
c) from those provided in this standard, and when none of the above are available
d) from those provided by the implementor and registered as non-proprietary definitions in the appropriate arc of the ISO registration tree.

Specific object class definitions from the first three sources are not available at this time for a number of services and protocols of the SAFENET profile. This edition of SAFENET does not list specific managed object class definitions for the following services: FTAM, Directory Services, ACSE, ROSE, CMISE, Lightweight Application Services, Presentation Services, Session Services, CL Transport, XTP, ES/IS, IS/IS, and LLC. When required to meet system needs and not yet available from one of the first two sources given above, the definition of these is a local matter as provided in d) above. Any object class definitions used shall be non-proprietary, provided in the form specified in ISO/IEC 10165-4, and registered in an appropriate arc of the ISO registration tree. Where feasible, these shall be derived from a standard object class definition such as "equipment" and "computerSystem" found in the NIST Special Pub 500-202 MIL.

The object class definitions shall be instantiated in each SAFENET node in accordance with the profile used for at least the managed object classes required by the base standards instantiated or specified in 5.5.2 and 5.5.3. Other object class definitions may be instantiated to meet the needs of a specific system. In the tables given in 5.5.2 and 5.5.3 of NIST Special Pub 500-202, MIL refers to the Management Information Library included in part 18 of NIST Special Pub 500-202.

5.5.1 SAFENET specified managed object classes. Object class definitions unique to the SAFENET profile are detailed in Appendix B.

5.5.2 Object classes mandatory in all profiles. The following object classes are required in all profiles.

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Service/Protocol	Object Class	Where Defined
CLNP	clNetworkProtocolLayerEntity	NIST Special Pub 500-202 MIL
FDDI	fddiSMT	FDDI SMT
	fddiPATH	FDDI SMT
	fddiMAC	FDDI SMT
	fddiPORT	FDDI SMT
Time Services		Appendix B

5.5.3 Object classes mandatory in the OSI profile only. The following object classes are required in the OSI profile.

Service/Protocol	Object Class	Where Defined
SMASE	computerSystem	NIST Special Pub 500-202 MIL
	alarmRecord	ISO/IEC 10165-2
	attributeValueChangeRecord	ISO/IEC 10165-2
	discriminator	ISO/IEC 10165-2
	eventForwardingDiscriminator	ISO/IEC 10165-2
	eventLogRecord	ISO/IEC 10165-2
	log	ISO/IEC 10165-2
	logRecord	ISO/IEC 10165-2
	objectCreationRecord	ISO/IEC 10165-2
	objectDeletionRecord	ISO/IEC 10165-2
	relationshipChangeRecord	ISO/IEC 10165-2
	stateChangeRecord	ISO/IEC 10165-2
Connection Oriented Transport	coTransportProtocolLayerEntity	NIST Special Pub 500-202 MIL
	transportConnection	NIST Special Pub 500-202 MIL

5.6 Time services. All requirements for the SAFENET Time Service (STS) are given below. All SAFENET implementations shall comply with these requirements.

5.6.1 Network Clock requirements. Each SAFENET node shall implement a Network Clock to provide time values to that node and to synchronize to a common reference time. The definition of reference time is beyond the scope of this standard, but some guidance is provided in MIL-HDBK-818-1. The physical implementation of the network clock is a local matter. Guidelines for the implementation of a Network Clock are found in NTP V3 and in MIL-HDBK-818-1. While it is expected that many systems will use additional hardware for implementation of the

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Network Clock in order to meet system performance requirements, the STS does not require any special hardware beyond the oscillator and/or registers commonly used by any computer to provide a source of time.

5.6.1.1 Time format. All implementations of a Network Clock shall maintain values of time in a single format, defined in 5.6.1.1.3.1 as Global Time. This is the time format used by the clock synchronization service and provided to the user of the time service. In addition, a general time period or interval of time is defined as TimeInterval and formally specified in 5.6.1.1.3.2.

5.6.1.1.1 SAFENET epoch. The GlobalTime format represents the amount of time (in seconds and the fraction of a second) since January 1, 1970. The choice of January 1, 1970 as an epoch is a modification of NTP V3 (see 5.6.2.4.2). This epoch was chosen for compatibility with the POSIX operating system standard.

5.6.1.1.2 SAFENET timescale. Leap seconds, when they occur, shall not be incorporated into GlobalTime. GlobalTime represents a continuous timescale and as such does not include the concept of leap seconds. GlobalTime begins on January 1, 1970 and moves forward continuously from that point. This timescale intersects UTC at the initiation of UTC (January 1, 1972). However, it gradually diverges from UTC because of the occurrence of leap seconds in UTC. This choice of a continuous timescale is a modification to NTP V3 (see 5.6.2.4.2). Leap second information is maintained by the Time Manager for use by routines that perform conversions to other time formats and from external references. These routines are outside the scope of this document. Additional discussion on timescales and leap seconds is found in NTP V3 and in MIL-HDBK-818-1.

5.6.1.1.3 Formal definitions. The formal portion of these definitions is based on Abstract Syntax Notation, Version 1 (ASN.1) (see ISO 8824 and 8825).

5.6.1.1.3.1 GlobalTime definition.

```
GlobalTime ::= BIT STRING (SIZE (64))
  -- GlobalTime is an unsigned 64-bit fixed-point number
  -- that represents a time value in seconds and the binary
  -- fraction of a second.  If the most significant bit is
  -- labeled 0 and the least significant bit 63, then the
  -- binary point is located between bits 31 and 32.  Bits
  -- 0-31 represent the whole number of seconds since 0
  -- hours, 0 minutes, and 0 seconds on January 1, 1970 at
```


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```
    -- the Greenwich meridian.  Bits 32-63 represent the
    -- fraction of a second beyond that point.  A GlobalTime
    -- value in which all 64 bits are zero is defined to be
    -- invalid and indicates an uninitialized or faulty
    -- Network Clock.  GlobalTime is aligned with the UTC
    -- timescale at the beginning of the UTC era resulting in
    -- the GlobalTime value of 63,072,000 seconds representing
    -- 0 hours, 0 minutes, and 0.0 seconds on January 1, 1972.
    -- GlobalTime shall not include leap second adjustments
    -- when they occur and as a result will diverge from UTC
    -- over time.
```

5.6.1.1.3.2 TimeInterval definition.

```
TimeInterval ::= BIT STRING (SIZE (64))
    -- TimeInterval is an unsigned 64-bit fixed-point number
    -- that represents a time value in seconds and the binary
    -- fraction of a second.  If the most significant bit is
    -- labeled 0 and the least significant bit 63, then the
    -- binary point is located between bits 31 and 32.  Bits
    -- 0-31 represent the integer number of seconds, and bits
    -- 32-63 represent the fraction of a second beyond the
    -- number of seconds.
```

5.6.1.2 Clock granularity. All implementations of a Network Clock shall maintain a granularity such that the smallest time increment maintained is no larger than 1 binary millisecond (1/1024 of a second), containing at least ten significant bits in the fractional portion of GlobalTime. Implementations may define a smaller granularity. The smallest possible clock granularity provided for by the GlobalTime format is approximately 230 picoseconds.

5.6.1.3 Clock monotonicity. All implementations of a valid Network Clock shall maintain time values such that they are always monotonically increasing. A Network Clock is not considered valid, and therefore can be reset without violating the monotonicity requirement, until it has been initialized and is performing in a correct manner.

5.6.1.4 Clock error bounds. In order to provide information on the correctness of the Network Clock, the value of the maximum error of the Network Clock shall be maintained at all times. A maximum error value of all ones shall be considered an invalid (nonfunctioning) clock. In addition, the estimated error of the Network Clock shall be maintained to provide a more accurate description of the statistical performance of the Network Clock.

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5.6.1.5 Clock performance. SAFENET establishes no performance requirements for the individual Network Clocks in a network. It is assumed that the system engineer will use components of sufficient quality to meet the requirements of the particular system. The quality of the components used will have a direct impact on the level of synchronization achieved. Additional guidance is provided in MIL-HDBK-818-1.

5.6.2 Clock synchronization service requirements. The clock synchronization service synchronizes the Network Clocks in all nodes of a SAFENET network to a time standard or ensemble timescale. The clock synchronization service shall also provide for selection of the best clock in a given set, detection of a clock that has failed, distribution of an indication that a leap second is going to occur, and basic reconfiguration of the time synchronization system when a clock acting as a server has failed.

5.6.2.1 Clock phase and frequency. Time values maintained by a Network Clock have two components: phase (i.e., apparent time) and frequency. The clock synchronization service shall synchronize the Network Clocks by performing corrections to both their phase and frequency so as to ensure greater clock stability. By synchronizing frequency as well as phase, the impact of the synchronization service on the underlying resources is reduced and the highest quality of timing information is maintained.

5.6.2.2 Synchronization within SAFENET. The clock synchronization service shall synchronize the value of Global Time contained in each Network Clock to a time standard or ensemble timescale. These values are synchronized through the use of the time synchronization protocol defined in 5.6.2.3.

5.6.2.3 Synchronization to an external reference. The STS shall permit the synchronization of a Network Clock in a SAFENET node to an external reference. This synchronization function needs to account for whether or not the external reference considers leap seconds. Also, each external source needs to specify appropriate performance metrics in order for the protocol to select reliable peers. All implementation aspects of external time references, access mechanisms, and methods and frequencies of synchronization to external time are left as system design matters.

5.6.2.4 Time synchronization protocol. All implementations of SAFENET shall use the Network Time Protocol (NTP) for time synchronization. It shall directly use the data transfer services of the OSI CL transport protocol.

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5.6.2.4.1 Use of NTP with OSI CL transport protocol. The following modifications to NTP V3 shall be made for operation of NTP with the OSI CL transport protocol.

a. The transport address used by NTP shall be an ISO Transport Service Access Point (TSAP) consisting of an NSAP and a T-selector.
b. The peer addresses and host addresses used within the NTP protocol machine shall be Network Service Access Point (NSAP) addresses in compliance with ISO 8348 (see 5.2.4).
c. The peer port and host port used within the NTP protocol machine shall be the T-selector values specified in 5.2.3.
d. The Reference Identifier field in the NTP PDU shall be 20 octets in length. For stratum 0 or stratum 1, this field shall contain a left-justified zero-padded ASCII string as specified in NTP V3. For stratum 2 and greater, the reference clock identifier shall be the NSAP address of the peer selected for synchronization. The concept of stratum as used in NTP is discussed in MIL-HDBK-818-1.

5.6.2.4.2 Additional NTP requirements. The following requirements shall be observed in any SAFENET implementation of the NTP.

a. It shall support all five modes of operation specified in NTP V3. Selection of modes may be done during system configuration or through the time manager.
b. It shall support the clock selection algorithms as defined in Chapter 4 of NTP V3.
c. It shall support the clock filtering algorithms as defined in Chapter 4 of NTP V3.
d. During normal operation, parameter default values shall be those specified in Table 4 of NTP V3. The time manager shall be capable of modifying these values to better accommodate special system requirements.
e. The SAFENET timescale shall have an epoch of January 1, 1970.
f. The SAFENET timescale shall be continuous and therefore shall not include leap seconds.

5.6.2.5 Clock synchronization performance. The STS imposes no requirements on the performance of the clock synchronization service. However, this is a critical area that will require attention during system design. The performance requirements of the system will need to be specified and appropriate

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configuration parameters determined. A discussion of this subject is found in MIL-HDBK-818-1.

5.6.3 User time service requirements. The user time service shall provide time values to users in the form of Global Time. The ST-GET-TIME service defined below shall be implemented to provide this capability. The user time service may also be implemented to provide time values in other formats, and to provide time-related services other than those described below. The conversion from Global Time to any additional time formats is a local matter.

5.6.3.1 ST-GET-TIME service. This service provides a user with access to the current value of Global Time maintained in the local Network Clock. The STS shall not provide access to the Network Clock values in remote nodes except through management. The time manager shall be capable of gaining access to the Network Clock values in remote nodes; however, this method will in all likelihood be quite inefficient and inaccurate.

5.6.3.1.1 ST-GET-TIME primitives. A service primitive is an abstract, implementation independent representation of an interaction between a service user and the service provider. A primitive has associated with it a direction, which may be from a service user to the service provider or from the service provider to a service user, and one or more parameters. The primitives used to describe the ST-GET-TIME service shall be request and confirm. Their interactions are shown in Figure 18.

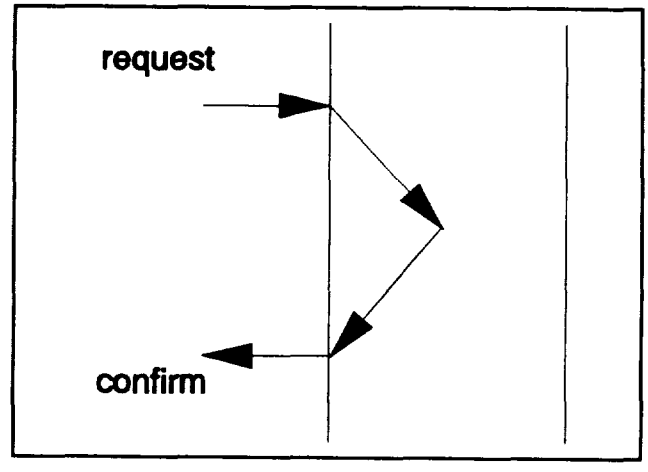


Figure 18. Time Sequence Diagram for ST-GET-TIME Service

a. Request: An interaction in which a service user requests some action of the service provider. The direction associated with this primitive type is from service user to service provider.

b. Confirm: An interaction in which the service provider reports to a service user the completion of some action previously initiated by a request primitive. The

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direction associated with this primitive type is from service provider to service user.

5.6.3.1.2 ST-GET-TIME parameters. The ST-GET-TIME service parameters and their applicabilities shall be as listed below. See 5.6.3.1.3 for the parameter definitions.

Parameter Name	Request	Confirm
StsGlobalTimeValue	NA	M
StsCorrectnessInterval	NA	U
StsExpectedError	NA	U
StsUpdateTime	NA	U
StsOperationSuccess	NA	M

Associated with each parameter used in service primitives is an applicability which describes that parameter's status with respect to the primitive in which it is named. Applicabilities relevant to the ST-GET-TIME primitive are listed below:

a. M (indicates the parameter is mandatory)
b. U (indicates the parameter is user optional)

5.6.3.1.3 ST-GET-TIME parameter definitions. The definitions of the parameters used in the ST-GET-TIME primitives are given below. The formal portion of each definition is based on ASN.1 syntax. The definitions for GlobalTime and TimeInterval used below are found in 5.6.1.1.3.

5.6.3.1.3.1 StsGlobalTimeValue. This parameter is the GlobalTime value contained in the local Network Clock in a SAFENET node at the time that a request for GlobalTime was issued. A value of all zeros indicates that the Network Clock is not synchronized. This could be because it is either faulty or uninitialized.

StsGlobalTimeValue ::= GlobalTime

5.6.3.1.3.2 StsCorrectnessInterval. This parameter represents the interval in which the value returned in StsGlobalTimeValue is guaranteed to be valid, (StsGlobalTimeValue - StsCorrectnessInterval < correct time < StsGlobalTimeValue + StsCorrectnessInterval). The value of this parameter shall be positive, and a value of all ones indicates that no assertions can be made about the correctness of the Network Clock. This is the synchronization distance defined in NTP V3.

StsCorrectnessInterval ::= TimeInterval

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5.6.3.1.3.3 StsExpectedError. This parameter provides an indication of the expected quality of the local Network Clock. It is an estimate of the interval over which the value of time returned in the StsGlobalTimeValue parameter is valid. This is the root dispersion defined in NTP V3.

StsExpectedError ::= TimeInterval

5.6.3.1.3.4 StsUpdateTime. This parameter represents the most recent time at which the Network Clock was updated. This is the reference timestamp in NTP V3. If the Network Clock has not been synchronized the value shall be zero.

StsUpdateTime ::= GlobalTime

5.6.3.1.3.5 StsOperationSuccess. This parameter indicates whether or not the ST-GET-TIME request was successful. A value of "true" indicates that the request was completed successfully. A value of "false" indicates that the request could not be performed and the data returned, if any, should be disregarded.

StsOperationSuccess ::= BOOLEAN

5.6.4 Time management service requirements. The time management service currently includes only a definition of the managed objects required for management of the STS. The entity responsible for management of the STS is referred to as the time manager. The specifics of the time manager are left to the system designer. Guidance on the functions needed in a time manager and a discussion of some of the critical issues for system designers is contained in MIL-HDBK-818-1.

For SAFENET implementations of the OSI or combined profiles, the time manager shall use the OSI network management protocols to exchange information and execute the necessary functions. In SAFENET implementations with only the lightweight profile, there are currently no defined system network management protocol or agent functions. In these cases, the time manager shall include all the necessary functions. Guidance for implementors using only the lightweight profile is provided in Appendix B of NTP V3.

The management information base entries for the STS are defined in Appendix B. The STS layer management entity shall permit access to these objects for management of the STS. It shall also participate in overall network management by providing access to status information and reporting events.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Documents conforming to the requirements of this standard are intended for use as military standardization documents and are listed in the DODISS. The purpose of this standard is to define standards for a set of components which can be used in communication subsystems of mission critical computer resources.

6.2 Tailoring guidance. To ensure proper application of this standard, invitations for bids, requests for proposals, and contractual statements of work should tailor the requirements in sections 4 and 5 of this standard to include only those requirements which apply to the specific SAFENET profile which is to be implemented. To assist in this tailoring, those requirement paragraphs which apply to each of the profile classes permitted in SAFENET are listed below.

6.2.1 OSI profile. The following requirement paragraphs apply to any implementation of the OSI profile: 4.2.1.1, 4.2.2.2, 4.2.2.4, 4.2.3, 4.5.1.1 - 4.5.1.5, 4.5.3.1 - 4.5.3.4, 4.6.1, 4.8, 4.8.1, 4.9, 5.1.1.1, 5.1.1.1.1 - 5.1.1.1.9, 5.1.2.2, 5.1.2.2.1, 5.1.2.2.1.1, 5.1.2.2.1.2, 5.1.2.4, 5.1.2.4.1, 5.1.2.4.1.1, 5.1.2.4.2, 5.1.2.4.2.1, 5.1.2.4.2.3, 5.1.2.4.2.4, 5.1.2.4.3, 5.1.2.4.3.1, 5.1.3.1, 5.1.3.1.1, 5.1.3.2, 5.1.3.2.1, 5.1.3.3, 5.1.3.4, 5.1.3.4.4, 5.1.3.4.5, 5.1.3.4.5.1, 5.1.3.4.5.2, 5.2.1, 5.2.2, 5.2.3, 5.2.3.1, 5.2.3.2, 5.2.4, 5.2.4.1 - 5.2.4.3, 5.2.5, 5.2.6, 5.2.6.1, 5.2.6.2, 5.2.6.2.1, 5.2.6.2.2, 5.4.1.1, 5.4.1.2, 5.4.1.2.2, 5.4.1.2.3, 5.4.1.3, 5.4.1.4, 5.4.1.4.1 - 5.4.1.4.3, 5.4.1.5, 5.4.1.6, 5.4.2.1, 5.4.2.2, 5.5, 5.5.1, 5.5.2, 5.5.3, 5.6.1, 5.6.1.1, 5.6.1.1.1, 5.6.1.1.2, 5.6.1.1.3, 5.6.1.1.3.1, 5.6.1.1.3.2, 5.6.1.2 - 5.6.1.5, 5.6.2, 5.6.2.1, 5.6.2.2 - 5.6.2.4, 5.6.2.4.1, 5.6.2.4.2, 5.6.3, 5.6.3.1, 5.6.3.1.1 - 5.6.3.1.3, 5.6.3.1.3.1 - 5.6.3.1.3.5, 5.6.4.

The following requirement paragraphs may apply to an implementation of the OSI profile under conditions described within this standard: 4.2.2.1, 4.3, 4.5.2, 5.1.2.1, 5.1.2.4.2.2, 5.1.3.4.1, 5.1.3.4.2, 5.1.3.4.2.1, 5.1.3.4.2.2, 5.1.3.4.3, 5.1.3.4.3.1 - 5.1.3.4.3.4, 5.3, 5.4.1.2.1, 5.4.1.7.

6.2.2 Lightweight profile. The following requirement paragraphs apply to any implementation of the lightweight profile: 4.2.1.2, 4.2.2.3, 4.2.2.4, 4.2.3, 4.5.1.1 - 4.5.1.5,

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4.5.3.1 - 4.5.3.4, 4.6.2, 4.8, 4.8.1, 4.9, 5.1.1.2, 5.1.1.2.1 - 5.1.1.2.3, 5.1.2.3.1, 5.1.2.3.1.1, 5.1.2.4, 5.1.2.4.1, 5.1.2.4.1.1, 5.1.2.4.2, 5.1.2.4.2.1, 5.1.2.4.2.3, 5.1.2.4.2.4, 5.1.2.4.3, 5.1.2.4.3.1, 5.1.3.1, 5.1.3.1.1, 5.1.3.2, 5.1.3.2.1, 5.1.3.3, 5.1.3.4, 5.1.3.4.4, 5.1.3.4.5, 5.1.3.4.5.1, 5.1.3.4.5.2, 5.2.1, 5.2.2, 5.2.3, 5.2.3.1 - 5.2.3.3, 5.2.4, 5.2.4.1 - 5.2.4.3, 5.2.5, 5.2.6, 5.2.6.1, 5.2.6.2, 5.2.6.2.1, 5.2.5.2.2, 5.4.1.1, 5.4.1.2, 5.4.1.2.2, 5.4.1.2.3, 5.4.1.3, 5.4.1.4, 5.4.1.4.1 - 5.4.1.4.3, 5.4.1.5, 5.4.1.6, 5.4.2.1, 5.4.2.2, 5.5, 5.5.1, 5.5.2, 5.6.1, 5.6.1.1, 5.6.1.1.1, 5.6.1.1.2, 5.6.1.1.3, 5.6.1.1.3.1, 5.6.1.1.3.2, 5.6.1.2 - 5.6.1.5, 5.6.2, 5.6.2.1, 5.6.2.2 - 5.6.2.4, 5.6.2.4.1, 5.6.2.4.2, 5.6.3, 5.6.3.1, 5.6.3.1.1 - 5.6.3.1.3, 5.6.3.1.3.1 - 5.6.3.1.3.5, 5.6.4.

The following requirement paragraphs may apply to an implementation of the lightweight profile under conditions described within this standard: 4.2.2.1, 4.3, 4.6.2, 5.1.2.1, 5.1.2.4.2.2, 5.1.3.4.1, 5.1.3.4.2, 5.1.3.4.2.1, 5.1.3.4.2.2, 5.1.3.4.3, 5.1.3.4.3.1 - 5.1.3.4.3.4, 5.3, 5.4.1.2.1, 5.4.1.7, 5.5.1.2.1.

6.2.3 Combined profile. The following requirement paragraphs apply to any implementation of the combined profile: 4.2.1.1, 4.2.1.2, 4.2.2.2, 4.2.2.3, 4.2.2.4, 4.2.3, 4.5.1.1 - 4.5.1.5, 4.5.3.1 - 4.5.3.4, 4.6.3, 4.8, 4.8.1, 4.9, 5.1.1.1, 5.1.1.1.1 - 5.1.1.1.9, 5.1.1.2, 5.1.1.2.1 - 5.1.1.2.3, 5.1.2.2, 5.1.2.2.1, 5.1.2.2.1.1, 5.1.2.2.1.2, 5.1.2.3.1, 5.1.2.3.1.1, 5.1.2.4, 5.1.2.4.1, 5.1.2.4.1.1, 5.1.2.4.2, 5.1.2.4.2.1, 5.1.2.4.2.3, 5.1.2.4.2.4, 5.1.2.4.3, 5.1.2.4.3.1, 5.1.3.1, 5.1.3.1.1, 5.1.3.2, 5.1.3.2.1, 5.1.3.3, 5.1.3.4, 5.1.3.4.4, 5.1.3.4.5, 5.1.3.4.5.1, 5.1.3.4.5.2, 5.2.1, 5.2.2, 5.2.3, 5.2.3.1 - 5.2.3.3, 5.2.4, 5.2.4.1 - 5.2.4.3, 5.2.5, 5.2.6, 5.2.6.1, 5.2.6.2, 5.2.6.2.1, 5.2.6.2.2, 5.4.1.1, 5.4.1.2, 5.4.1.2.2, 5.4.1.2.3, 5.4.1.3, 5.4.1.4, 5.4.1.4.1 - 5.4.1.4.3, 5.4.1.5, 5.4.1.6, 5.4.2.1, 5.4.2.2, 5.5, 5.5.1, 5.5.2, 5.6.1, 5.6.1.1, 5.6.1.1.1, 5.6.1.1.2, 5.6.1.1.3, 5.6.1.1.3.1, 5.6.1.1.3.2, 5.6.1.2 - 5.6.1.5, 5.6.2, 5.6.2.1, 5.6.2.2 - 5.6.2.4, 5.6.2.4.1, 5.6.2.4.2, 5.6.3, 5.6.3.1, 5.6.3.1.1 - 5.6.3.1.3, 5.6.3.1.3.1 - 5.6.3.1.3.5, 5.6.4.

The following requirement paragraphs may apply to an implementation of the combined profile under conditions described within this standard: 4.2.2.1, 4.3, 4.5.2, 5.1.2.1, 5.1.2.4.2.2, 5.1.3.4.1, 5.1.3.4.2, 5.1.3.4.2.1, 5.1.3.4.2.2, 5.1.3.4.3, 5.1.3.4.3.1 - 5.1.3.4.3.4, 5.3, 5.4.1.2.1, 5.4.1.7, 5.5.1.2.1.

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6.3 Subject term listing.

Computer
Fiber optics
Multicast
XTP
Network
Survivability
FDDI
SAFENET
Lightweight profile
OSI profile

6.4 International standardization agreements. Certain provisions of this standard are the subject of international standardization agreement, including all the applicable ISO documents. When change notice, revision, or cancellation of this standard is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

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PROTOCOL IMPLEMENTATION CONFORMANCE STATEMENT PROFORMA

10.0 SCOPE

This appendix provides the PICS proforma for SAFENET in compliance with the relevant requirements, and in accordance with the relevant guidance, given in ISO/IEC 9646-1. Included in this appendix is a Protocol Implementation Conformance Statement (PICS) proforma which shall be completed as the initial step in certification of an implementation.

This appendix is a mandatory part of the standard. Requirements contained herein shall be applied to any implementation prior to certification as conformant to SAFENET.

20.0 APPLICABLE DOCUMENTS

ISO/IEC 9646-1 - Information technology - Open Systems Interconnections - Conformance Testing Methodology and Framework - Part 1: General Concepts. 1990

ISO/IEC 9646-2 - Information technology - Open Systems Interconnections - Conformance Testing Methodology and Framework - Part 2: Abstract Test Suite Specification. 1990

30.0 DEFINITIONS

30.1 Acronyms used in this appendix. The acronyms used in this appendix are defined as follows:

PICS - Protocol Implementation Conformance Statement

30.2 Special symbols. The following symbols are used in the PICS proforma:

M	mandatory field/function
O.#	optional field/function
C#	conditional
X	prohibited
#	refers to note # below the column

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40.0 INTRODUCTION

To evaluate conformance of a particular implementation, it is necessary to have a statement of which capabilities and options have been implemented for a given protocol. Such a statement is called a Protocol Implementation Conformance Statement (PICS).

40.1 The PICS proforma. The requirements and uses of the PICS proforma are outlined in ISO/IEC 9646 parts 1 and 2. The PICS proforma in this appendix has been prepared under the guidance of these documents. A completed PICS proforma is the PICS for the implementation in question. The PICS can have a number of uses, including:

a. Use by the protocol tester, as the basis for selecting appropriate tests against which to assess conformance of the implementation.

b. Use by the protocol implementor, as a checklist to reduce the risk of failure of conforming to the standard through oversight.

c. Use by the supplier and acquirer - or potential acquirer - of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma.

d. Use by the user - or potential user - of the implementation, as a basis for initially assessing the possibility of interworking with another implementation.

The PICS proforma identifies major characteristics of the standard in a questionnaire form. The PICS does not represent a complete description of the standard, but provides a condensed representation suitable for the conformance test process.

50.0 INSTRUCTIONS FOR SUBMISSION TO CONFORMANCE TEST

The supplier of a protocol implementation which is claimed to conform to SAFENET is required to complete a copy of the PICS proforma provided in this appendix and to provide the information necessary to identity both the supplier and the implementation.

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Where an implementation is capable of being configured in more than one way, a single PICS may be able to describe all such configurations. However, the supplier also has the choice of providing more than one PICS if that provides a better presentation of information. Each may cover any subset of the implementation's configuration capabilities.

50.1 Instructions for completing the PICS proforma. The first part of the PICS proforma, 60.0, shall be completed as indicated. This information is necessary to identify both the supplier and the implementation.

The main portion of the PICS proforma is section 70.0. This consists of a columnar questionnaire to describe the implementation. The first column (Item No.) is used by this appendix and the conformance test lab to identify requirements and the associated test. The second and third columns (Description and Reference) identify the requirement with a brief description and a reference to the base standard. The last two columns (Status and Support) describe the status of the requirement and provide space to respond.

The supplier may also provide further information, categorized as either Additional Information or Exceptional Information. When present, each shall be labeled A<i> or X<i> respectively for cross-referencing purposes, where "i" shall be an unambiguous identification (usually numerical) of an explanation which shall be included with the final PICS.

The paragraphs below provide a more detailed description of the Status and Support columns and of Additional and Exceptional Information.

50.1.1 Status column.

50.1.1.1 Mandatory status. Items marked with a status 'M' are mandatory. If N[] is marked for this item, an exception, X<i>, shall be reported.

50.1.1.2 Optional status. Items marked with 'O' or 'O.x' are optional. No exception need be generated for marking the N[] for an optional item. In cases with 'O.x', the '.x' refers to a grouping. At least one of the items with the same value of '.x' must be selected. In some cases, additional guidance is provided at the bottom of the column.

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50.1.1.3 Conditional status. Items marked with a status 'C' are conditional. If N[] is marked for this item, an exception, X<i>, shall be reported.

50.1.1.4 Prohibited status. Items marked with a status 'X' are prohibited.

50.1.2 Additional information. Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PICS. These shall be marked by writing in 'A<i>' in the support column in addition to the normal response.

There are no restrictions on presentation of Additional Information. If the information is brief, it may be annotated with 'A<i>' and written next to the column when space is available. If more detailed information is required, it shall be labeled with the 'A<i>' and attached to the PICS. It is not intended or expected that a large quantity will be supplied, and the PICS can be considered complete without any such information.

50.1.3 Exceptional information. It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. The supplier is required to write into the Support column an X<i> reference to an item of Exceptional Information, and to provide the appropriate rationale on the exception item itself.

There are no restrictions on presentation of Exceptional Information. If the information is brief, it may be annotated with 'X<i>' and written next to the column when space is available. If more detailed information is required, it shall be labeled with the 'X<i>' and attached to the PICS.

A possible reason for an Exception is a defect in the standard, a correction for which is expected to change the requirement not met by the implementation.

In other instances, the supplier may be aware of non-conformance and is presenting the PICS consistent with the goals stated in section 40.1 or as a submission for preliminary conformance testing.

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60.0 IMPLEMENTATION IDENTIFICATION

SUPPLIER	
CONTACT POINT FOR QUERIES ABOUT THE PICS	
IMPLEMENTATION NAMES(s) AND VERSIONS(s)	
OTHER INFORMATION NECESSARY FOR FULL IDENTIFICATION - e.g., NAME(s) AND VERSION(s) FOR MACHINES AND/OR OPERATING SYSTEMS; SYSTEM NAMES(s)	

NOTES:

1. Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirements for full identification.

2. The terms Name and Version should be interpreted appropriately to correspond with a suppliers terminology (e.g., Type, series, Model).

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70.0 PICS PROFORMA FOR SAFENET

70.1 Protocol requirement. Specify implemented protocol. SAFENET may be implemented by the OSI profile, the lightweight profile, or the combined profile.

Item No.	Protocol Requirements	Reference	Status	Support
SN100.1	OSI Profile	4.1.2	O.1	Y[] N[]
SN100.2	Lightweight Profile	4.1.2	O.1	Y[] N[]
SN100.3	Combined Profile	4.1.2	O.1	Y[] N[]

O.1: Only one item shall be implemented.

70.2 OSI profile. This section shall only be completed if SN100.1 or SN100.3 is supported.

70.2.1 OSI user services.

70.2.1.1 Private Communications interface. Are all mandatory capabilities implemented in accordance with MAP 3.0, Appendix 7, Attachment 4? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming.

SN211: Yes[] No[] (5.1.1.1.1)

70.2.1.2 File Transfer, Access, and Management (FTAM). Are all mandatory capabilities implemented in accordance with ISO 8571 parts 1 through 4, and as further specified in the NIST Special Pub 500-202 parts 9 and 10? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the FTAM PICS proforma, ISO 8571 part 5, shall be obtained from ANSI, completed and returned with this PICS.

SN212: Yes[] No[] (5.1.1.1.2)

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Item No.	FTAM	Reference	Status	Support
SN212.1	Limited	5.1.1.1.2.a	O.1	Y[] N[]
SN212.2	Full	5.1.1.1.2.b	O.1	Y[] N[]
SN212.3	Extended	5.1.1.1.2.c	O.1	Y[] N[]

O.1: Only one item shall be implemented.

70.2.1.3 Directory services. Are all mandatory capabilities implemented in accordance with ISO/IEC 9594 parts 1 through 8, and as further specified in the NIST Special Pub 500-202 part 11? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO/IEC 9594 parts 1 through 8 shall be obtained from ANSI, completed and returned with this PICS.

SN213: Yes[] No[] (5.1.1.1.3)

Item No.	Directory Services	Reference	Status	Support
SN213.1	Directory Service Agent (DSA)	5.1.1.1.3.a	O	Y[] N[]
SN213.2	Directory User Agent (DUA)	5.1.1.1.3.b	M	Y[] N[]

70.2.1.4 Association Control Service Element (ACSE). Are all mandatory capabilities of the ACSE and its protocol implemented in accordance with ISO 8649 and ISO 8650, and as further specified in NIST Special Pub 500-202 part 5, sections 5 and 13? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8649 and ISO 8650 shall be obtained from ANSI, completed and returned with this PICS.

SN214: Yes[] No[] (5.1.1.1.4)

70.2.1.5 Remote Operations Service Element (ROSE). Are all mandatory capabilities of the ROSE and its protocol implemented

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as defined in ISO/IEC 9072 parts 1 and 2, and as further defined in the NIST Special Pub 500-202 part 5, sections 6 and 13? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 9072 shall be obtained from ANSI, completed and returned with this PICS.

SN215: Yes[] No[] (5.1.1.1.5)

70.2.1.6 System Management Application Service Element (SMASE). Are all mandatory capabilities of the SMASE and its protocols implemented as defined in ISO/IEC 10164 parts 1 through 6, and as further defined in the NIST Special Pub 500-202 part 18, section 5? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO/IEC 10164 parts 1 through 6 shall be obtained from ANSI, completed and returned with this PICS.

SN216: Yes[] No[] (5.1.1.1.6)

70.2.1.7 Common Management Information Service Element (CMISE). Are all mandatory capabilities of the CMISE and its protocol implemented as defined in ISO/IEC 9595 and ISO/IEC 9596 part 1, and as further defined in the NIST Special Pub 500-202 part 18, section 6? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO/IEC 9595 and ISO/IEC 9596 part 1 shall be obtained from ANSI, completed and returned with this PICS.

SN217: Yes[] No[] (5.1.1.1.7)

70.2.1.8 Presentation layer. Are all mandatory capabilities of the presentation layer protocol implemented as defined in ISO 8822 and ISO 8823, and as further defined in the NIST Special Pub 500-202 part 5, sections 8 and 13? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8822 and ISO 8823 shall be obtained from ANSI, completed and returned with this PICS.

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SN218: Yes[] No[] (5.1.1.1.8)

70.2.1.9 Session layer. Are all mandatory capabilities of the session layer protocol implemented as defined in ISO 8326 and ISO 8327, and as further defined in the NIST Special Pub 500-202 part 5, sections 9 and 13? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8326 and ISO 8327 shall be obtained from ANSI, completed and returned with this PICS.

SN219: Yes[] No[] (5.1.1.1.9)

70.2.2 TP4 requirements. Are all mandatory capabilities of the Connection Oriented Transport Protocol (TP4) implemented as defined in the ISO 8073 with Addendum 2? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8073 shall be obtained from ANSI, completed and returned with this PICS.

SN220: Yes[] No[] (5.1.2.2.1)

Item No.	TP4 Requirements	Reference	Status	Support
SN220.1	Protocol classes other than 4 rejected	5.1.2.2.1.1.a	M	Y[] N[]
SN220.2	Sequence number size	5.1.2.2.1.1.b	M	Y[] N[]
SN220.3	Use or non-use of checksums	5.1.2.2.1.1.c	M	Y[] N[]
SN220.4	Acknowledgment time parameter	5.1.2.2.1.1.d	M	Y[] N[]
SN220.5	No user data in CR TPDU or CC TPDUs	5.1.2.2.1.1.e	M	Y[] N[]

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SN220.6	Ignore unknown parameter in CR TPDUs	5.1.2.2.1.1.f	M	Y[] N[]
SN220.7	Protocol accepts a DR or DC TPDU	5.1.2.2.1.1.g	M	Y[] N[]
SN220.8	Reason code in DR TPDUs	5.1.2.2.1.1.h	M	Y[] N[]
SN220.9	Ignore unrecognized or not applicable bits	5.1.2.2.1.1.i	M	Y[] N[]
SN220.10	Disconnect on maximum connections	5.1.2.2.1.1.j	M	Y[] N[]
SN220.11	TSAP ID encoding	5.1.2.2.1.1.k	M	Y[] N[]
SN220.12	Invalid TSAP parameter	5.1.2.2.1.2	M	Y[] N[]

70.2.3 OSI profile protocol objects.

70.2.3.1 SMASE protocol objects.

Item No.	SMASE Protocol Objects	Reference	Status	Support
SN231.1	computerSystem	5.5.3	M	Y[] N[]
SN231.2	alarmRecord	5.5.3	M	Y[] N[]
SN231.3	attributeValue ChangeRecord	5.5.3	M	Y[] N[]
SN231.4	discriminator	5.5.3	M	Y[] N[]
SN231.5	eventForwarding Discriminator	5.5.3	M	Y[] N[]
SN231.6	eventLogRecord	5.5.3	M	Y[] N[]
SN231.7	log	5.5.3	M	Y[] N[]
SN231.8	logRecord	5.5.3	M	Y[] N[]

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SN231.9	objectCreation Record	5.5.3	M	Y[] N[]
SN231.10	objectDeletion Record	5.5.3	M	Y[] N[]
SN231.11	relationshipChange Record	5.5.3	M	Y[] N[]
SN231.12	stateChangeRecord	5.5.3	M	Y[] N[]

70.2.3.2 Connection Oriented Transport protocol objects.

Item No.	CO Transport Protocol Objects	Reference	Status	Support
SN232.1	coTransport ProtocolLayer Entity	5.5.3	M	Y[] N[]
SN232.2	transport Connection	5.5.3	M	Y[] N[]

70.2.4 OSI addressing requirements.

Item No.	Addressing Requirements	Reference	Status	Support
SN240.1	Presentation	5.2.1	M	Y[] N[]
SN240.2	Session	5.2.2	M	Y[] N[]
SN240.3	Connection Oriented Transport	5.2.3	M	Y[] N[]
SN240.4	T-selector value	5.2.3.2.a	M	Y[] N[]

70.3 Lightweight profile. This section shall only be completed if SN100.2 or SN100.3 is supported.

70.3.1 XTP requirement. Are all mandatory capabilities of the XTP implemented as defined in the Xpress Transfer Protocol, Revision 3.6? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the

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implementation is non-conforming. A copy of the PICS proforma for Xpress Transfer Protocol shall be obtained from the certifying test center, completed and returned with this PICS.

SN310: Yes[] No[] (5.1.2.3.1)

Item No.	XTP Requirements	Reference	Status	Support
SN310.1	XTP SAFENET Address Format	5.1.2.3.1.1.a	M	Y[] N[]
SN310.2	Sort Service	5.1.2.3.1.1.b	M	Y[] N[]
SN310.3	FIFO sort policy	5.1.2.3.1.1.c	M	Y[] N[]
SN310.4	zero length data denoted by BTAG field with 1's	5.1.2.3.1.1.d	M	Y[] N[]
SN310.5	isochronous stream service	5.1.2.3.1.1.e	X	Y[] N[]
SN310.6	ROUTE, MAINT, and MGMT packets ignored and not generated	5.1.2.3.1.1.f	M	Y[] N[]
SN310.7	Context lookups on <SRC NSAP,key> pairs	5.1.2.3.1.1.g	M	Y[] N[]
SN310.8	MAC address in multicast CNTL packets XKEY and XROUTE fields	5.1.2.3.1.1.j	M	Y[] N[]
SN310.9	Multicast packets passed to LLC. Unicast packets passed to CLNP	5.1.2.3.1.1.k	M	Y[] N[]

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70.3.2 Lightweight addressing requirements.

Item No.	Addressing Requirements	Reference	Status	Support
SN320.1	Transport	5.2.3	M	Y[] N[]
SN320.2	T-selector values	5.2.3.2.a-j	M	Y[] N[]
SN320.3	XTP	5.2.3.3	M	Y[] N[]

70.4 Protocols common to all profiles.

70.4.1 Connectionless Transport Protocol. Are all mandatory capabilities of the Connectionless Transport Protocol implemented as defined in ISO 8602? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8602 shall be obtained from ANSI, completed and returned with this PICS.

SN410: Yes[] No[] (5.1.2.4.1)

Item No.	CL Transport Protocol	Reference	Status	Support
SN410.1	TSAP ID size	5.1.2.4.1.1	M	Y[] N[]
SN410.2	TSAP ID for multicast	5.1.2.4.1.1	M	Y[] N[]
SN410.3	TSAP ID for point to point transfer	5.1.2.4.1.1	M	Y[] N[]
SN410.4	MAC group address for multicast	5.1.2.4.1.1	M	Y[] N[]
SN410.5	Inactive subset of ISO CLNP for multicast	5.1.2.4.1.1	M	Y[] N[]

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70.4.2 Network protocol requirements. Are all mandatory capabilities of the Connectionless Network Protocol implemented as defined in ISO 8473 and of the Network Layer Service as defined in ISO 8348/ADD2? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8473 shall be obtained from ANSI, completed and returned with this PICS.

SN420: Yes[] No[] (5.1.2.4.2)

Item No.	Network Protocol Requirements	Reference	Status	Support
SN420.1	Non-segmenting subset not used	5.1.2.4.2.1.a	M	Y[] N[]
SN420.2	Inactive subset not used TP4	5.1.2.4.2.1.b	M	Y[] N[]
SN420.3	Deliver all inactive subset PDUs	5.1.2.4.2.1.c	M	Y[] N[]
SN420.4	Fragmentation of CL TPDUs	5.1.2.4.2.1.d	X	Y[] N[]
SN420.5	Selection of use or non-use of checksums	5.1.2.4.2.1.e	M	Y[] N[]
SN420.6	Maintenance QOS parameter for routing	5.1.2.4.2.1.f	X	Y[] N[]
SN420.7	Security QOS parameter for routing	5.1.2.4.2.1.f	X	Y[] N[]
SN420.8	Priority QOS for routing	5.1.2.4.2.1.f	X	Y[] N[]

70.4.3 ES/IS requirements. Are all mandatory capabilities of the ES/IS Routing Exchange protocol implemented as defined in ISO 9542? Answering "No" to this question indicates non-

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conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 9542 shall be obtained from ANSI, completed and returned with this PICS.

SN430: Yes[] No[] (5.1.2.4.2.3)

Item No.	ES/IS Requirements	Reference	Status	Support
SN430.1	Support of any valid SAP format	5.1.2.4.2.4.a	M	Y[] N[]
SN430.2	Configuration information	5.1.2.4.2.4.b	M	Y[] N[]
SN430.3	Route redirection	5.1.2.4.2.4.b	M	Y[] N[]
SN430.4	Configuration timer setting mechanism	5.1.2.4.2.4.c	M	Y[] N[]
SN430.5	Holding timer setting mechanism	5.1.2.4.2.4.c	M	Y[] N[]
SN430.6	Selection of use or non-use of checksum	5.1.2.4.2.4.d	M	Y[] N[]
SN430.7	Maintenance QOS parameter for routing	5.1.2.4.2.4.e	X	Y[] N[]
SN430.8	Security QOS parameter for routing	5.1.2.4.2.4.e	X	Y[] N[]
SN430.9	Priority QOS parameter for routing	5.1.2.4.2.4.e	X	Y[] N[]
SN430.10	Enabling the configuration notification	5.1.2.4.2.4.f	M	Y[] N[]

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SN430.11	Disabling the configuration notification	5.1.2.4.2.4.f	M	Y[] N[]
SN430.12	LSAP address	5.1.2.4.2.4.g	M	Y[] N[]
SN430.13	Error report flag	5.1.2.4.2.4.h	M	Y[] N[]

70.4.4 IS/IS requirements. Are all mandatory capabilities of the IS/IS Routing Exchange protocol implemented as defined in ISO 10589? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 10589 shall be obtained from ANSI, completed and returned with this PICS.

SN440: Yes[] No[] (5.1.2.4.2.2)

Item No.	IS/IS Requirements	Reference	Status	Support
SN440.1	IS/IS	5.1.2.4.2.2	C1	Y[] N[]

C1: If router per paragraph 5.3 then M else X.

70.4.5 Logical link control (LLC) type 1. Are all mandatory capabilities of the LLC Type 1 protocol implemented as defined in ISO 8802-2? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 8802-2 shall be obtained from ANSI, completed and returned with this PICS.

SN450: Yes[] No[] (5.1.2.4.3)

70.4.6 MAC protocol requirements. Are all mandatory capabilities of the MAC protocol implemented as defined in ISO 9314-2 FDDI MAC for dual MAC stations? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 9314-2 shall be obtained from ANSI, completed and returned with this PICS.

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SN460: Yes[] No[] (5.1.3.1.1.3.e)

Item No.	MAC Protocol Requirements	Reference	Status	Support
SN460.1	48 bit address field	5.1.3.1.1.a	M	Y[] N[]
SN460.2	Minimum number of group addresses	5.1.3.1.1.b	M	Y[] N[]
SN460.3	Synchronous bandwidth	5.1.3.1.1.c	M	Y[] N[]
SN460.4	Asynchronous bandwidth	5.1.3.1.1.d	M	Y[] N[]
SN460.5	Use of restricted token	5.1.3.1.1.e	X	Y[] N[]
SN460.6	Use of immediate mode	5.1.3.1.1.f	X	Y[] N[]

70.4.7 FDDI station management (SMT). Are all mandatory capabilities of the FDDI Station Management Protocol implemented as defined in ANSI X3T9.5/84-49? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ANSI X3T9.5/84-49 shall be obtained from ANSI, completed and returned with this PICS.

SN470: Yes[] No[] (5.1.3.2)

Item No.	FDDI SMT	Reference	Status	Support
SN470.1	Bypass switch control	5.1.3.2.1	C.1	Y[] N[]

C.1: If station uses trunk connection then M else O.

70.4.8 Physical layer protocol (PHY). Are all mandatory capabilities of FDDI PHY implemented as defined in ISO 9314-1? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is

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non-conforming. A copy of the PICS proforma for ISO 9314-1 shall be obtained from ANSI, completed and returned with this PICS.

SN480: Yes[] No[] (5.1.3.3)

70.4.9 Physical layer medium dependent (PMD). Are all mandatory capabilities of FDDI PMD implemented as defined in ISO 9314-3, with the exception of power levels as specified in MIL-STD-2204? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for ISO 9314-3 shall be obtained from ANSI, completed and returned with this PICS.

SN490: Yes[] No[] (5.1.3.4)

Item No.	Optical interface class	Reference	Status	Support
SN490.1	class 1	5.1.3.4.1	O.1	Y[] N[]
SN490.2	class 2	5.1.3.4.2	O.1	Y[] N[]
SN490.3	class 3	5.1.3.4.3	O.1	Y[] N[]

O.1: Only one item shall be implemented.

70.4.9.1 Optical signal.

Item No.	Average Signal Power	Reference	Status	Support
SN491.1	FDDI output signal	5.1.3.4.1	C1	Y[] N[]
SN491.2	FDDI input signal	5.1.3.4.1	C2	Y[] N[]
SN491.3	Class 3 output signal	5.1.3.4.3.1	C3	Y[] N[]
SN491.4	Class 3 input signal	5.1.3.4.3.2	C4	Y[] N[]
SN491.5	Class 3 signal detect function assert power (Pa)	5.1.3.4.3.3 .a	C4	Y[] N[]

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SN491.6	Class 3 signal detect function deassert power (Pd)	5.1.3.4.3.3 .b	C4	Y[] N[]
SN491.7	Class 3 transmit disable function	5.1.3.4.3.4	C4	Y[] N[]

C1: If SN490.1 or SN490.2 then M else X.
C2: If SN490.1 then M else X.
C3: If SN490.3 then M else X.
C4: If SN490.2 or SN490.3 then M else X.

70.4.9.2 Electrical signal interface.

Item No.	Electrical Signal Interface	Reference	Range Min	Range Max	Status	Support
SN492.1	Insertion signal	5.1.3.4.4	4.7	5.3 Vdc	M	Y[] N[]
SN492.2	Bypass signal	5.1.3.4.4	0.0	1.0 Vdc	M	Y[] N[]
SN492.3	Max. Rise time	5.1.3.4.4	-	0.2 ms	M	Y[] N[]
SN492.4	Max. Fall time	5.1.3.4.4	-	0.2 ms	M	Y[] N[]
SN492.5	Min. Current capacity	5.1.3.4.4	340	- ma	M	Y[] N[]

70.4.9.3 Electrical interface connector.

Item No.	Electrical Interface Connector	Reference	Status	Support
SN493.1	Military Specification MIL-C-26482	5.4.2.2	M	Y[] N[]
SN493.2	Military Standard MS3476	5.4.2.2	M	Y[] N[]

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SN493.3	Mates with Part Number MS3475 W 14 - 4 P	5.4.2.2	M	Y[] N[]

70.4.9.4 Fiber optic interface connector.

Item No.	Fiber Optic Interface Connector	Reference	Status	Support
SN494.1	MIL-C-28876 sheet #1 or #11	5.4.1.6	M	Y[] N[]
SN494.2	Max mated power loss	5.4.1.6	M	Y[] N[]
SN494.3	Mates with Part Number B1S1	5.4.1.6	M	Y[] N[]
SN494.4	Pin definition	5.4.1.6	M	Y[] N[]

70.4.10 Time service requirements.

70.4.10.1 Network clock requirements.

Item No.	Network clock requirements	Reference	Status	Support
SN4A1.1	Global time format	5.6.1.1	M	Y[] N[]
SN4A1.2	Clock granularity	5.6.1.2	M	Y[] N[]
SN4A1.3	Clock monotonicity	5.6.1.3	M	Y[] N[]
SN4A1.4	Clock error bounds	5.6.1.4	M	Y[] N[]

70.4.10.2 NTP requirements. Are all mandatory capabilities of the Network Time Protocol implemented as defined in NTP V3? Answering "No" to this question indicates non-conformance to the protocol specification. Unsupported mandatory capabilities shall be identified, with an explanation of why the implementation is non-conforming. A copy of the PICS proforma for NTP V3 shall be obtained from the certifying test center, completed and returned with this PICS.

SN4A2: Yes[] No[] (5.6.2.4)

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Item No.	NTP requirements	Reference	Status	Support
SN4A2.1	Transport address	5.6.2.4.1.a	M	Y[] N[]
SN4A2.2	Peer and host addresses	5.6.2.4.1.b	M	Y[] N[]
SN4A2.3	Peer and host ports	5.6.2.4.1.c	M	Y[] N[]
SN4A2.4	Reference identifier field	5.6.2.4.1.d	M	Y[] N[]
SN4A2.5	All five modes of operation	5.6.2.4.2.a	M	Y[] N[]
SN4A2.6	Clock selection algorithms	5.6.2.4.2.b	M	Y[] N[]
SN4A2.7	Clock filtering algorithms	5.6.2.4.2.c	M	Y[] N[]
SN4A2.8	Use of default parameter value	5.6.2.4.2.d	M	Y[] N[]

70.4.11 OSI connectionless network protocol objects.

Item No.	OSI CLNP Objects	Reference	Status	Support
SN4B0.1	clNetworkProtocol LayerEntity	5.5.2	M	Y[] N[]

70.4.12 Common addressing requirements.

Item No.	Addressing Requirements	Reference	Status	Support
SN4C0.1	Connectionless transport	5.2.3	M	Y[] N[]
SN4C0.2	Transport multicast	5.2.3.1	M	Y[] N[]
SN4C0.3	T-selector values	5.2.3.2.b-d	M	Y[] N[]
SN4C0.4	Network layer	5.2.4	M	Y[] N[]

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SN4C0.5	Network service access point selectors	5.2.4.1	M	Y[] N[]
SN4C0.6	Network entity titles	5.2.4.2	M	Y[] N[]
SN4C0.7	Inactive subset	5.2.4.3	M	Y[] N[]
SN4C0.8	Logical link control	5.2.5	M	Y[] N[]
SN4C0.9	Media access control	5.2.6	M	Y[] N[]
SN4C0.10	Individual media access control	5.2.6.1	M	Y[] N[]
SN4C0.11	group media access control	5.2.6.2	M	Y[] N[]
SN4C0.12	standard group	5.2.6.2.1	M	Y[] N[]
SN4C0.13	SAFENET group address space	5.2.6.2.2	M	Y[] N[]

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APPENDIX B

SAFENET MANAGED OBJECTS

10.0 SCOPE

The purpose of this appendix is to define managed objects which are unique to SAFENET.

This appendix is a mandatory part of the standard. The information contained herein defines managed object classes which are required to support SAFENET and are not defined elsewhere.

20.0 APPLICABLE DOCUMENTS

ISO/IEC 10165-4	Information Processing Systems - Open Systems Interconnection - Structure of Management Information - Part 4 - Guidelines for the Definition of Managed Objects. 1991

30.0 DEFINITIONS

This section is not applicable to this appendix.

40.0 SAFENET TIME SERVICE MANAGEMENT INFORMATION LIBRARY

The SAFENET Management Information Library (MIL) defined in this section contains definitions of object classes associated with the management of the STS. Each managed object class and each of the attributes of each object class has an associated object identifier. These object identifiers are not registered under an external registration authority and can not be considered to be globally unique. These identifiers are unique only in the context of the SAFENET MIL.

40.1 Object class definitions.

40.1.1 NTP protocol profile object class. This object provides access to the basic parameters of the time synchronization protocol. There shall be exactly one of these managed objects within a SAFENET system. All instances of the protocol shall use the same parameter values.

```
ntpProtocolProfile MANAGED OBJECT CLASS
     DERIVED FROM top
```


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```
     CHARACTERIZED BY ntpProtocolProfilePackage PACKAGE
          BEHAVIOR DEFINITIONS
               The ntpProtocolProfile object provides general
          information about the time service in a SAFENET system.
          ATTRIBUTES
               ntpPPIndex GET,
               ntpTSelect GET,
               ntpMaxStratum GET-REPLACE,
               ntpMaxClockAge GET-REPLACE,
               ntpMaxSkew GET-REPLACE,
               ntpMaxDistance GET-REPLACE,
               ntpMinPollInterval GET-REPLACE,
               ntpMaxPollInterval GET-REPLACE,
               ntpMinSelectClock GET-REPLACE,
               ntpMaxSelectClock GET-REPLACE,
               ntpMinDispersion GET-REPLACE,
               ntpMaxDispersion GET-REPLACE,
               ntpReachRegSize GET-REPLACE,
               ntpFilterSize GET-REPLACE,
               ntpFilterWeight GET-REPLACE,
               ntpSelectWeight GET-REPLACE,
               ntpCumLeapSeconds GET-REPLACE,
               ntpLeapIndication GET-REPLACE;
          ACTIONS
               ntpPPLeapSecondAction
               ntpPPResetAction;
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)};
```

40.1.2 SAFENET Network Clock Object Class. This object provides information concerning the dynamic state of the STS and the Network Clock in a particular LAN node. An instantiation of this object is required for every Network Clock managed by the STS in a SAFENET system.

```
ntpNetClock MANAGED OBJECT CLASS
     DERIVED FROM top
     CHARACTERIZED BY ntpNetClockPackage PACKAGE
          BEHAVIOR
               The ntpNetClock object provides access to and
               information about the SAFENET time service and the
               Network Clock managed by the service in a
               particular SAFENET node.
          ATTRIBUTES
               ntpNetClockIndex GET,
               ntpVersion GET,
               ntpNetClockStatus GET,
```


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```
                ntpNetClockValue GET,
                ntpNetClockPrecision GET-REPLACE,
                ntpNetClockAddress GET-REPLACE,
                ntpPeerClocksAddresses GET-REPLACE,
                ntpCurrSynchSourceAddress GET-REPLACE,
                ntpNetClockCurrStratum GET,
                ntpNetClockAdjInterval GET-REPLACE,
                ntpNetClockMaxAperture GET,
                ntpNetClockRootDelay GET,
                ntpNetClockRootDispersion GET
                ntpSystemEventCounter GET-REPLACE,
                ntpSystemEventCode GET,
                ntpSystemEventTime GET;
          ACTIONS
                ntpNetClockResetAction;
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)};
```

40.1.3 NTP Peer Association Object Class. This object provides the status of each peer association maintained in a particular SAFENET node. An instantiation of this object class is required for each peer association about which state information is maintained in an instance of the STS.

```
ntpPeerAssocStatus MANAGED OBJECT CLASS
     DERIVED FROM top
     CHARACTERIZED BY ntpPeerAssocStatusPackage PACKAGE
          BEHAVIOR
                The ntpPeerAssocStatus object provides access to
                and information about each association maintained
                by a particular instance of the STS.
          ATTRIBUTES
                ntpPeerAssocStatusIndex GET,
                ntpHostMode GET-REPLACE,
                ntpPeerMode GET,
                ntpPeerStratum GET,
                ntpPeerPrecision GET,
                ntpPeerReachability GET,
                ntpPeerOffset GET,
                ntpPeerDelay GET,
                ntpPeerDispersion GET,
                ntpPeerPollInterval GET,
                ntpPeerRootDelay GET,
                ntpPeerRootDispersion GET,
                ntpPeerEventCode GET;
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)};
```


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40.1.4 SAFENET Reference Clock Object Class. This object provides information about a particular reference clock or external source. An instantiation of this object class is required for each reference clock or external source in a SAFENET system.

```
ntpRefClock MANAGED OBJECT CLASS
     DERIVED FROM top
     CHARACTERIZED BY ntpRefClockPackage PACKAGE
          BEHAVIOR
               The ntpRefClock object provides information about
               each external reference clock available in a
               SAFENET system.
          ATTRIBUTES
               ntpRefClockIndex GET,
               ntpRefClockStatus GET,
               ntpRefClockValue GET,
               ntpRefClockPrecision GET,
               ntpRefClockAddress GET,
               ntpRefClockIdentifier GET,
               ntpRefClockCurrStratum GET,
               ntpRefClockRootDispersion GET,
               ntpRefClockEventCode GET,
               ntpRefClockEventTime GET;
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)};
```

40.2 Attribute Definitions.

40.2.1 NTP Protocol Profile Attributes.

```
ntpPPIndex ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     INTEGER (0..65535)
     BEHAVIOR
          DEFINED AS      Index attribute for uniquely identifying
                          the ntpProtocolProfile object instances.

REGISTERED AS  {safenet MIL(2204) sts(1) ntpProtocolProfile(1)
               ntpPPIndex(0)};

ntpTSelect ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.TSelect
     BEHAVIOR
          DEFINED AS      This attribute specifies the ISO T-
                          selector value for the Network Time
                          Protocol.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpTSelect(1)};

ntpMaxStratum ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Stratum
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         stratum value that can be encoded as a
                         packet variable, also interpreted as
                         network unreachable.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxStratum(2)};

ntpMaxClockAge ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.MaxClockAge
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         interval without an update that a
                         reference clock will be considered
                         valid.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxClockAge(3)};

ntpMaxSkew ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.MaxSkew
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         offset error caused by the skew of a
                         local clock over the interval specified
                         by ntpMaxClockAge.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxSkew(4)};

ntpMaxDistance ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.MaxDistance
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         acceptable synchronization distance.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxDistance(5)};

ntpMinPollInterval ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.PollInterval
     BEHAVIOR
```


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```
          DEFINED AS     This attribute specifies the minimum
                         polling interval allowable in the
                         SAFENET system.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMinPollInterval(6)};

ntpMaxPollInterval ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.PollInterval
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         polling interval allowable in the
                         SAFENET system.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxPollInterval(7)};

ntpMinSelectClock ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.MinSelectClock
     BEHAVIOR
          DEFINED AS     This attribute specifies the minimum
                         number of peers acceptable for
                         synchronization.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMinSelectClock(8)};

ntpMaxSelectClock ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.MaxSelectClock
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         number of peers considered for
                         selection.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxSelectClock(9)};

ntpMinDispersion ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Dispersion
     BEHAVIOR
          DEFINED AS     This attribute specifies the minimum
                         dispersion increment for each stratum
                         level.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMinDispersion(10)};

ntpMaxDispersion ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Dispersion
     BEHAVIOR
```


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```
          DEFINED AS     This attribute specifies the maximum
                         dispersion increment allowable, and
                         specifies the dispersion assumed for
                         missing data.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpMaxDispersion(11)};

ntpReachRegSize ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.ReachRegSize
     BEHAVIOR
          DEFINED AS     This attribute specifies the size of the
                         reachability register.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpReachRegSize(12)};

ntpFilterSize ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.FilterSize
     BEHAVIOR
          DEFINED AS     This attribute specifies the size of the
                         clock filter shift register.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpFilterSize(13)};

ntpFilterWeight ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.FilterWeight
     BEHAVIOR
          DEFINED AS     This attribute specifies the weight used
                         to compute the filter dispersion.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpFilterWeight(14)};

ntpSelectWeight ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.SelectWeight
     BEHAVIOR
          DEFINED AS     This attribute specifies the weight used
                         to compute the selection dispersion.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpSelectWeight(15)};

ntpCumLeapSeconds ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PP.CumLeapSeconds
     BEHAVIOR
          DEFINED AS     This attribute specifies the cumulative
                         number of leap seconds that have
                         occurred since January 1, 1972.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpCumLeapSeconds(16)};

ntpLeapIndication ATTRIBUTE
     WITH ATTRIBUTE SYNTAX     NTP-PP.LeapIndication
     BEHAVIOR
          DEFINED AS      This attribute indicates that a leap
                          second is going to occur at the end of
                          the current day.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpLeapIndication(17)};
```

40.2.2 SAFENET Network Clock Attributes.

```
ntpNetClockIndex ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     INTEGER (0..65535)
     BEHAVIOR
          DEFINED AS      Index attribute for uniquely identifying
                          the ntpNetClock object instances.
REGISTERED AS  {safenetMIL(2204) sts(1) ntpNetClock(2)
               ntpNetClockIndex(0)};

ntpVersion ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-NC.Version
     BEHAVIOR
          DEFINED AS      This attribute indicates the version of
                          NTP being used.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpVersion(1)};

ntpNetClockStatus ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-NC.Status
     BEHAVIOR
          DEFINED AS      This attribute indicates the current
                          status of the Network Clock in this
                          SAFENET node.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockStatus(2)};

ntpNetClockValue ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-NC.ClockValue
     BEHAVIOR
          DEFINED AS      This attribute indicates the current
                          time of the Network Clock in this
                          SAFENET node.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockValue(3)};

ntpNetClockPrecision ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Precision
     BEHAVIOR
          DEFINED AS     This attribute indicates the precision
                         of the Network Clock in this SAFENET
                         node.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockPrecision(4)};

ntpNetClockAddress ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.ClockAddress
     BEHAVIOR
          DEFINED AS     This attribute indicates the network
                         address of this SAFENET node.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockAddress(5)};

ntpPeerClocksAddresses ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-NC.PeerClocksAddresses
     BEHAVIOR
          DEFINED AS     This attribute lists the network
                         addresses of the peers currently being
                         maintained by this SAFENET node.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpPeerClocksAddresses(6)};

ntpCurrSynchSourceAddress ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-NC.CurrSynchSourceAddress
     BEHAVIOR
          DEFINED AS     This attribute specifies the network
                         address or the reference clock type of
                         the current synchronization source for
                         this SAFENET node.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpCurrSynchSourceAddress(7)};

ntpNetClockCurrStratum ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Stratum
     BEHAVIOR
          DEFINED AS     This attribute indicates the current
                         stratum value for the Network Clock in
                         this SAFENET node.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockCurrStratum(8)};

ntpNetClockAdjInterval ATTRIBUTE
     WITH ATTRIBUTE SYNTAX     NTP-NC.AdjInterval
     BEHAVIOR
          DEFINED AS     This attribute specifies the interval
                         over which gradual phase adjustments to
                         the Network Clock are to be applied.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockAdjInterval(9)};

ntpNetClockMaxAperture ATTRIBUTE
     WITH ATTRIBUTE SYNTAX     NTP-NC.MaxAperture
     BEHAVIOR
          DEFINED AS     This attribute specifies the maximum
                         aperature of the Network Clock.  An
                         adjustment larger than this value would
                         require a step adjustment of the Network
                         Clock.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockMaxAperture(10)};

ntpNetClockRootDelay ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM. Delay
     BEHAVIOR
          DEFINED AS     This attribute specifies the delay
                         between this SAFENET Network Clock and
                         the root of the synchronization subnet.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockRootDelay(11)};

ntpNetClockRootDispersion ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Dispersion
     BEHAVIOR
          DEFINED AS     This attribute specifies the dispersion
                         between this SAFENET Network Clock and
                         the root of the synchronization subnet.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockRootDispersion(12)};

ntpSystemEventCounter ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-NC.SystemEventCounter
     BEHAVIOR
          DEFINED AS     This attribute specifies a counter
                         indicating the number of system
```


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```
                        exception events that have occurred
                        since the last time the counter was
                        checked and cleared.
REGISTERED AS   {safenetMIB(2204) sts(1) ntpNetClock(2)
                ntpSystemEventCounter(13)};

ntpSystemEventCode ATTRIBUTE
     WITH ATTRIBUTE SYNTAX      NTP-COM.SystemEventCode
     BEHAVIOR
          DEFINED AS      This attribute identifies the latest
                          system exception event.
REGISTERED AS   {safenetMIB(2204) sts(1) ntpNetClock(2)
                ntpSystemEventCode(14)};

ntpSystemEventTime ATTRIBUTE
     WITH ATTRIBUTE SYNTAX      NTP-COM.SystemEventTime
     BEHAVIOR
          DEFINED AS      This attribute indicates the time at
                          which the latest system exception event
                          occurred.
REGISTERED AS   {safenetMIB(2204) sts(1) ntpNetClock(2)
                ntpSystemEventTime(15)};
```

40.2.3 NTP Peer Association Attributes.

```
ntpPeerAssocStatusIndex ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX      INTEGER (0..65535)
     BEHAVIOR
          DEFINED AS      Index attribute for uniquely identifying
                          the ntpPeerAssocStatus object instances.
REGISTERED AS   {safenetMIL(2204) sts(1) ntpPeerAssocStatus(3)
                ntpPASIndex (0)};

ntpHostMode ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX      NTP-COM.Mode
     BEHAVIOR
          DEFINED AS      This attribute specifies the mode of the
                          host (host being the local entity) in
                          this association.
REGISTERED AS   {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
                ntpHostMode(1)};

ntpPeerMode ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX      NTP-COM.Mode
     BEHAVIOR
```


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```
          DEFINED AS     This attribute specifies the mode of the
                         peer (peer being the remote entity) in
                         this association.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerMode(2)};

ntpPeerStratum ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Stratum
     BEHAVIOR
          DEFINED AS     This attribute specifies the stratum
                         value of the peer entity.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerStratum(3)};

ntpPeerPrecision ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Precision
     BEHAVIOR
          DEFINED AS     This attribute specifies the precision
                         of the Network Clock being used by the
                         peer entity.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerPrecision(4)};

ntpPeerReachability ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-PA.Reachability
     BEHAVIOR
          DEFINED AS     This attribute indicates the
                         reachability status of the peer entity.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerReachability(5)};

ntpPeerOffset ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Offset
     BEHAVIOR
          DEFINED AS     This attribute indicates the latest
                         offset value determined between the peer
                         entity and the host entity.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerOffset(6)};

ntpPeerDelay ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Delay
     BEHAVIOR
          DEFINED AS     This attribute indicates the latest
                         delay value calculated between the peer
                         entity and the host entity.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerDelay(7)};

ntpPeerDispersion ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Dispersion
     BEHAVIOR
          DEFINED AS     This attribute indicates the latest
                         dispersion value calculated between the
                         peer entity and the host entity.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerDispersion(8)};

ntpPeerPollInterval ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.PollInterval
     BEHAVIOR
          DEFINED AS     This attribute indicates the polling
                         interval being used with this
                         association.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerPollInterval(9)};

ntpPeerRootDelay ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Delay
     BEHAVIOR
          DEFINED AS     This attribute indicates the total
                         roundtrip delay to the primary reference
                         source at the root of the
                         synchronization subnet.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerRootDelay(10)};

ntpPeerRootDispersion ATTRIBUTE
     WITH ATTRIBUTE SYNTAX     NTP-COM.Dispersion
     BEHAVIOR
          DEFINED AS     This attribute indicates the maximum
                         error relative to the primary reference
                         source at the root of the
                         synchronization subnet.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerRootDispersion(11)};

ntpPeerEventCode ATTRIBUTE
     WITH ATTRIBUTE SYNTAX     NTP-PA.PeerEventCode
     BEHAVIOR
          DEFINED AS     This attribute indicates the last peer
                         exception event.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpPeerAssocStatus(3)
               ntpPeerEventCode(12)};
```

40.2.4 Reference Clock Attributes.

```
ntpRefClockIndex ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     INTEGER (0..65535)
     BEHAVIOR
          DEFINED AS     Index attribute for uniquely identifying
                         the ntpRefClock object instances.
REGISTERED AS  {safenetMIL(2204) sts(1) ntpRefClock(4)
               ntpRefClockIndex(0)};

ntpRefClockStatus ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Status
     BEHAVIOR
          DEFINED AS     This attribute indicates the current
                         status of the reference clock or
                         external source.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockStatus(1)};

ntpRefClockValue ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.ClockValue
     BEHAVIOR
          DEFINED AS     This attribute indicates the current
                         time of the reference clock or external
                         source.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockValue(2)};

ntpRefClockPrecision ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Precision
     BEHAVIOR
          DEFINED AS     This attribute indicates the precision
                         of the reference clock or external
                         source.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockPrecision(3)};

ntpRefClockAddress ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.ClockAddress
     BEHAVIOR
          DEFINED AS     This attribute indicates the network
                         address of the reference clock or
                         external source.
```


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```
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockAddress(4)};

ntpRefClockIdentifier ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.ReferenceClockType
     BEHAVIOR
          DEFINED AS     This attribute specifies the type of
                         reference clock or external source that
                         this object represents.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockIdentifier(5)};

ntpRefClockCurrStratum ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Stratum
     BEHAVIOR
          DEFINED AS     This attribute indicates the current
                         stratum for this reference clock or
                         external source.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockCurrStratum(6)};

ntpRefClockRootDispersion ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.Dispersion
     BEHAVIOR
          DEFINED AS     This attribute specifies the dispersion
                         for the reference clock or external
                         reference.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockRootDispersion(7)};

ntpRefClockEventCode ATTRIBUTE,
     WITH ATTRIBUTE SYNTAX     NTP-COM.SystemEventCode
     BEHAVIOR
          DEFINED AS     This attribute identifies the latest
                         system exception event.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockEventCode(8)};

ntpRefClockEventTime ATTRIBUTE
     WITH ATTRIBUTE SYNTAX     NTP-COM.SystemEventTime
     BEHAVIOR
          DEFINED AS     This attribute indicates the time at
                         which the latest system exception event
                         occurred.
REGISTERED AS  {safenetMIB(2204) sts(1) ntpRefClock(4)
               ntpRefClockEventTime(9)};
```


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40.3 Action Definitions.

40.3.1 NTP Protocol Profile Actions.

```
ntpPPLeapSecondAction ACTION
     BEHAVIOR
          The behavior of this action is undefined in this
          document.  It provides the capability to distribute an
          indication that a leap second is about to occur.  It
          includes mechanisms to set the appropriate parameters
          in the protocol.
     MODE CONFIRMED and NON-CONFIRMED
     WITH INFORMATION SYNTAX
     WITH REPLY SYNTAX
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpPPLeapSecondAction(20)};

ntpPPResetAction
     BEHAVIOR
          The BEHAVIOR of this action is undefined in this
          document. It provides the capability to distribute an
          indication to all instances of the SAFENET time service
          to restart the time synchronization protocol.
     MODE CONFIRMED and NON-CONFIRMED
     WITH INFORMATION SYNTAX
     WITH REPLY SYNTAX
REGISTERED AS  {safenetMIB(2204) sts(1) ntpProtocolProfile(1)
               ntpPPResetAction(21)};
```

40.3.2 Network Clock Actions.

```
ntpNetClockResetAction
     BEHAVIOR
          The BEHAVIOR of this action is undefined in this
          document. It provides the capability to distribute an
          indication to all instances of the SAFENET time service
          to restart the time synchronization protocol.
     MODE CONFIRMED and NON-CONFIRMED
     WITH INFORMATION SYNTAX
     WITH REPLY SYNTAX
REGISTERED AS  {safenetMIB(2204) sts(1) ntpNetClock(2)
               ntpNetClockResetAction(20)};
```

40.4 ASN.1 Definition Modules.

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The ASN.1 definitions for GlobalTime and TimeInterval used below are found in 5.7.1.1.3.

40.4.1 Common Definitions.

```
NTP-COM
     DEFINITIONS ::= BEGIN

          GlobalTime ::= BIT STRING (SIZE (64))
          -- This type is fully specified in 5.7.1.1.1.

          TimeInterval ::= BIT STRING (SIZE (64))
          -- This type is fully specified in 5.7.1.1.2.

          ReferenceClockType ::= ENUMERATED {
                                 unspecifiedOrUnknown     (0),
                                 calibratedAtomicClock    (1),
                                 radioVLForLF             (2),
                                 radioHF                  (3),
                                 radioUHF                 (4),
                                 localNet                 (5),
                                 synchNTP                 (6),
                                 wallclock                (7),
                                 telephoneModem           (8),
                                 other                    (9)}

          Stratum ::= INTEGER
          -- A value of zero means that the stratum is not
               specified. A value of one indicates a primary
               reference. Values from 2 to 255 indicate secondary
               references of increasing distance from the root of
               the synchronization subnet.

          PollInterval ::= TimeInterval
          -- This field represents the polling interval in
               seconds and can only contain positive values.

          Precision ::= TimeInterval
          -- This field represents precision in seconds and can
               only contain positive values.

          Delay ::= TimeInterval
          -- This field represents the delay in seconds.

          Dispersion ::= TimeInterval
```


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```
        -- This field represents the dispersion in seconds and
             can only contain positive values.

        ClockAddress ::= OCTET STRING SIZE (20)
        -- The ISO Network Address (NSAP) as defined for use in
             SAFENET.

        Status ::= ENUMERATED {
                             operatingWithinNominals  (0),
                             replyTimeout             (2),
                             badReplyFormat           (3),
                             hardwareSoftwareFault    (4),
                             propagationFailure       (5),
                             badDateFormatOrValue     (6),
                             badTimeFormatOrValue     (7)}

        ClockValue ::= GlobalTime

        Mode ::= BIT STRING (SIZE (3)) {
                             unspecified              (000),
                             symmetricActive          (001),
                             symmetricPassive         (010),
                             client                   (011),
                             server                   (100),
                             broadcast                (101),
                             reserved                 (110),
                             reserved                 (111)}
        -- A value of 110 is reserved for control messages.
        -- A value of 111 is reserved for private messages.

        SystemEventCode ::= ENUMERATED {
                             unspecified              (0),
                             restart                  (1),
                             systemOrHardwareFault    (2),
                             newStatusWord            (3),
                             newSynchSourceOrStratum  (4),
                             systemClockReset         (5),
                             systemInvalidTimeOrDate  (6),
                             systemClockException     (7)}

        SystemEventTime ::= GlobalTime

END -- End of NTP COMMON syntax definitions.
```

40.4.2 NTP Protocol Profile Definitions.

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```
NTP-PP
     DEFINITIONS ::= BEGIN

          TSelect ::= OCTET STRING (SIZE (4))
          -- The T-selector for NTP as defined for use in
               SAFENET.

          MaxClockAge ::= TimeInterval

          MaxSkew ::= TimeInterval

          MaxDistance ::= TimeInterval

          MinSelectClock ::= INTEGER (0 .. 255)

          MaxSelectClock ::= INTEGER (0 .. 255)

          ReachRegSize ::= INTEGER (0 .. 32)

          FilterSize ::= INTEGER (0 .. 32)

          FilterWeight ::= REAL (0.000 .. 1.000)

          SelectWeight ::= REAL (0.000 .. 1.000)

          CumLeapSeconds ::= INTEGER (0 .. 255)

          LeapIndication ::= BIT STRING (SIZE(2)) {
                               no warning                    (00),
                               last minute has 61 seconds    (01),
                               last minute has 59 seconds    (10),
                               alarm condition               (11)}

END -- End of NTP-PP syntax definitions
```

40.4.3 SAFENET Network Clock Definitions.

```
NTP-NC
     DEFINITIONS ::= BEGIN

          Version ::= BIT STRING (SIZE (3)) {
                               unspecified    (000),
                               version1       (001),
                               version2       (010),
                               version3       (011)}
```


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```
        SinglePeerClock ::= SEQUENCE {
                            assocNum [0] INTEGER,
                            assocClock [1] ClockAddress }

        PeerClocksAddresses ::= SET OF SinglePeerClock

        CurrSynchSourceAddress ::= CHOICE {
                            refPeerAssoc [0] ClockAddress,
                            refClockID [1] ReferenceClockType}
        -- This indicates either the remote Network Clock with
             which the local Network Clock has chosen to
             synchronize or the type of external reference to
             which the local Network Clock is synchronized.

        AdjInterval ::= TimeInterval

        MaxAperature ::= TimeInterval

        SystemEventCounter ::= INTEGER (0 ... 255)

        SystemEventCode ::= BIT STRING (SIZE (4)) {
                            unspecified               (0000),
                            restart                   (0001),
                            systemOrHardwareFault     (0010),
                            newStatusWord             (0011),
                            newSynchSourceOrStratum   (0100),
                            systemClockReset          (0101),
                            systemInvalidTimeOrDate   (0110),
                            systemClockException      (0111)}
        -- The values 1000 to 1111 are reserved.

        SystemEventTime ::= GlobalTime

END -- End of NTP-NC syntax definitions
```

40.4.4 NTP Peer Association Definitions.

```
NTP-PA
     DEFINITIONS ::= BEGIN

          PeerReachability ::= BIT STRING
          -- This acts as a shift register of length determined
               by the NTP-PP.ReachRegSize attribute.  Each time a
               response is received a "1" is shifted in.  When a
               response is not received, a "0" is shifted in.  A
```


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```
            peer is considered reachable as long as there is a
            "1" somewhere in the bit pattern.

        PeerEventCode ::= BIT STRING {
                            unspecified                 (0000),
                            errorCLNP                   (0001),
                            authenticationFailure       (0010),
                            statusChangedToUnreachable  (0011),
                            statusChangedToReachable    (0100),
                            peerClockException          (0101)}
        -- The values 0110 through 1111 are reserved.

END -- End of NTP-PA syntax definitions
```


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